

A comprehensive view on Trigonometric Function

Bridging HKDSE, HKALE, and University Mathematics from First Principles

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Abstract

This series of lecture notes is designed for secondary school students (HKDSE M1/M2 and HKALE) transitioning to university-level mathematical rigor. Starting from foundational synthetic geometry and coordinate systems, this curriculum systematically unifies right-angled triangle trigonometry with the unit circle, complex numbers on the Argand plane, compound angle identities, and non-Euclidean hyperbolic and spherical projections. By bridging graphical intuition with algebraic rigor, these notes provide a self-contained pathway from F.4 mathematics to advanced mathematical analysis.

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1 Polygons

1.1 Points, Lines and Angles

We begin our exploration of space from the absolute beginning. Imagine a world with no size, no width, and no depth. To understand geometry, we must build it piece by piece, starting from objects with no dimensions at all, and gradually constructing richer structures.

Definition 1.1.1 (Point). A **point** is a 0-dimensional geometric object that specifies a unique, precise position in space. It possesses no length, width, depth, or area. We typically denote a point using a capital English letter, such as A , B , or C .

Remark 1.1.1. Because a point has zero dimensions, it cannot be divided into smaller pieces. It acts as the fundamental, indivisible “particle” of our geometric universe.

Example 1.1.1 (Numerical Test on Position). Suppose we assign numerical positions to points along a single straight path (a number line). Let Point A be located at position $+3$ and Point B be located at position -2 . Find the physical distance separating these two points.

Solution. The distance between two positions on a number line is found by subtracting the lesser value from the greater value. Here, $3 > -2$.

$$\text{Distance} = 3 - (-2) = 3 + 2 = 5 \text{ units.}$$

Thus, the two points are separated by an interval of 5 units. $\square$

Example 1.1.2 (Algebraic Expression Derivation). Let P be a point represented on a coordinate grid by the algebraic variables (x, y) . If the location of P is restricted such that its horizontal distance from the vertical axis is always twice its height from the horizontal axis, write an algebraic relation representing the coordinates of P . Assume x and y are positive.

Solution. The horizontal distance from the vertical axis is represented by x . The height from the horizontal axis is represented by y . The given condition states that the horizontal distance is twice the height:

$$x = 2y.$$

This algebraic equation defines the mathematical relationship constraint on the point’s position. $\square$

Example 1.1.3 (Algebraic Expression Application). A point C is moving along a straight path. Its position at any time t (in seconds) is given by the formula $S(t) = 3t + 4$ meters. Determine the specific time t when the point reaches the position 22 meters.

Solution. We are given the position equation $S(t) = 3t + 4$. We need to find t such that $S(t) = 22$. Setting up the linear equation:

$$3t + 4 = 22.$$

Subtracting 4 from both sides:

$$3t = 18.$$

Dividing both sides by 3:

$$t = 6.$$

Thus, the point C reaches the position 22 meters at $t = 6$ seconds. $\square$

Example 1.1.4 (Abstract Application to Problems). Show that if three distinct points A , B , and C lie on a flat plane, the set of points containing only A and B cannot cover or contain C .

Solution. By definition, a point is a unique position. Since A , B , and C are distinct, their positions are mutually exclusive. Let $S = \{A, B\}$ be the set containing points A and B . Since $C \neq A$ and $C \neq B$, it follows by the definition of set membership that $C \notin S$. Thus, the set containing only the two positions A and B cannot contain the position of C . $\square$

Example 1.1.5 (Failure of the Concept). Explain why a physical pencil dot drawn on a sheet of paper is not, in a strict mathematical sense, a point.

Solution. A physical dot drawn by a pencil has a measurable diameter (usually about 0.5 mm), an area, and a microscopic thickness of graphite. In contrast, a mathematical point is strictly 0-dimensional, meaning it has exactly zero width, zero height, and zero area. Therefore, a physical dot is merely a visual representation, whereas the true mathematical point is an idealized location with no physical dimensions. $\square$

* * *

Having established the concept of static positions (points), we now discuss what happens when we connect these positions. This leads us directly to the concept of lines.

Definition 1.1.2 (Line). A **line** is a 1-dimensional geometric object formed by an infinite, continuous set of points.

- A line is **simple** if it does not cross or intersect itself.
- In a standard flat 2-dimensional space (which we call a **plane**), a **straight line** is the path representing the shortest possible distance connecting any two of its points.

Axiom 1.1.1. *Through any two distinct points in a plane, there exists exactly one unique straight line passing through them.*

Example 1.1.6 (Numerical Test on Straight Lines). Consider two points P_1 at position 1 and P_2 at position 7 on a straight track. If a traveler walks directly from P_1 to P_2 along the straight line, and then back to a point P_3 at position 4, find the total distance traveled.

Solution. The distance from $P_1(1)$ to $P_2(7)$ along a straight path is:

$$|7 - 1| = 6 \text{ units.}$$

The distance from $P_2(7)$ back to $P_3(4)$ along the straight path is:

$$|4 - 7| = |-3| = 3 \text{ units.}$$

The total distance traveled is the sum of these two straight intervals:

$$\text{Total Distance} = 6 + 3 = 9 \text{ units.}$$

$\square$

Example 1.1.7 (Algebraic Expression Derivation). The equation of a straight line in a plane is given by the algebraic expression $y = mx + c$, where m is the slope and c is a constant. If a straight line passes through the points $(0, 2)$ and $(1, 5)$, find the values of m and c .

Solution. Substitute the coordinates of the first point $(0, 2)$ into the straight line equation:

$$2 = m(0) + c \implies c = 2.$$

Now, substitute the second point $(1, 5)$ and the value of $c = 2$ into the equation:

$$5 = m(1) + 2 \implies m = 3.$$

Thus, the algebraic equation of the unique straight line passing through these two points is $y = 3x + 2$. $\square$

Example 1.1.8 (Algebraic Expression Application). A simple line is represented by the algebraic curve $y = x^2 - 4x$. A straight line is represented by $y = 2x - 9$. Find the coordinates of the point where these two lines meet (intersect).

Solution. To find the meeting point, we set the two algebraic expressions equal to one another:

$$x^2 - 4x = 2x - 9.$$

Rearranging all terms to one side of the equation:

$$x^2 - 6x + 9 = 0.$$

Factoring the quadratic expression:

$$(x - 3)^2 = 0 \implies x = 3.$$

Substitute $x = 3$ back into the straight line equation to find the corresponding value of y :

$$y = 2(3) - 9 = -3.$$

Thus, the two lines intersect at the single point $(3, -3)$. $\square$

Example 1.1.9 (Abstract Application to Problems). Prove abstractly that two distinct straight lines in a plane can intersect at most at one point.

Solution. We prove this by contradiction. Suppose two distinct straight lines, L_1 and L_2 , intersect at two different points, A and B . This means that both A and B lie on L_1 , and both A and B also lie on L_2 .

However, by our fundamental geometric axiom, there exists exactly one unique straight line that can pass through any two distinct points A and B . This means L_1 and L_2 must be the exact same line, which contradicts our starting assumption that L_1 and L_2 are distinct. Thus, two distinct straight lines can intersect at most at one point. $\square$

Example 1.1.10 (Failure of the Concept (Non-Simple Line)). Explain why the path described by the algebraic parametric coordinates $(t^2 - 1, t^3 - t)$ for $t \in \mathbb{R}$ is not a simple line.

Here, a parametric curve is where x and y are both functions of another variable. This time, it is t . You may or may not be able to write the explicit equation relating x and y , but both can be calculated according to t .

Solution. By definition, a simple line is one that never crosses or meets itself. Let us check if this path passes through the same point for two different times t_1 and t_2 . We set the coordinates equal:

$$t_1^2 - 1 = t_2^2 - 1 \implies t_1^2 = t_2^2 \implies t_1 = -t_2 \quad (\text{since } t_1 \neq t_2).$$

Now substitute this relation into the second coordinate:

$$t_1^3 - t_1 = t_2^3 - t_2.$$

Using $t_2 = -t_1$:

$$t_1^3 - t_1 = (-t_1)^3 - (-t_1) \implies t_1^3 - t_1 = -t_1^3 + t_1 \implies 2(t_1^3 - t_1) = 0.$$

This has non-zero solutions at $t_1 = 1$ and $t_2 = -1$. Evaluating the coordinates at these times: For $t = 1$: $(1^2 - 1, 1^3 - 1) = (0, 0)$. For $t = -1$: $((-1)^2 - 1, (-1)^3 - (-1)) = (0, 0)$. Because the line meets itself at the origin $(0, 0)$ for two distinct parameters, it intersects itself and is therefore not a simple line. $\square$

* * *

When lines meet, they do not merely cross; they split the surrounding flat space. To measure this splitting, we introduce the concept of parallel lines and angles.

Definition 1.1.3 (Parallel Lines and Angles). Let L_1 and L_2 be two straight lines in the same plane.

- L_1 and L_2 are **parallel** (written as $L_1 \parallel L_2$) if they never intersect, no matter how far they are extended.
- If they are not parallel, they intersect at a unique **point of intersection**. At this point, the divergence between the directions of the two lines is called an **angle**.
- A complete, full rotation around a single point is defined to have a measure of 360° (degrees).
- An angle forming a straight line is called a **straight angle** and measures 180° .
- An angle of exactly 90° is called a **right angle**. Two lines meeting at a right angle are said to be **perpendicular** (written as $L_1 \perp L_2$).
- An angle θ is **acute** if $0^\circ < \theta < 90^\circ$, and **obtuse** if $90^\circ < \theta < 180^\circ$.

Theorem 1.1.1 (Angles on a Straight Line). *If a straight line is divided into two adjacent angles θ_1 and θ_2 by a ray originating from a point on the line, then $\theta_1 + \theta_2 = 180^\circ$.*

Proof. $\square$

Example 1.1.11 (Numerical Test on Angles). Three adjacent angles lie on a single straight line. Their measures are $2x$, $3x$, and 40° . Find the numerical value of x .

Solution. By the adjacent angles on a straight line theorem, the sum of these angles must equal 180° :

$$2x + 3x + 40^\circ = 180^\circ.$$

Combining like terms:

$$5x + 40^\circ = 180^\circ \implies 5x = 140^\circ \implies x = 28^\circ.$$

$\square$

Example 1.1.12 (Algebraic Expression Derivation). Suppose two straight lines intersect at a point, forming four angles around the intersection point. Let these angles in clockwise order be α , β , γ , and δ . Prove algebraically that the opposite (vertically opposite) angles are equal, i.e., $\alpha = \gamma$.

Solution. Since the lines are straight, α and β lie on a straight line. Therefore:

$$\alpha + \beta = 180^\circ \implies \alpha = 180^\circ - \beta.$$

Similarly, β and γ lie on a straight line. Therefore:

$$\beta + \gamma = 180^\circ \implies \gamma = 180^\circ - \beta.$$

Since both α and γ are equal to $180^\circ - \beta$, we have:

$$\alpha = \gamma.$$

□

Example 1.1.13 (Algebraic Expression Application). An angle measures $(3y + 15)^\circ$ and its adjacent angle on a straight line measures $(2y - 10)^\circ$. Determine if the first angle is acute, right, or obtuse.

Solution. Since they are adjacent angles on a straight line, their sum is 180° :

$$(3y + 15) + (2y - 10) = 180 \implies 5y + 5 = 180 \implies 5y = 175 \implies y = 35.$$

Now, evaluate the measure of the first angle:

$$\text{First Angle} = 3(35) + 15 = 105 + 15 = 120^\circ.$$

Since $90^\circ < 120^\circ < 180^\circ$, the first angle is **obtuse**.

□

Example 1.1.14 (Abstract Application to Problems). Let L_1 and L_2 be two perpendicular lines. If a third line L_3 is drawn such that the angle it makes with L_1 is 30° , show that the acute angle it makes with L_2 must be 60° .

Solution. Since $L_1 \perp L_2$, the angle between them is 90° . Let the direction of L_1 be represented as our base line (0°). Since $L_2 \perp L_1$, L_2 is at 90° .

The line L_3 makes an angle of 30° with L_1 . Thus, the direction of L_3 can be written as 30° . The remaining angle between L_3 and L_2 is:

$$90^\circ - 30^\circ = 60^\circ.$$

Since 60° is less than 90° , it is indeed the acute angle between L_3 and L_2 .

□

Example 1.1.15 (Failure of the Concept (Curved Space)). On the surface of a globe (sphere), two longitude lines start from the North Pole. They both go straight down to the Equator. Explain how they can both be perpendicular (90°) to the Equator and yet intersect each other at the North Pole, violating flat-plane parallel behavior.

Solution. In flat-plane geometry, if two lines are perpendicular to a third line, they must be parallel to each other and never meet. However, on a curved surface like a sphere, the rules of 2D flat planes do not apply.

The two longitude lines meet the Equator at 90° angles, which would imply they are parallel near the Equator. Yet, they meet and intersect at the North Pole. This occurs because the surface of a sphere is non-Euclidean, showing that our flat-plane definitions of parallel lines and constant angles fail on curved surfaces.

□

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Problems for Section 1.1

1. A point P is at position -12 on a number line, and point Q is at position 18 . Find the distance between P and Q .
2. If the distance between a point $A(x)$ on a number line and $B(5)$ is 13 units, find all possible numerical values for x .
3. A point moves along a straight path with position equation $s(t) = t^2 - 2t + 5$. Find the two times t when the point is at position 13 .
4. Write the algebraic equation of the straight line passing through the points $(1, 4)$ and $(3, 10)$.
5. Determine if the three points $A(1, 2)$, $B(3, 6)$, and $C(5, 10)$ lie on the exact same straight line.
6. Two lines $y = 3x - 5$ and $y = -2x + 10$ meet at point P . Find the coordinates of P .
7. An angle is three times the size of its adjacent angle on a straight line. Find the size of both angles.
8. Three angles lie on a straight line: x° , $(2x - 10)^\circ$, and $(x + 30)^\circ$. Solve for x .
9. If two lines intersect to form an angle θ such that $\theta = 3x + 12$ and its vertically opposite angle is $5x - 8$, solve for x and find the angle.
10. Two lines meet at a point. If one of the angles formed is 90° , prove algebraically that all four angles formed at the intersection must be 90° .
11. Prove that if line A is parallel to line B , and line B is parallel to line C , then line A must be parallel to line C in a flat plane.
12. Let θ be an angle. If the difference between its supplement (the angle that adds with it to 180°) and its complement (the angle that adds with it to 90°) is evaluated, show that this difference is always 90° , regardless of the value of θ .
13. If a line $y = mx + 4$ is perpendicular to $y = 2x - 1$, find the value of m . (Recall that perpendicular lines have slopes multiplying to -1).
14. Let three lines intersect at a single point. Show that the sum of all six angles formed around this point is exactly 360° .
15. A simple line curve is given by $y = x^2 - x - 6$. Find the points where it intersects the straight horizontal line $y = 0$.
16. A traveler moves on a coordinate plane from point $A(0, 0)$ along a straight line to $B(4, 3)$. Use the Pythagorean theorem to calculate this straight-line distance.
17. Show that the parametric path (t^3, t^6) for $t \in \mathbb{R}$ is a simple line, whereas $(\sin t, \cos(2t))$ is not.
18. Two intersecting lines form an acute angle of $(5x - 15)^\circ$. If this angle is half the size of its adjacent obtuse angle, solve for x .
19. If a straight line is split into four equal angles by three rays, what is the measure of each angle?
20. Explain abstractly why a point cannot have a “slope” or a “direction” associated with it.

1.2 Properties of Parallel Lines

In the previous subsection, we established the fundamental nature of points and straight lines, and we observed that two straight lines in a flat plane either meet at a single point of intersection or never meet at all. When they never meet, we call them parallel.

To explore the deeper properties of parallel lines, we must introduce a third line that intersects them. This act of crossing is highly intuitive: imagine a straight railway track with two parallel rails, crossed diagonally by a straight road. This road acts as a **transversal**, and the angles formed at the two separate intersection points are beautifully locked in harmony.

Definition 1.2.1 (Transversal). A **transversal** is a straight line that intersects two or more other straight lines in the same plane at distinct points.

Example 1.2.1 (Numerical Test on Transversals). Suppose a transversal line L_3 intersects line L_1 at point P and line L_2 at point Q . If the distance along the transversal between P and Q is 10 cm, and another transversal L_4 crosses them such that its intersection points are R on L_1 and S on L_2 with a distance of 15 cm, determine the total number of intersection points formed by these four lines, assuming none of them are parallel.

Solution. We have four distinct lines: L_1, L_2, L_3 , and L_4 . If no two lines are parallel and no three lines meet at the same point, the maximum number of intersection points among n lines is given by the combination formula:

$$\text{Number of Intersections} = \frac{n(n-1)}{2}.$$

For $n = 4$:

$$\text{Number of Intersections} = \frac{4(4-1)}{2} = \frac{12}{2} = 6.$$

Thus, there are 6 unique intersection points in this system. $\square$

Example 1.2.2 (Algebraic Expression Derivation). Let line L_1 be represented by the equation $y = 2$, and line L_2 be represented by $y = 5$. Let a transversal L_3 be represented by the equation $y = 3x - 1$. Derive the coordinates of the two distinct intersection points P and Q where the transversal cuts L_1 and L_2 respectively.

Solution. To find the intersection point P of L_1 and L_3 , we set their y -values equal:

$$2 = 3x - 1 \implies 3x = 3 \implies x = 1.$$

So, P is located at $(1, 2)$.

To find the intersection point Q of L_2 and L_3 , we set their y -values equal:

$$5 = 3x - 1 \implies 3x = 6 \implies x = 2.$$

So, Q is located at $(2, 5)$. $\square$

Example 1.2.3 (Algebraic Expression Application). A transversal L_3 cuts two lines L_1 and L_2 . The intersection points are $A(a, 3a)$ and $B(4, 12)$. If the transversal is a straight line passing through the origin $(0, 0)$, show that a can be any non-zero real number, and write the algebraic equation of the transversal.

Solution. Since the transversal passes through the origin $(0, 0)$ and the point $B(4, 12)$, its slope m is:

$$m = \frac{12 - 0}{4 - 0} = 3.$$

Thus, the equation of the transversal is $y = 3x$. For any point $A(a, 3a)$ to lie on this transversal line, we substitute its coordinates into the equation:

$$3a = 3(a).$$

This statement is an identity that is true for all real values of a . Since A and B must be distinct points for L_3 to be a proper transversal cutting two separate lines, $A \neq B$, meaning $a \neq 4$. $\square$

Example 1.2.4 (Abstract Application to Problems). Let L_1 and L_2 be two parallel lines. Let L_3 be a transversal that is perpendicular to L_1 . Show abstractly that L_3 must also be perpendicular to L_2 .

Solution. By definition, L_1 and L_2 are parallel, meaning they share the same spatial direction. The transversal L_3 intersects L_1 at a right angle (90°). Because the directional orientation of L_2 is identical to L_1 , the angle of divergence between the transversal L_3 and L_2 must be identical to the angle of divergence between L_3 and L_1 . Therefore, the intersection angle between L_3 and L_2 is also 90° , making $L_3 \perp L_2$. $\square$

Example 1.2.5 (Pure Approach to Understanding). Visualize a transversal as a straight blade slicing through two parallel wooden planks. Explain in plain, non-technical language why the geometric patterns of the corners created at both planks must look identical.

Solution. Because the two wooden planks are parallel, they point in the exact same direction. When the straight blade cuts through both planks at a constant slant, it does not twist or bend. As a result, the blade meets the second plank at the exact same slant angle that it met the first plank. The shape of the corners, the sharpness of the splits, and the arrangement of the open spaces around the cut are identical copy-pastes of each other, shifted along the path of the blade. $\square$

* * *

When a transversal cuts two parallel lines, several pairs of angles are formed that share special relationships. We classify these relationships into three primary categories: **corresponding angles**, **alternate angles**, and **interior angles**.

Theorem 1.2.1 (Parallel Line Angle Theorems). *If two parallel lines are cut by a transversal, then:*

1. **Corresponding angles are equal** (*corr. $\angle$ s, $\parallel$ lines*). These are angles in the same relative position at each intersection.
2. **Alternate angles are equal** (*alt. $\angle$ s, $\parallel$ lines*). These are angles on opposite sides of the transversal and inside the parallel lines.
3. **Interior angles are supplementary** (*int. $\angle$ s, $\parallel$ lines*). These are angles on the same side of the transversal and inside the parallel lines, summing to 180° .

Proof. $\square$

Example 1.2.6 (Numerical Test on Parallel Angles). In the figure below, two parallel lines $L_1 \parallel L_2$ are cut by a transversal L_3 . If one of the acute corresponding angles is measured to be 72° , calculate the measure of its adjacent interior angle on the same side of the transversal.

Solution. Let the given corresponding angle be $\alpha = 72^\circ$. By the properties of parallel lines, the interior angle β on the same side of the transversal is supplementary to α . Therefore:

$$\alpha + \beta = 180^\circ.$$

Substituting the known value:

$$72^\circ + \beta = 180^\circ \implies \beta = 180^\circ - 72^\circ = 108^\circ.$$

Thus, the interior angle measures 108° . □

Example 1.2.7 (Algebraic Expression Derivation). Two parallel lines are intersected by a transversal. A pair of alternate angles are represented by the algebraic expressions $(5x - 30)^\circ$ and $(2x + 15)^\circ$. Derive the value of x and determine the actual size of these angles.

Solution. Since the lines are parallel, their alternate angles must be equal:

$$5x - 30 = 2x + 15.$$

Subtract $2x$ from both sides:

$$3x - 30 = 15.$$

Add 30 to both sides:

$$3x = 45 \implies x = 15.$$

Now substitute $x = 15$ back into either expression to find the angle:

$$\text{Angle} = 2(15) + 15 = 30 + 15 = 45^\circ.$$

Both alternate angles measure 45° . □

Example 1.2.8 (Algebraic Expression Application). A transversal cuts two parallel lines. A pair of interior angles on the same side of the transversal are represented by $(3y + 20)^\circ$ and $(y - 12)^\circ$. Find the value of y and determine if either of these interior angles is obtuse.

Solution. Since the lines are parallel, the interior angles on the same side of the transversal must add up to 180° :

$$(3y + 20) + (y - 12) = 180.$$

Combine like terms:

$$4y + 8 = 180.$$

Subtract 8 from both sides:

$$4y = 172 \implies y = 43.$$

Now, evaluate the two interior angles:

$$\text{First Angle} = 3(43) + 20 = 129 + 20 = 149^\circ.$$

$$\text{Second Angle} = 43 - 12 = 31^\circ.$$

Since $149^\circ > 90^\circ$, the first interior angle is indeed obtuse. □

Example 1.2.9 (Abstract Application to Problems). Let $L_1 \parallel L_2$. A transversal cuts them, forming an alternate angle pair θ_1 and θ_2 , and a corresponding angle pair θ_1 and θ_3 . Using only the axiom that corresponding angles are equal, show abstractly that alternate angles must also be equal ($\theta_1 = \theta_2$).

Solution. Let θ_1 be an interior angle formed by the transversal cutting L_1 . By the axiom of corresponding angles, the angle in the same relative position at the intersection with L_2 is θ_3 , so:

$$\theta_1 = \theta_3.$$

Now observe the relationship between θ_3 and θ_2 (which is the alternate angle to θ_1). At the second intersection, θ_3 and θ_2 are vertically opposite angles. We proved in Subsection 1.1 that vertically opposite angles are always equal:

$$\theta_3 = \theta_2.$$

Combining these two equations:

$$\theta_1 = \theta_3 = \theta_2 \implies \theta_1 = \theta_2.$$

Thus, alternate angles are equal. □

Example 1.2.10 (Failure of the Theorem). Suppose two non-parallel lines M_1 and M_2 are intersected by a transversal. If one angle is 80° , can we conclude that its alternate angle is also 80° ? Explain why.

Solution. No, we cannot. The equality of alternate angles relies entirely on the condition that the two lines being crossed are parallel ($M_1 \parallel M_2$). When the lines are not parallel, they point in different directions. The transversal will slice through them at different angles of inclination, meaning the alternate angles will have different values. Thus, the theorem fails when the parallel constraint is removed. □

* * *

In geometry, we often need to go backward: if we do not know whether two lines are parallel, how can we prove they are? The converse of our parallel line theorems provides the tools to do exactly this.

Theorem 1.2.2 (Converse Parallel Line Theorems). *Let L_1 and L_2 be two straight lines cut by a transversal.*

1. *If any pair of corresponding angles are equal, then $L_1 \parallel L_2$ (corr. $\angle$ s equal).*
2. *If any pair of alternate angles are equal, then $L_1 \parallel L_2$ (alt. $\angle$ s equal).*
3. *If any pair of interior angles on the same side are supplementary, then $L_1 \parallel L_2$ (int. $\angle$ s supp.).*

Proof. By contradiction. □

Example 1.2.11 (Numerical Test on Proving Parallel). A surveyor measures the angles formed by a transversal cutting two fences. The angle on the upper fence is 115° . The alternate angle on the lower fence is measured to be 115° . Are the two fences parallel?

Solution. The surveyor has measured a pair of alternate angles, and both are exactly 115° . According to the converse of the alternate angle theorem, if alternate angles are equal, the two lines must be parallel. Since:

$$115^\circ = 115^\circ,$$

the two fences are indeed parallel. □

Example 1.2.12 (Algebraic Expression Derivation). Two lines are cut by a transversal. The corresponding angles are given by $A = x^2 - 4x$ and $B = 2x + 16$. Derive the values of x that would make the two lines parallel.

Solution. To make the lines parallel, the corresponding angles must be equal:

$$x^2 - 4x = 2x + 16.$$

Rearranging into a standard quadratic form:

$$x^2 - 6x - 16 = 0.$$

Factoring the quadratic equation:

$$(x - 8)(x + 2) = 0.$$

Thus, the values are $x = 8$ or $x = -2$. Either of these values will make the corresponding angles equal and therefore guarantee the lines are parallel. $\square$

Example 1.2.13 (Algebraic Expression Application). Determine if the line L_1 and line L_2 are parallel if the interior angles on the same side of a transversal are given by $u = 5k + 10$ and $v = 3k - 6$, where $k = 22$.

Solution. First, we evaluate the measures of the two angles u and v using $k = 22$:

$$u = 5(22) + 10 = 110 + 10 = 120^\circ.$$

$$v = 3(22) - 6 = 66 - 6 = 60^\circ.$$

Next, we check if these two interior angles on the same side are supplementary (add up to 180°):

$$u + v = 120^\circ + 60^\circ = 180^\circ.$$

Since the sum is exactly 180° , by the converse interior angle theorem, the two lines L_1 and L_2 are parallel. $\square$

Example 1.2.14 (Abstract Application to Problems). Let L_1, L_2 , and L_3 be three lines in a plane. If $L_1 \perp L_3$ and $L_2 \perp L_3$, use the converse parallel theorems to prove that $L_1 \parallel L_2$.

Solution. The line L_3 acts as a transversal crossing both L_1 and L_2 . Since $L_1 \perp L_3$, the angle of intersection between L_1 and L_3 is 90° . Since $L_2 \perp L_3$, the angle of intersection between L_2 and L_3 is also 90° .

These two 90° angles lie in corresponding positions at each intersection. Since the corresponding angles are equal ($90^\circ = 90^\circ$), we apply the converse corresponding angles theorem. Therefore, L_1 must be parallel to L_2 . $\square$

Example 1.2.15 (Analytic Discussion on Parallel Proofs). A student attempts to prove two lines L_1 and L_2 are parallel by showing that an acute angle at L_1 is 50° and an obtuse angle at L_2 is 130° . Analyze whether this information is sufficient to prove the lines are parallel.

Solution. Let us analyze the position of these angles. If the 50° acute angle and the 130° obtuse angle are interior angles on the same side of the transversal, their sum is:

$$50^\circ + 130^\circ = 180^\circ.$$

Since they are supplementary, this would prove $L_1 \parallel L_2$.

However, if they are not interior angles on the same side (for instance, if they are alternate interior and exterior angles, or if they are in positions that do not correspond), their supplementary nature does not automatically satisfy the converse theorems without establishing their relative locations first. Therefore, the numerical values alone are **only** sufficient if their specific relative positions as interior angles on the same side are explicitly given. $\square$

Problems for Section 1.2

1. Two parallel lines are cut by a transversal. If one interior angle is 105° , find the other interior angle on the same side of the transversal.
2. In a diagram where $AB \parallel CD$, a transversal cuts them. If a pair of corresponding angles are $(3x - 15)^\circ$ and $(2x + 10)^\circ$, solve for x .
3. Two lines are cut by a transversal. The alternate interior angles are $(4y + 20)^\circ$ and $(6y - 10)^\circ$. If the lines are parallel, find y .
4. Two interior angles on the same side of a transversal crossing parallel lines are in the ratio $2 : 3$. Find the measure of the smaller angle.
5. A transversal cuts two parallel lines. If the sum of a pair of alternate interior angles is 160° , find the measure of each angle.
6. If $L_1 \parallel L_2$ and a transversal cuts them, prove that the bisectors of any pair of alternate interior angles are parallel to each other.
7. In a coordinate plane, line L_1 has the equation $y = 4x + 2$. Line L_2 has the equation $y = 4x - 7$. Explain why these two lines will never meet, and state the name of their geometric relationship.
8. A transversal cuts two lines. If the corresponding angles are $x^2 + 50$ and $15x$, find all possible values of x that make the lines parallel.
9. Three lines L_1, L_2, L_3 are cut by a transversal. If $L_1 \parallel L_2$ and $L_2 \parallel L_3$, and an angle on L_1 is 110° , find the corresponding angle on L_3 .
10. In a given figure, line AB is cut by a transversal EF at P , and line CD is cut by EF at Q . If $\angle APQ = 75^\circ$ and $\angle CQD = 105^\circ$, determine whether $AB \parallel CD$.
11. A lightning bolt cuts across two parallel electrical wires. If the bolt makes an alternate angle of $(x^2 - 10)^\circ$ with the first wire and $(3x + 18)^\circ$ with the second wire, solve for the physical angle x .
12. Prove that if a transversal cuts two parallel lines, the bisectors of the interior angles on the same side meet at a right angle (90°).
13. If two parallel lines are cut by a transversal, show that the sum of any pair of alternate exterior angles is not necessarily 180° , but they are always equal.
14. Two lines are cut by a transversal. If the interior angles on the same side are $(11k - 5)^\circ$ and $(7k + 5)^\circ$, find the value of k that makes the lines parallel.
15. Let $L_1 \parallel L_2$. If a transversal T cuts them, and we know that the sum of three of the eight angles formed is 300° , find all eight angles.
16. Prove by contradiction that if a transversal intersects two lines such that the interior angles on the same side are not supplementary, then the two lines must eventually meet.
17. In a classic Z-shape configuration, the top horizontal line is parallel to the bottom horizontal line. If the top-left angle is 42° , find the bottom-right angle of the Z-shape.

18. Two parallel lines are cut by a transversal. One angle is x° and another angle (not vertically opposite, adjacent, or alternate) is $(3x)^\circ$. List all possible pairs of these angles and solve for x .
19. A drafting tool has two parallel rulers. A diagonal bar connects them. If the angle between the diagonal bar and the top ruler is increased by 5° , by how many degrees must the angle on the bottom ruler change to keep them parallel?
20. If two lines L_1 and L_2 are cut by a transversal T such that the corresponding angles are supplementary, under what condition are these lines parallel?

1.3 Polygons

In the previous subsections, we studied individual points, the straight lines that connect them, and the angles formed when a third line crosses them. However, our geometric world has so far remained “open”—our lines stretch out infinitely, and our angles are unbound. To create shapes that can enclose space, we must allow multiple lines to intersect and link together in a closed loop.

This brings us to the concept of a **polygon**. Just as a point is the simplest zero-dimensional object, and a line segment is the simplest one-dimensional link, a **triangle** is the absolute minimal closed shape we can form in a two-dimensional flat plane. We cannot enclose any area with only one or two straight lines; we require at least three. Therefore, the triangle serves as the primary building block for all other multi-sided shapes.

Definition 1.3.1 (Polygon). A **polygon** is a closed, two-dimensional flat shape bounded by three or more straight line segments connected end-to-end.

- The points where the line segments meet are called **vertices** (singular: **vertex**).
- The line segments themselves are called the **sides** of the polygon.
- A polygon is **simple** if its boundary does not cross itself.
- The **minimal polygon** in a plane is a **triangle**, which has exactly three vertices and three sides.

Example 1.3.1 (Numerical Test on Polygon Vertices). If a simple polygon has n vertices, and we are told that $n < 5$ and $n > 2$, determine the unique geometric name of this polygon and state its number of sides.

Solution. The number of vertices n must be an integer. The only integer strictly between 2 and 5 is:

$$n = 3.$$

By definition, a polygon with exactly 3 vertices is a **triangle**. A polygon always has the same number of sides as vertices, so this shape has exactly 3 sides. $\square$

Example 1.3.2 (Algebraic Expression Derivation). The perimeter of a polygon is the total length of its boundary. Let a triangle have three side lengths represented algebraically by x , $2x - 3$, and $x + 5$. Derive an algebraic expression for the perimeter $P(x)$ of this triangle, and determine the domain restriction on x to ensure all side lengths are physically positive.

Solution. The perimeter $P(x)$ is the sum of the three side lengths:

$$P(x) = x + (2x - 3) + (x + 5).$$

Combining like terms:

$$P(x) = 4x + 2.$$

To ensure all sides have positive physical lengths, each individual expression must be greater than zero:

- 1) $x > 0$
- 2) $2x - 3 > 0 \implies x > 1.5$
- 3) $x + 5 > 0 \implies x > -5$

To satisfy all three inequalities simultaneously, we must have $x > 1.5$. Thus, the domain restriction is $x \in (1.5, \infty)$. $\square$

Example 1.3.3 (Algebraic Expression Application). A student is constructing a regular polygon where every side has a length of $(k + 2)$ cm. If the perimeter of this 8-sided polygon (an octagon) is measured to be 56 cm, solve for the parameter k .

Solution. Since the polygon is regular, all 8 sides are equal in length. The formula for the perimeter is:

$$P = 8 \times \text{side length} = 8(k + 2).$$

We are given that the perimeter $P = 56$ cm. Setting up the equation:

$$8(k + 2) = 56.$$

Dividing both sides by 8:

$$k + 2 = 7 \implies k = 5.$$

$\square$

Example 1.3.4 (Abstract Application to Problems). Show abstractly that you cannot form a simple closed polygon in a flat plane using only two straight line segments.

Solution. Let us have two distinct straight line segments, S_1 connecting points A and B , and S_2 connecting points C and D .

For these segments to form a closed loop, the ends of the segments must meet. This requires pairing the vertices: let A merge with C at a single point, and let B merge with D at another single point.

This means both S_1 and S_2 are straight line segments connecting the exact same two points (let us call them P_1 and P_2). However, by our fundamental geometric axiom, there exists exactly one unique straight line passing through any two distinct points. Therefore, S_1 and S_2 must lie directly on top of each other along the same path. They cannot bow out or separate to enclose any open region or space. Thus, a minimum of three line segments is required to form a closed polygon. $\square$

Example 1.3.5 (Failure of the Concept). Explain why the boundary of a semi-circle (consisting of a straight diameter and a curved arc) is not a polygon.

Solution. By definition, all boundary segments of a polygon must be straight line segments. While a semi-circle is a closed two-dimensional shape, its boundary contains a curved arc. Since this curved arc is not a straight line segment, the semi-circle fails the definition and is not classified as a polygon. $\square$

* * *

Now that we have defined triangles and general polygons, we turn our attention to their interior angles. The behavior of these angles is beautifully consistent and forms the foundation of all polygon analysis.

Definition 1.3.2 (Triangles by Angles). We classify triangles based on their interior angles:

- An **acute triangle** is a triangle where all three interior angles are acute ($< 90^\circ$).
- A **right-angled triangle** is a triangle containing exactly one right angle (90°).
- An **obtuse triangle** is a triangle containing exactly one obtuse angle ($> 90^\circ$).

Theorem 1.3.1 (Angle Sum of a Triangle). *The sum of the three interior angles of any triangle in a flat plane is exactly 180° ($\angle$ sum of $\triangle$).*

Proof. Let $\triangle ABC$ be a triangle with interior angles $\alpha = \angle BAC$, $\beta = \angle ABC$, and $\gamma = \angle BCA$. Through vertex A , construct a straight line L parallel to side BC (by Euclid's Parallel Postulate). The line L forms a straight angle of 180° at vertex A . This straight angle is partitioned into three adjacent angles: the interior angle α , a left angle θ_1 , and a right angle θ_2 , such that:

$$\theta_1 + \alpha + \theta_2 = 180^\circ.$$

Since line L is parallel to side BC , segment AB acts as a transversal. By the Alternate Angles Theorem, $\theta_1 = \beta$. Similarly, segment AC acts as a transversal, giving $\theta_2 = \gamma$. Substituting these alternate angles into our straight-line equation yields:

$$\beta + \alpha + \gamma = 180^\circ \implies \angle A + \angle B + \angle C = 180^\circ.$$

□

Theorem 1.3.2 (Angle Sum of a Polygon). *The sum of the interior angles of a simple polygon with n sides is given by the formula:*

$$S_n = (n - 2) \times 180^\circ.$$

Proof. Take G to be the center of the polygon, and divide the polygon into n pieces of triangles. Each triangle has a sum of 180° , and altogether $180^\circ n$. Subtracting the central 360° and we conclude the sum of interior angles in an n -sided polygon is $(n - 2)180^\circ$. □

Example 1.3.6 (Numerical Test on Interior Angles). A quadrilateral (a 4-sided polygon) has three interior angles measuring 70° , 110° , and 120° . Find the measure of the fourth interior angle.

Solution. Using the polygon angle sum theorem for $n = 4$:

$$\text{Total Sum} = (4 - 2) \times 180^\circ = 2 \times 180^\circ = 360^\circ.$$

Let the fourth angle be θ . Setting up the sum:

$$70^\circ + 110^\circ + 120^\circ + \theta = 360^\circ.$$

$$300^\circ + \theta = 360^\circ \implies \theta = 60^\circ.$$

The fourth interior angle is 60° . □

Example 1.3.7 (Algebraic Expression Derivation). The interior angles of a pentagon (a 5-sided polygon) are represented by the arithmetic sequence x° , $(x + 20)^\circ$, $(x + 40)^\circ$, $(x + 60)^\circ$, and $(x + 80)^\circ$. Derive the value of x .

Solution. For a 5-sided polygon ($n = 5$), the sum of the interior angles is:

$$S_5 = (5 - 2) \times 180^\circ = 3 \times 180^\circ = 540^\circ.$$

Adding the five algebraic angle terms together:

$$x + (x + 20) + (x + 40) + (x + 60) + (x + 80) = 540.$$

Combine like terms:

$$5x + 200 = 540.$$

Subtract 200 from both sides:

$$5x = 340 \implies x = 68.$$

The first angle x is 68° . □

Example 1.3.8 (Algebraic Expression Application). In a right-angled triangle, one of the acute angles is twice the size of the other acute angle. Find the size of both acute angles.

Solution. Let the smaller acute angle be a° . The second acute angle is therefore $2a^\circ$. Since the triangle is right-angled, the third angle is exactly 90° .

Using the triangle angle sum theorem:

$$a + 2a + 90 = 180.$$

Combine terms:

$$3a + 90 = 180 \implies 3a = 90 \implies a = 30.$$

The smaller acute angle is 30° , and the larger acute angle is $2(30^\circ) = 60^\circ$. □

Example 1.3.9 (Abstract Application to Problems). Prove abstractly that a triangle can never have more than one right angle.

Solution. We prove this by contradiction. Suppose a triangle has two right angles, say $\angle A = 90^\circ$ and $\angle B = 90^\circ$. Let the third angle be $\angle C$.

By the triangle angle sum theorem, the sum of all three angles must equal 180° :

$$\angle A + \angle B + \angle C = 180^\circ.$$

Substitute the assumed values:

$$90^\circ + 90^\circ + \angle C = 180^\circ \implies 180^\circ + \angle C = 180^\circ.$$

Subtracting 180° from both sides yields:

$$\angle C = 0^\circ.$$

An angle of 0° means that the third vertex lies directly on the line connecting the first two vertices, so no actual closed two-dimensional shape can be formed. This contradicts the definition of a triangle having three distinct non-zero angles. Thus, a triangle can never have more than one right angle. □

Example 1.3.10 (Analytic Discussion on Polygon Decomposition). Explain how the polygon angle sum formula $(n - 2) \times 180^\circ$ is derived analytically by dividing any simple n -sided polygon into triangles.

Solution. Choose any single vertex of an n -sided polygon. Draw straight lines (diagonals) from this chosen vertex to all other non-adjacent vertices.

The polygon has n vertices. The chosen vertex cannot draw a diagonal to itself, nor can it draw diagonals to its two immediate neighbors (since those lines are the outer edges of the polygon). This leaves $n - 3$ diagonals that can be drawn.

These $n - 3$ diagonals split the interior space of the polygon into exactly $n - 2$ separate triangles. Since the interior angles of these triangles perfectly cover the interior angles of the polygon, and each triangle has an angle sum of 180° , the total interior angle sum of the polygon must be $(n - 2) \times 180^\circ$. $\square$

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Problems for Section 1.3

1. Find the sum of the interior angles of a decagon (a 10-sided polygon).
2. If the sum of the interior angles of a polygon is 1440° , how many sides does the polygon have?
3. Two of the angles in a triangle are 47° and 83° . Find the third angle.
4. In a triangle, the three interior angles are in the ratio $1 : 3 : 5$. Find the size of the largest angle.
5. Find the size of each interior angle of a regular hexagon (a 6-sided polygon with equal angles).
6. If a regular polygon has an interior angle of 144° , how many sides does it have?
7. The interior angles of a hexagon are x° , $2x^\circ$, $3x^\circ$, $4x^\circ$, $5x^\circ$, and $6x^\circ$. Solve for x .
8. Prove that an equilateral triangle (all sides equal) must have three interior angles that each measure 60° .
9. A triangle has interior angles $(x + 10)^\circ$, $(2x - 5)^\circ$, and $(x + 15)^\circ$. Determine if this triangle is acute, right-angled, or obtuse.
10. If one interior angle of a triangle is equal to the sum of the other two interior angles, prove that the triangle must be right-angled.
11. An exterior angle of a polygon is formed by extending one of its sides. If the sum of an interior angle and its adjacent exterior angle is always 180° , prove that the sum of the exterior angles of any simple polygon is always 360° .
12. In a quadrilateral $ABCD$, $\angle A = 2x^\circ$, $\angle B = 3x^\circ$, $\angle C = 4x^\circ$, and $\angle D = 3x^\circ$. Find all four angles.
13. Prove that a triangle can have at most one obtuse angle.
14. A heptagon (7-sided polygon) has five interior angles that each measure 130° . If the remaining two interior angles are equal to each other, find their measure.
15. Find the number of diagonals that can be drawn in a nonagon (a 9-sided polygon). (Hint: Think about how many diagonals can be drawn from each vertex).

16. The perimeter of an isosceles triangle (a triangle with two equal sides) is 32 cm. If the unequal side is 10 cm, find the lengths of the two equal sides.
17. If the interior angle of a regular polygon is five times its exterior angle, find the number of sides of this polygon.
18. Let ABC be a triangle. If we extend the side BC to a point D , prove that the exterior angle $\angle ACD$ is equal to the sum of the two opposite interior angles, $\angle CAB + \angle ABC$.
19. A polygon has n sides. If we double the number of sides to $2n$, by how much does the sum of its interior angles increase?
20. Explain abstractly why we cannot have a regular polygon with an interior angle of exactly 130° . (Hint: The number of sides n must be an integer).

1.4 The Trigonometry

In the previous subsections, we explored the qualitative nature of polygons, focusing on how lines meet to form closed boundaries and how their interior angles are strictly constrained. However, geometry is not merely a study of shapes and angles; it is also a study of size, distance, and measurement. To fully analyze these shapes, we must bridge the gap between angle measurements and side lengths. This bridge is called **trigonometry** (literally, “triangle measurement”).

We begin by establishing the concept of a **metric**—the physical distance or length of a line segment. By focusing on the simplest closed polygon, the right-angled triangle, we can define three fundamental ratio functions. Because any two right-angled triangles sharing the same acute angle are mathematically similar, the ratios of their side lengths are completely locked to the size of that angle, independent of the triangle’s actual physical size.

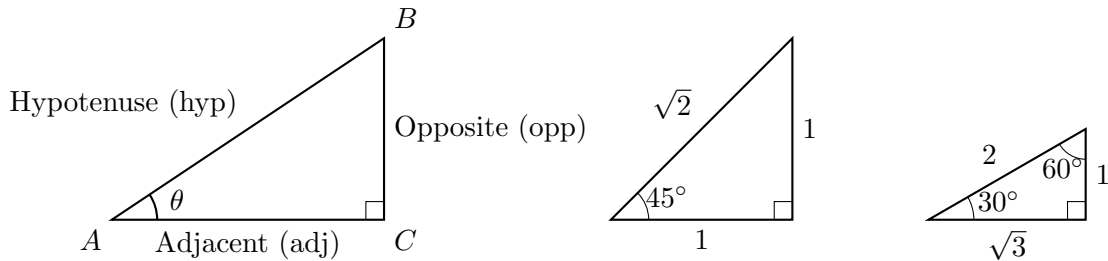


Figure 1: Trigonometric ratio definitions and standard reference triangles for special angles.

Definition 1.4.1 (Trigonometric Ratios). Let $\triangle ABC$ be a right-angled triangle with $\angle C = 90^\circ$. Let θ be the acute angle $\angle BAC$.

- The side opposite to the right angle, AB , is the **hypotenuse** (hyp).
- The side opposite to θ , BC , is the **opposite** side (opp).
- The side adjacent to θ (excluding the hypotenuse), AC , is the **adjacent** side (adj).

We define the three basic trigonometric ratios of θ as:

$$\begin{aligned}\sin \theta &= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB} \\ \cos \theta &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AC}{AB} \\ \tan \theta &= \frac{\text{opposite}}{\text{adjacent}} = \frac{BC}{AC}\end{aligned}$$

Proposition 1.4.1 (Special Angles). *By constructing specific geometric figures (an isosceles right-angled triangle and an equilateral triangle cut in half), we define the exact values for the special acute angles 30° , 45° , and 60° :*

θ	30°	45°	60°
$\sin \theta$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos \theta$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan \theta$	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

Proof. By construction. □

Example 1.4.1 (Numerical Test on Ratios). Let $\triangle ABC$ be a right-angled triangle with sides of length 3 cm, 4 cm, and 5 cm. If the side of length 3 cm is opposite to the acute angle θ , calculate the exact values of $\sin \theta$, $\cos \theta$, and $\tan \theta$.

Solution. By the Pythagorean theorem, since $3^2 + 4^2 = 9 + 16 = 25 = 5^2$, the side of length 5 cm is the hypotenuse (hyp = 5). Since the side opposite to θ is 3 (opp = 3), the adjacent side must be 4 (adj = 4).

Applying the definitions:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{3}{5}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{4}{5}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}$$

□

Example 1.4.2 (Algebraic Expression Derivation). Derive the algebraic identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

using the definitions of the trigonometric ratios and the Pythagorean theorem $a^2 + b^2 = c^2$.

Solution. Let $a = \text{opp}$, $b = \text{adj}$, and $c = \text{hyp}$ in a right-angled triangle. By the Pythagorean theorem, we have:

$$a^2 + b^2 = c^2.$$

Now, write the sum of the squares of the sine and cosine ratios:

$$\sin^2 \theta + \cos^2 \theta = \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = \frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{a^2 + b^2}{c^2}.$$

Substitute $a^2 + b^2 = c^2$ into the numerator:

$$\sin^2 \theta + \cos^2 \theta = \frac{c^2}{c^2} = 1.$$

This completes the derivation of the fundamental Pythagorean trigonometric identity. □

Example 1.4.3 (Algebraic Expression Application). Using the algebraic identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin^2 \theta + \cos^2 \theta = 1$, express $\cos \theta$ purely in terms of $\tan \theta$, assuming θ is an acute angle.

Solution. Divide the identity $\sin^2 \theta + \cos^2 \theta = 1$ by $\cos^2 \theta$ (since θ is acute, $\cos \theta \neq 0$):

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta} \implies \tan^2 \theta + 1 = \frac{1}{\cos^2 \theta}.$$

Taking the reciprocal of both sides:

$$\cos^2 \theta = \frac{1}{1 + \tan^2 \theta}.$$

Taking the positive square root (since $\cos \theta > 0$ for acute angles):

$$\cos \theta = \frac{1}{\sqrt{1 + \tan^2 \theta}}.$$

□

Example 1.4.4 (Abstract Application to Problems). A surveyor stands at a distance d from the base of a vertical tower. Looking up at the top of the tower, the angle of elevation is θ . If the surveyor walks back to a distance $2d$, show abstractly that the new angle of elevation ϕ must satisfy $\tan \phi = \frac{1}{2} \tan \theta$.

Solution. Let the height of the vertical tower be h . The tower forms a right angle with the horizontal ground. From the first position, we have a right-angled triangle with opposite side h and adjacent side d . Thus:

$$\tan \theta = \frac{h}{d}.$$

From the second position, the adjacent side is now $2d$. Thus:

$$\tan \phi = \frac{h}{2d}.$$

We can rewrite $\tan \phi$ as:

$$\tan \phi = \frac{1}{2} \left(\frac{h}{d} \right) = \frac{1}{2} \tan \theta.$$

Thus, the tangent of the new angle of elevation is exactly half of the original tangent value. □

Example 1.4.5 (Failure of the Definition). Explain why the basic right-angled triangle definitions of $\sin \theta$, $\cos \theta$, and $\tan \theta$ fail when the angle θ is equal to 0° , 90° , or greater.

Solution. The basic definitions of trigonometric ratios rely strictly on the existence of a physical, non-degenerate right-angled triangle.

If $\theta = 0^\circ$, the triangle collapses into a single straight line, and the opposite side length becomes zero, meaning we no longer have a 2D shape.

If $\theta = 90^\circ$, the sum of the two angles would be $90^\circ + 90^\circ = 180^\circ$, leaving 0° for the third angle, which also collapses the triangle.

If $\theta > 90^\circ$, a triangle cannot contain both a right angle and an obtuse angle, because its angle sum would exceed 180° . Therefore, the simple geometric ratio-of-sides definition fails for these boundary cases, motivating the need for an algebraic system (which we will introduce in Section 3). □

* * *

We now extend our trigonometric tools beyond right-angled triangles to general triangles. By examining the altitude (height) of a triangle, we can establish formulas for its area and discover the elegant relationship between its sides and angles.

Theorem 1.4.1 (Area of a Triangle). *For any triangle $\triangle ABC$ with sides a , b , and c opposite to angles A , B , and C respectively, the area $\mathcal{A}$ is given by:*

$$\mathcal{A} = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B.$$

Proof. Without loss of generality, we only prove the first equality.

From which, our definition of triangle, let h be the height perpendicular to AC , then

$$\text{Area} = \frac{1}{2}ah.$$

Consider the right-angled triangle with sides a and h , we found that in both acute and obtuse case, $\frac{h}{a} = \sin C$. Substituting into the area formula, we have

$$\text{Area} = \frac{1}{2}ab \sin C$$

as desired. □

Theorem 1.4.2 (The Sine Law). *For any triangle $\triangle ABC$, the ratio of the side lengths to the sine of their opposite angles is constant:*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Proof. Let $\triangle ABC$ have side lengths a, b, c opposite to angles A, B, C . Drop an altitude of length h from vertex C to the opposite side AB (or its extension). In right-angled triangle $\triangle ADC$ (where D is the foot of the altitude):

$$\sin A = \frac{h}{b} \implies h = b \sin A.$$

In right-angled triangle $\triangle BDC$:

$$\sin B = \frac{h}{a} \implies h = a \sin B.$$

Equating both expressions for height h :

$$b \sin A = a \sin B.$$

Dividing both sides by $\sin A \sin B$ (assuming non-degenerate angles where $\sin A, \sin B \neq 0$):

$$\frac{a}{\sin A} = \frac{b}{\sin B}.$$

By dropping an altitude from vertex A to side BC , an identical derivation establishes $\frac{b}{\sin B} = \frac{c}{\sin C}$, completing the chain of equalities. □

Example 1.4.6 (Direct Application of Sine Law). In $\triangle ABC$, angle $\angle A = 45^\circ$, angle $\angle B = 60^\circ$, and side $a = 10$ cm. Calculate the exact length of side b .

Solution. By the Sine Law, we set up the proportion between sides and the sines of their opposite angles:

$$\frac{a}{\sin A} = \frac{b}{\sin B} \implies \frac{10}{\sin(45^\circ)} = \frac{b}{\sin(60^\circ)}.$$

Substitute the special angle values $\sin(45^\circ) = \frac{\sqrt{2}}{2}$ and $\sin(60^\circ) = \frac{\sqrt{3}}{2}$:

$$\frac{10}{\frac{\sqrt{2}}{2}} = \frac{b}{\frac{\sqrt{3}}{2}} \implies \frac{20}{\sqrt{2}} = \frac{2b}{\sqrt{3}}.$$

Cross-multiply to isolate b :

$$2b\sqrt{2} = 20\sqrt{3} \implies b = \frac{10\sqrt{3}}{\sqrt{2}} = 5\sqrt{6} \text{ cm.}$$

□

Theorem 1.4.3 (The Cosine Law). *For any triangle $\triangle ABC$, the square of any side is related to the other two sides and the cosine of their included angle:*

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

Proof. Place triangle $\triangle ABC$ on the Cartesian coordinate plane such that vertex C is at the origin $(0, 0)$ and side CB lies along the positive x -axis, placing vertex B at $(a, 0)$. By the unit circle definition of trigonometry, the coordinates of vertex A are $(b \cos C, b \sin C)$. The length of side c is the straight-line distance between vertex $A(b \cos C, b \sin C)$ and vertex $B(a, 0)$. Applying the 2D Distance Formula:

$$c^2 = (b \cos C - a)^2 + (b \sin C - 0)^2.$$

Expanding the squared binomials:

$$c^2 = b^2 \cos^2 C - 2ab \cos C + a^2 + b^2 \sin^2 C.$$

Grouping the terms with common factor b^2 :

$$c^2 = a^2 + b^2(\cos^2 C + \sin^2 C) - 2ab \cos C.$$

Applying the fundamental Pythagorean identity $\cos^2 C + \sin^2 C = 1$:

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

□

Theorem 1.4.4 (Heron's Formula). *The area $\mathcal{A}$ of any triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$ can be calculated purely from its side lengths:*

$$\mathcal{A} = \sqrt{s(s-a)(s-b)(s-c)}.$$

Proof. Done in Exmaple 1.4.11

□

Example 1.4.7 (Numerical Test on Sine and Cosine Laws). A triangle has sides $b = 5$ cm, $c = 8$ cm, and an included angle $A = 60^\circ$. Calculate the exact length of side a .

Solution. Using the Cosine Law to find side a :

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

Substitute the known values:

$$a^2 = 5^2 + 8^2 - 2(5)(8) \cos(60^\circ).$$

Using our table of special angles, $\cos(60^\circ) = \frac{1}{2}$:

$$a^2 = 25 + 64 - 80 \left(\frac{1}{2} \right) = 89 - 40 = 49.$$

Taking the positive square root since a is a physical length:

$$a = 7 \text{ cm.}$$

□

Example 1.4.8 (Algebraic Expression Derivation). Using the Area of a Triangle theorem, derive the Sine Law.

Solution. The area $\mathcal{A}$ of any triangle $\triangle ABC$ is equal regardless of which base and height are chosen. Therefore, we can set the three area expressions equal to one another:

$$\frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B = \frac{1}{2}ab \sin C.$$

Multiply the entire equation by 2:

$$bc \sin A = ac \sin B = ab \sin C.$$

Now, divide all terms by the product of the three side lengths, abc :

$$\frac{bc \sin A}{abc} = \frac{ac \sin B}{abc} = \frac{ab \sin C}{abc}.$$

Simplifying each fraction by canceling common terms in the numerators and denominators yields:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$$

Taking the reciprocal of each term gives the standard form of the Sine Law:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

□

Example 1.4.9 (Algebraic Expression Application). Show how the Cosine Law collapses to the classic Pythagorean theorem $c^2 = a^2 + b^2$ when the angle C is a right angle (90°).

Solution. The Cosine Law states:

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

If $\triangle ABC$ is a right-angled triangle with $\angle C = 90^\circ$, we evaluate $\cos C$:

$$\cos(90^\circ) = 0.$$

Substituting this value back into the formula:

$$c^2 = a^2 + b^2 - 2ab(0) = a^2 + b^2.$$

Thus, the term representing the deviation from a right angle vanishes, and we are left with the exact Pythagorean relation. □

Example 1.4.10 (Abstract Application of Heron's Formula). A triangle has side lengths $a = 5$, $b = 6$, and $c = 7$. Calculate the exact area of this triangle using Heron's Formula.

Solution. First, find the semi-perimeter s :

$$s = \frac{a + b + c}{2} = \frac{5 + 6 + 7}{2} = \frac{18}{2} = 9.$$

Next, apply Heron's Formula:

$$\mathcal{A} = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{9(9-5)(9-6)(9-7)}.$$

Compute the values inside the square root:

$$\mathcal{A} = \sqrt{9(4)(3)(2)} = \sqrt{216}.$$

Simplify the radical expression:

$$\mathcal{A} = \sqrt{36 \times 6} = 6\sqrt{6} \text{ square units.}$$

□

Example 1.4.11 (Analytic Discussion on Heron's Proof). Discuss how Heron's Formula can be proven analytically by combining the identity $\sin^2 C = 1 - \cos^2 C$, the area formula $\mathcal{A} = \frac{1}{2}ab \sin C$, and the Cosine Law.

Solution. We start with the squared area formula:

$$\mathcal{A}^2 = \frac{1}{4}a^2b^2 \sin^2 C = \frac{1}{4}a^2b^2(1 - \cos^2 C) = \frac{1}{4}a^2b^2(1 - \cos C)(1 + \cos C).$$

From the Cosine Law, we isolate $\cos C$:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Substitute this term into our expression:

$$1 - \cos C = \frac{2ab - (a^2 + b^2 - c^2)}{2ab} = \frac{c^2 - (a - b)^2}{2ab} = \frac{(c - a + b)(c + a - b)}{2ab}.$$

Similarly:

$$1 + \cos C = \frac{2ab + a^2 + b^2 - c^2}{2ab} = \frac{(a + b)^2 - c^2}{2ab} = \frac{(a + b - c)(a + b + c)}{2ab}.$$

Substitute these back into the squared area equation:

$$\begin{aligned} \mathcal{A}^2 &= \frac{1}{4}a^2b^2 \left[\frac{(c - a + b)(c + a - b)}{2ab} \right] \left[\frac{(a + b - c)(a + b + c)}{2ab} \right] \\ &= \frac{(a + b + c)(a + b - c)(a - b + c)(-a + b + c)}{16}. \end{aligned}$$

Using $s = \frac{a+b+c}{2}$, we recognize:

$$a + b + c = 2s, \quad a + b - c = 2(s - c), \quad a - b + c = 2(s - b), \quad -a + b + c = 2(s - a).$$

Thus:

$$\mathcal{A}^2 = \frac{2s \cdot 2(s - c) \cdot 2(s - b) \cdot 2(s - a)}{16} = s(s - a)(s - b)(s - c).$$

Taking the square root yields Heron's Formula. □

—

Problems for Section 1.4

1. In a right-angled triangle, the hypotenuse is 13 cm and one of the sides is 5 cm. Find the sine, cosine, and tangent of the angle adjacent to the 5 cm side.
2. Find the exact numerical value of the expression $\sin(45^\circ)\cos(45^\circ) + \sin(30^\circ)$.
3. A right-angled triangle has an acute angle of 30° and an adjacent side of length $6\sqrt{3}$ cm. Find the length of the opposite side.
4. If $\tan \theta = \frac{5}{12}$ and θ is acute, find the value of $\sin \theta + \cos \theta$.
5. Calculate the exact area of a triangle with sides $a = 6$ cm, $b = 10$ cm, and included angle $C = 30^\circ$.
6. In $\triangle ABC$, $a = 12$, $A = 45^\circ$, and $B = 30^\circ$. Use the Sine Law to find the exact length of side b .
7. A triangle has side lengths $a = 4$ cm, $b = 5$ cm, and $c = 6$ cm. Calculate $\cos C$ using the Cosine Law.
8. Solve for the area of a triangle with side lengths $a = 7$, $b = 8$, and $c = 9$ using Heron's Formula.
9. Prove that in any right-angled triangle, the sine of one acute angle is equal to the cosine of the other acute angle.
10. Let $\triangle ABC$ be a triangle with sides $a = 10$ cm, $b = 10\sqrt{3}$ cm, and $c = 20$ cm. Show that this is a right-angled triangle and find the measure of angle A .
11. If the area of a triangle is $12\sqrt{3}$ cm² and two of its sides are 6 cm and 8 cm, find the two possible values for the included angle.
12. A ladder of length 8 meters leans against a vertical wall. If the ladder makes an angle of 60° with the ground, how high up the wall does it reach?
13. Prove that $\frac{\sin A}{a} = \frac{1}{2R}$, where R is the radius of the circle passing through all three vertices of $\triangle ABC$ (the circumcircle).
14. In a triangle, if $a^2 = b^2 + c^2 + bc$, find the exact measure of angle A .
15. An isosceles triangle has two sides of length 10 cm and a base of length 12 cm. Use Heron's Formula to find its area.
16. In $\triangle ABC$, if $\sin A = \sin B$, prove using the Sine Law that the triangle must be isosceles ($a = b$).
17. A triangle has sides in the ratio 3 : 5 : 7. Find the measure of its largest angle.
18. If $\sin \theta = k$, express $\tan \theta$ purely in terms of k for an acute angle θ .
19. Using the Cosine Law, prove that if $a^2 + b^2 > c^2$, the angle C must be acute.
20. Prove that the perimeter of any right-angled triangle with hypotenuse 1 is given by $1 + \sin \theta + \cos \theta$, where θ is one of the acute angles.

2 The Coordinate Geometry

2.1 Rectangular Coordinates

In Section 1, we analyzed geometric objects—points, lines, and polygons—as absolute entities floating in an empty space. This classical approach, known as synthetic geometry, is beautiful but has a major limitation: it relies on clever geometric tricks and constructions to solve problems. To make geometry systematic, quantitative, and easily calculable, we must assign numerical coordinates to our positions. This revolutionizes our study by turning geometric problems into algebraic equations.

We begin with the most common and intuitive way of positioning: the **Rectangular (or Cartesian) Coordinate System**. This provides a “bird’s-eye view” of our space. Imagine looking down at a modern city planned as a perfect grid of streets and avenues. Every location is defined globally and uniquely by two numbers: how far east/west (horizontal) you are, and how far north/south (vertical) you are.

Definition 2.1.1 (Rectangular Coordinates in 2D). A **rectangular coordinate system** in a 2-dimensional plane (the xy -plane) is constructed by drawing two perpendicular real number lines that intersect at their zero points.

- The horizontal number line is called the **x-axis**.
- The vertical number line is called the **y-axis**.
- Their point of intersection is called the **origin**, denoted by $O(0, 0)$.
- Any point P in the plane is uniquely identified by an ordered pair of real numbers (x, y) , representing its perpendicular distance and direction from the y -axis and x -axis respectively.

Remark 2.1.1. This grid representation is highly applicable to everyday digital tools. For example, a modern spreadsheet program (like Microsoft Excel) organizes data using columns (letters acting as horizontal markers) and rows (numbers acting as vertical markers). Any cell is uniquely located by a coordinate pair such as B5 or C12.

Definition 2.1.2 (Extension to 3D Space). To model our actual physical world, we can add a third mutually perpendicular axis, the **z-axis**, representing height or vertical position. A point in this 3D space is uniquely identified by an ordered triple (x, y, z) .

Theorem 2.1.1 (Distance Formula in the Plane). *The straight-line distance d between any two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ in a rectangular coordinate plane is given by:*

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Proof. By Pythagoras Theorem, we can take the orthogonal length as the difference in rectangular coordinates. The theorem follows. $\square$

Theorem 2.1.2 (Midpoint Formula). *The coordinates of the midpoint M of the line segment connecting $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are given by:*

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Proof. Trivial. $\square$

Example 2.1.1 (Numerical Test on Grid Distances). A delivery drone flies in a straight line from a warehouse located at $A(-3, 2)$ to a customer located at $B(5, 8)$ on a rectangular grid. Calculate the exact distance the drone travels.

Solution. Using the 2D Distance Formula with $(x_1, y_1) = (-3, 2)$ and $(x_2, y_2) = (5, 8)$:

$$d = \sqrt{(5 - (-3))^2 + (8 - 2)^2}.$$

Simplify the terms inside the parentheses:

$$d = \sqrt{(5 + 3)^2 + (6)^2} = \sqrt{8^2 + 6^2}.$$

Calculate the squares:

$$d = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ units.}$$

Thus, the drone travels exactly 10 grid units. $\square$

Example 2.1.2 (Algebraic Expression Derivation). Let $P(x, y)$ be a point whose distance from the origin $O(0, 0)$ is always exactly r units. Derive the algebraic equation representing the path (locus) of P .

Solution. Using the Distance Formula, the distance between $P(x, y)$ and $O(0, 0)$ is:

$$d = \sqrt{(x - 0)^2 + (y - 0)^2} = \sqrt{x^2 + y^2}.$$

Since this distance must equal r , we write:

$$\sqrt{x^2 + y^2} = r.$$

Squaring both sides of the equation yields:

$$x^2 + y^2 = r^2.$$

This algebraic derivation proves that the equation of a circle centered at the origin with radius r is $x^2 + y^2 = r^2$. $\square$

Example 2.1.3 (Algebraic Expression Application). A line segment has endpoints $A(2a, b - 3)$ and $B(4, 5)$. The midpoint of this segment is measured to be $M(5, 2)$. Find the values of the algebraic constants a and b .

Solution. Using the Midpoint Formula:

$$M = \left(\frac{2a + 4}{2}, \frac{(b - 3) + 5}{2} \right).$$

We are given that $M = (5, 2)$. Setting the corresponding coordinates equal:

$$\text{For the horizontal coordinate: } \frac{2a + 4}{2} = 5 \implies 2a + 4 = 10 \implies 2a = 6 \implies a = 3$$

$$\text{For the vertical coordinate: } \frac{b + 2}{2} = 2 \implies b + 2 = 4 \implies b = 2$$

Thus, the algebraic constants are $a = 3$ and $b = 2$. $\square$

Example 2.1.4 (Abstract Application to Problems). Let $A(0, 0)$, $B(a, 0)$, and $C(0, b)$ be three points on a rectangular plane, where $a, b > 0$. Show abstractly that the midpoint of the segment BC is equidistant from all three vertices A , B , and C .

Solution. First, find the coordinates of the midpoint M of BC :

$$M = \left(\frac{a+0}{2}, \frac{0+b}{2} \right) = \left(\frac{a}{2}, \frac{b}{2} \right).$$

Now, compute the distance from M to each vertex:

$$\begin{aligned} 1) \quad \text{Distance } AM &= \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - 0\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \frac{1}{2}\sqrt{a^2 + b^2} \\ 2) \quad \text{Distance } BM &= \sqrt{\left(\frac{a}{2} - a\right)^2 + \left(\frac{b}{2} - 0\right)^2} = \sqrt{\left(-\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} = \frac{1}{2}\sqrt{a^2 + b^2} \\ 3) \quad \text{Distance } CM &= \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - b\right)^2} = \sqrt{\left(\frac{a}{2}\right)^2 + \left(-\frac{b}{2}\right)^2} = \frac{1}{2}\sqrt{a^2 + b^2} \end{aligned}$$

Since $AM = BM = CM = \frac{1}{2}\sqrt{a^2 + b^2}$, the midpoint of the hypotenuse is indeed equidistant from all three vertices of this right-angled triangle. $\square$

Example 2.1.5 (Failure of the Concept). Explain why the rectangular coordinate distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ fails to measure the actual travel distance between two points in a city with a strict grid layout where streets cannot be crossed diagonally.

Solution. The distance formula measures the straight-line (Euclidean) distance as the crow flies.

However, in a real city with a grid layout of buildings, a car cannot drive diagonally through walls. It must travel horizontally along streets, then vertically along avenues. The actual walking or driving distance is given by the sum of horizontal and vertical changes:

$$d_{\text{taxi}} = |x_2 - x_1| + |y_2 - y_1|.$$

This is known as the Manhattan (or taxicab) metric. Therefore, the Euclidean formula fails to model actual path distances when diagonal movement is restricted. $\square$

—

Problems for Section 2.1

- Find the distance between the points $A(1, -3)$ and $B(7, 5)$.
- Find the coordinates of the midpoint of the line segment joining $P(-5, 12)$ and $Q(3, -4)$.
- A line segment has midpoint $M(2, -1)$. If one endpoint is $A(-3, 4)$, find the coordinates of the other endpoint B .
- Show that the triangle with vertices $A(1, 1)$, $B(4, 1)$, and $C(1, 5)$ is a right-angled triangle.
- If the distance between $P(1, k)$ and $Q(4, -2)$ is exactly 5 units, find all possible values of k .
- A point $P(x, y)$ is equidistant from $A(2, 0)$ and $B(6, 0)$. Derive the algebraic equation representing the locus of P .
- Find the distance of the point $A(3, -4, 12)$ from the origin $O(0, 0, 0)$ in 3D space.

8. If the midpoint of the line segment joining $(2x, 5)$ and $(4, y)$ is $(5, 8)$, solve for x and y .
9. Find the perimeter of the triangle with vertices $P(0, 0)$, $Q(3, 4)$, and $R(-3, 4)$.
10. Prove using coordinate geometry that the diagonals of a rectangle with vertices $(0, 0)$, $(a, 0)$, (a, b) , and $(0, b)$ are equal in length.
11. A circle has its center at $C(2, -3)$. If the point $A(5, 1)$ lies on the circle, find the radius and the equation of this circle.
12. Find the coordinates of the two points on the y -axis that are at a distance of 13 units from the point $(12, 5)$.
13. The vertices of a quadrilateral are $A(1, 1)$, $B(5, 1)$, $C(5, 4)$, and $D(1, 4)$. Calculate its area.
14. Show that the three points $A(-1, -1)$, $B(2, 2)$, and $C(5, 5)$ are collinear (lie on the same straight line) by comparing distances.
15. If a point P moves such that its distance from $A(0, 3)$ is always equal to its distance from $B(0, -3)$, find the algebraic equation of its path.
16. Find the coordinates of the point that divides the line segment joining $A(1, 2)$ and $B(7, 8)$ in the ratio $1 : 2$ internally.
17. In a 3D coordinate system, find the distance between $P(1, 2, 3)$ and $Q(4, 6, 8)$.
18. If the three vertices of a parallelogram are $A(0, 0)$, $B(4, 0)$, and $C(5, 3)$ in order, find the coordinates of the fourth vertex D .
19. Find the equation of the perpendicular bisector of the line segment joining $A(1, 3)$ and $B(5, 7)$.
20. Discuss why we define the axes to be perpendicular (90°) to each other instead of intersecting at some other angle.

2.2 Polar Coordinates

In the previous subsection, we established the rectangular coordinate system, which serves as an excellent global map of our geometric plane. It is a “bird’s-eye” perspective, describing locations using horizontal and vertical offsets from a fixed, neutral origin.

However, in many real-world scenarios, a global grid is highly unnatural. If you are standing in a dense forest or piloting an aircraft, you do not locate objects by their coordinates on a map of the Earth. Instead, you locate them relative to **yourself**. You want to know: “How far away is the target, and in what direction must I turn to look at it?”

This localized, self-centered view is modeled by the **Polar Coordinate System**. Instead of measuring along two perpendicular tracks, we measure a straight-line radial distance from a center point, along with an angular deviation from a fixed reference direction. This mirrors daily language, such as navigating at sea, tracking radar, or locating an object on a clock face (e.g., “target at 2 o’clock, distance 500 meters”).

Definition 2.2.1 (Polar Coordinates). A **polar coordinate system** in a flat plane is constructed using a fixed point of origin and a reference direction line.

- The fixed central reference point is called the **pole**, often denoted by O .
- The horizontal ray originating from the pole and extending to the right is called the **polar axis**.

- Any point P in the plane is uniquely identified by an ordered pair (r, θ) .
- The variable r is the **radial distance**, representing the straight-line distance from the pole O to the point P . By physical convention, $r \geq 0$.
- The variable θ is the **angular coordinate** (or polar angle), representing the angle swept counterclockwise from the positive polar axis to the ray OP .

Example 2.2.1 (Plotting Basic Polar Coordinates). Identify the quadrant and physical location of the points $P_1(4, 0^\circ)$, $P_2(3, 90^\circ)$, and $P_3(5, 210^\circ)$ on the polar plane.

Solution. 1. For $P_1(4, 0^\circ)$: The angle is 0° , meaning we move along the positive polar axis (to the right) by a distance of 4 units. 2. For $P_2(3, 90^\circ)$: A 90° counterclockwise rotation points straight up along the vertical axis. We move 3 units upward from the pole. 3. For $P_3(5, 210^\circ)$: Since $180^\circ < 210^\circ < 270^\circ$, this rotation lands in **Quadrant III**. The point lies 5 units from the origin along the ray making a 30° angle below the negative horizontal axis. $\square$

Remark 2.2.1. In polar coordinates, angles are measured in degrees (or radians, which we introduce in Section 3.5). Angles swept counterclockwise are defined as positive, while angles swept clockwise are defined as negative.

Example 2.2.2 (Numerical Test on Polar Plotting). A vessel is detected on a radar screen at the polar position $P(4, 120^\circ)$. Plot the location of this vessel and find its supplementary angle position relative to the negative polar axis.

Solution. To locate $P(4, 120^\circ)$: 1. Start at the pole O . 2. Rotate counterclockwise from the horizontal polar axis by 120° . Since $90^\circ < 120^\circ < 180^\circ$, this ray points into the upper-left quadrant. 3. Move a distance of 4 units outward along this ray.

The straight horizontal path directly opposite to the polar axis represents 180° . The angle between our ray and this negative polar axis is:

$$180^\circ - 120^\circ = 60^\circ.$$

Thus, the point lies 60° north of the negative polar direction, at a distance of 4 units. $\square$

Example 2.2.3 (Algebraic Expression Derivation). The distance d between two points

$$P_1(r_1, \theta_1) \text{ and } P_2(r_2, \theta_2)$$

in polar coordinates can be derived using the Cosine Law from Section 1.4. Derive the explicit algebraic equation for d .

Solution. Consider the triangle formed by the pole O , point P_1 , and point P_2 . * The distance from O to P_1 is r_1 . * The distance from O to P_2 is r_2 . * The angle between the ray OP_1 and OP_2 is the difference between their individual polar angles, which is $|\theta_2 - \theta_1|$.

Using the Cosine Law on $\triangle OP_1P_2$, the side opposite to the angle $|\theta_2 - \theta_1|$ is the distance d :

$$d^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1).$$

Taking the square root of both sides gives the polar distance formula:

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}.$$

$\square$

Example 2.2.4 (Algebraic Expression Application). Using the polar distance formula, calculate the exact distance between the points $A(3, 45^\circ)$ and $B(4, 105^\circ)$.

Solution. Assign the coordinates: $r_1 = 3, \theta_1 = 45^\circ$ and $r_2 = 4, \theta_2 = 105^\circ$. Calculate the difference between the angles:

$$\theta_2 - \theta_1 = 105^\circ - 45^\circ = 60^\circ.$$

Now substitute these values into the polar distance formula:

$$d = \sqrt{3^2 + 4^2 - 2(3)(4)\cos(60^\circ)}.$$

We know from our table of special angles that $\cos(60^\circ) = \frac{1}{2}$:

$$d = \sqrt{9 + 16 - 24\left(\frac{1}{2}\right)} = \sqrt{25 - 12} = \sqrt{13} \text{ units.}$$

Thus, the exact distance between A and B is $\sqrt{13}$ units. □

Example 2.2.5 (Abstract Application to Problems). Let $r = a$ represent an algebraic equation in polar coordinates, where $a > 0$ is a constant. Show abstractly that this equation represents a circle, and determine its center and radius.

Solution. The polar equation is $r = a$.

By definition, r represents the radial distance of any coordinate point $P(r, \theta)$ from the pole O . Since $r = a$ is completely independent of the angle θ , the angle θ can take any real value from 0° to 360° (and beyond) while the distance remains locked at a .

The set of all points that are at a constant distance a from a fixed central point O is, by definition, a circle. The fixed center is the pole O , and the radius of this circle is the constant value a . □

Example 2.2.6 (Failure of the Concept). In rectangular coordinates, every point has a unique representation (x, y) . Explain why this uniqueness fails in polar coordinates, providing examples.

Solution. In polar coordinates, a single physical position does not have a unique numerical representation. This failure of uniqueness occurs for two primary reasons:

1. ****Angular Periodicity:**** A full rotation around the pole is 360° . Therefore, rotating a point's angle by 360° returns it to the exact same position. For example, the coordinates $(3, 45^\circ)$ and $(3, 405^\circ)$ represent the exact same point:

$$45^\circ + 360^\circ = 405^\circ.$$

2. ****The Pole Representation:**** At the pole O , the radial distance $r = 0$. Since you are standing directly at the origin, it does not matter which direction you face; you are still at the center. Therefore, $(0, 0^\circ)$, $(0, 90^\circ)$, and $(0, -120^\circ)$ all represent the exact same point: the pole O . □

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Problems for Section 2.2

1. Plot the point $P(5, 30^\circ)$ on a coordinate plane, and list two other valid coordinate pairs representing this exact same point.
2. A point has polar coordinates $(3, 150^\circ)$. If you rotate this point clockwise around the pole by 90° , what are its new coordinates?
3. Find the exact polar distance between the points $A(2, 15^\circ)$ and $B(2, 75^\circ)$.
4. A radar system tracks a target at $P(10, 240^\circ)$. Find the target's distance and direction angle relative to the positive polar axis if approached clockwise.
5. Show that the polar equation $r = 2 \cos \theta$ is equivalent to a circle by plotting points at $\theta = 0^\circ, 30^\circ, 45^\circ, 60^\circ$, and 90° .
6. Find the polar distance between $A(5, 0^\circ)$ and $B(12, 90^\circ)$.
7. If two points are given by $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$, under what algebraic condition on their angles is the distance between them simply $|r_1 - r_2|$?
8. A point is located at $(4, -60^\circ)$. Find the equivalent coordinate representation where the angle is positive and restricted to $0^\circ \leq \theta < 360^\circ$.
9. Let $A(6, 45^\circ)$ and $B(6, 225^\circ)$ be two points. Show that the pole O is the exact midpoint of the line segment connecting them.
10. Explain abstractly why the polar equation $\theta = 45^\circ$ represents a straight line. Describe its shape in detail.
11. Calculate the exact distance between $P_1(\sqrt{2}, 15^\circ)$ and $P_2(2\sqrt{2}, 60^\circ)$.
12. A physical particle is moving according to the polar equation $r(\theta) = 3\theta$. If the angle θ is doubled from 30° to 60° , find the factor by which its distance from the pole increases.
13. An aircraft is traveling at a constant speed along the path $r = 8$. Describe the path of the aircraft to someone who only understands rectangular grids.
14. Find the distance between the two points $A(8, 300^\circ)$ and $B(6, 120^\circ)$.
15. Explain why, in a physical polar navigation system, we rarely allow the radial distance variable r to be negative.
16. Let $A(4, \theta)$ be a point. If we increase θ by 180° , describe geometrically where the new point lies relative to A .
17. Find the perimeter of the triangle with vertices $O(0, 0)$, $P_1(5, 10^\circ)$, and $P_2(5, 70^\circ)$.
18. Three points form an equilateral triangle. If the pole $O(0, 0)$ is the centroid, and one vertex is at $A(4, 0^\circ)$, find the polar coordinates of the other two vertices.
19. If $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$ lie on the same ray originating from the pole, what is the relation between θ_1 and θ_2 ?
20. Discuss how a clock's hands represent a physical, mechanical implementation of a polar coordination coordinate system.

2.3 The Main Issues: Coordinate Bridging

In the previous two subsections, we explored two wildly different methods of mapping the plane.

- **The Rectangular Coordinate System** (xy -plane) is globally consistent. No matter where you move on the grid, the horizontal direction is always parallel to the x -axis, and the vertical direction is always parallel to the y -axis. However, it lacks intuitive significance to an observer standing directly in the scene. If you are sailing a boat, knowing that a rock is at grid coordinates (154, 230) is not immediately useful without active translation.
- **The Polar Coordinate System** ($r\theta$ -plane) is locally significant. It focuses entirely on your current position (the pole) and tells you exactly how far to look (r) and in which direction to turn (θ). However, it is globally inconsistent. If two observers stand at different positions, their poles and reference axes are different, meaning they will assign completely different coordinates to the exact same physical object.

To solve this conflict, we must establish a mathematical bridge between the two systems. We must be able to translate the global bird's-eye view of the rectangular grid into the local, direct view of the polar coordinate system, and vice versa. This translation is precisely where **trigonometry** displays its ultimate algebraic power.

Theorem 2.3.1 (Coordinate Conversion Formulas). *Let a point P have rectangular coordinates (x, y) and polar coordinates (r, θ) relative to an origin and polar axis aligned with the positive x -axis. The mathematical relationships between these coordinates are given by:*

$$\begin{aligned} \text{Polar to Rectangular: } & x = r \cos \theta, \quad y = r \sin \theta \\ \text{Rectangular to Polar: } & r = \sqrt{x^2 + y^2}, \quad \tan \theta = \frac{y}{x} \quad (x \neq 0) \end{aligned}$$

Proof. Trivial. □

Example 2.3.1 (Numerical Test on Conversion). A ship's radar reports a target at polar coordinates $P(6, 60^\circ)$. Convert these coordinates into standard rectangular coordinates (x, y) .

Solution. We are given the polar coordinates with $r = 6$ and $\theta = 60^\circ$. Using the Polar to Rectangular conversion formulas:

$$x = r \cos \theta = 6 \cos(60^\circ)$$

$$y = r \sin \theta = 6 \sin(60^\circ)$$

Using our table of special angles, we substitute $\cos(60^\circ) = \frac{1}{2}$ and $\sin(60^\circ) = \frac{\sqrt{3}}{2}$:

$$x = 6 \left(\frac{1}{2} \right) = 3$$

$$y = 6 \left(\frac{\sqrt{3}}{2} \right) = 3\sqrt{3}$$

Thus, the rectangular coordinates of the target are $(3, 3\sqrt{3})$. □

Example 2.3.2 (Algebraic Expression Derivation). Using the rectangular to polar conversion formulas, derive the polar equation of the vertical line whose rectangular equation is $x = k$, where $k > 0$ is a constant.

Solution. We begin with the rectangular equation of the vertical line:

$$x = k.$$

We substitute the polar conversion expression for x , which is $x = r \cos \theta$:

$$r \cos \theta = k.$$

To express r as a function of θ , we divide both sides of the equation by $\cos \theta$ (assuming $\theta \neq 90^\circ$ and $\theta \neq 270^\circ$):

$$r = \frac{k}{\cos \theta}.$$

Using the reciprocal trigonometric identity $\sec \theta = \frac{1}{\cos \theta}$, we can write the completed derivation as:

$$r = k \sec \theta.$$

This polar equation represents a vertical line in the polar plane. □

Example 2.3.3 (Algebraic Expression Application). Convert the rectangular equation of a circle $x^2 + y^2 - 4y = 0$ into its equivalent polar equation, and simplify the expression to express r purely in terms of θ .

Solution. First, identify the rectangular-to-polar relations:

$$x^2 + y^2 = r^2 \quad \text{and} \quad y = r \sin \theta.$$

Substitute these terms directly into the circle equation:

$$r^2 - 4(r \sin \theta) = 0.$$

Factor out the common term r :

$$r(r - 4 \sin \theta) = 0.$$

This gives two possible solutions: $r = 0$ (which represents the pole) or:

$$r = 4 \sin \theta.$$

Since the circle passes through the pole, the single polar equation $r = 4 \sin \theta$ perfectly describes the entire circle. □

Example 2.3.4 (Abstract Application to Problems). Show abstractly that the distance between two points in a rectangular plane, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, is mathematically identical to the polar distance formula $d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}$ under coordinate conversion.

Solution. Let $x_1 = r_1 \cos \theta_1$, $y_1 = r_1 \sin \theta_1$ and $x_2 = r_2 \cos \theta_2$, $y_2 = r_2 \sin \theta_2$. Substitute these into the squared rectangular distance formula:

$$d^2 = (r_2 \cos \theta_2 - r_1 \cos \theta_1)^2 + (r_2 \sin \theta_2 - r_1 \sin \theta_1)^2.$$

Expand both squared binomial terms:

$$\begin{aligned} d^2 &= (r_2^2 \cos^2 \theta_2 - 2r_1r_2 \cos \theta_1 \cos \theta_2 + r_1^2 \cos^2 \theta_1) \\ &\quad + (r_2^2 \sin^2 \theta_2 - 2r_1r_2 \sin \theta_1 \sin \theta_2 + r_1^2 \sin^2 \theta_1). \end{aligned}$$

Group the terms by common coefficients r_1^2 and r_2^2 :

$$d^2 = r_1^2(\cos^2 \theta_1 + \sin^2 \theta_1) + r_2^2(\cos^2 \theta_2 + \sin^2 \theta_2) - 2r_1r_2(\cos \theta_2 \cos \theta_1 + \sin \theta_2 \sin \theta_1).$$

Using the Pythagorean identity $\cos^2 \alpha + \sin^2 \alpha = 1$, and the trigonometric subtraction identity $\cos(\theta_2 - \theta_1) = \cos \theta_2 \cos \theta_1 + \sin \theta_2 \sin \theta_1$, we substitute:

$$d^2 = r_1^2(1) + r_2^2(1) - 2r_1r_2 \cos(\theta_2 - \theta_1).$$

Taking the square root proves that the two formulas are mathematically identical under coordinate conversion:

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}.$$

□

Example 2.3.5 (Failure of the Concept). Explain why converting the rectangular point $(-3, -3)$ to polar coordinates using the simple formula

$$\theta = \arctan(y/x)$$

can lead to a calculation error if the quadrant is not verified.

Solution. If we directly substitute $x = -3$ and $y = -3$ into the tangent formula:

$$\tan \theta = \frac{-3}{-3} = 1 \implies \theta = \arctan(1).$$

A standard calculator will return $\arctan(1) = 45^\circ$.

However, looking at the rectangular coordinates, $(-3, -3)$ lies in the third quadrant (both x and y are negative). An angle of 45° points into the first quadrant.

The inverse tangent function ($\arctan$) has a restricted range and cannot distinguish between $\frac{-y}{-x}$ and $\frac{y}{x}$. To resolve this failure of the simple formula, we must manually adjust for the quadrant. Since the point is in the third quadrant, we must add 180° to the principal value:

$$\theta = 45^\circ + 180^\circ = 225^\circ.$$

Thus, the correct polar coordinates are $(3\sqrt{2}, 225^\circ)$, showing that the unadjusted conversion formula fails for points in the second and third quadrants. □

—

Problems for Section 2.3

1. Convert the polar point $P(8, 30^\circ)$ into rectangular coordinates.
2. Convert the rectangular point $Q(5, -5)$ into polar coordinates, choosing a positive r and $0^\circ \leq \theta < 360^\circ$.
3. Find the polar equation of the horizontal line $y = 3$.
4. Convert the polar equation $r = 5$ into a rectangular equation.
5. Convert the rectangular point $A(-\sqrt{3}, 1)$ into polar coordinates.
6. Express the polar equation $r = 6 \cos \theta$ in rectangular form.
7. Convert the rectangular equation $x^2 + y^2 = 9$ into its polar form.

8. A straight line passes through the pole and makes an angle of 60° with the polar axis. Write its equation in both polar and rectangular systems.
9. Convert the polar point $M(4, 180^\circ)$ into rectangular coordinates.
10. Find the rectangular equation equivalent to the polar equation $r = 2 \csc \theta$.
11. Convert the rectangular equation $y = 2x$ into a polar equation, expressing θ as a constant angle.
12. Convert the polar equation $r^2 \sin(2\theta) = 4$ into rectangular coordinates. (Hint: Use the double angle identity $\sin(2\theta) = 2 \sin \theta \cos \theta$).
13. A path is described in polar coordinates by $r = \frac{6}{2 \cos \theta + 3 \sin \theta}$. Find its equivalent rectangular equation and show it represents a straight line.
14. Convert the rectangular point $B(0, -7)$ into polar coordinates.
15. Show that the polar equation $r = a \cos \theta + b \sin \theta$ represents a circle passing through the origin in the rectangular plane.
16. A laser tracking system measures a target at rectangular coordinates $(-4, 4\sqrt{3})$. Convert this immediately to polar coordinates for a steerable mirror mount.
17. Find the rectangular coordinates of the polar point $C(2, 270^\circ)$.
18. Convert the rectangular equation $x^2 - y^2 = 1$ (a hyperbola) into its equivalent polar equation.
19. If $P(x, y)$ is converted to $P(r, \theta)$, show that the slope of the line segment connecting the origin to P is equal to $\tan \theta$.
20. Discuss how a computer graphic rendering system uses coordinate conversion to draw circular dials on a rectangular pixel grid.

3 The Trigonometric function

3.1 The Radian Measure

Throughout our journey in geometry and trigonometry, we have measured angles using degrees ($^\circ$), defining a full rotation as 360° . While highly intuitive and deeply rooted in history (likely originating from ancient Babylonian solar calendars), the degree is an arbitrary unit of measurement. There is no intrinsic mathematical reason why a circle should be divided into exactly 360 parts rather than 100 or 1000.

To perform advanced mathematical analysis, calculus, and university-level physics, we need a natural, unit-free way to measure rotation. Since we have already mapped our angles to the unit circle (radius $r = 1$), we can assign to each angle an intrinsic physical quantity: the **arc length** it subtends on that unit circle. This leads us directly to the definition of the **radian**. By expressing angles as real numbers representing arc lengths, we free our trigonometric functions from arbitrary units and allow them to be analyzed purely as functions of real variables.

Definition 3.1.1 (Radian Measure). One **radian** (abbreviated as rad, or often written without any unit) is defined as the measure of a central angle of a circle that subtends an arc equal in length to the radius of the circle.

- Since the circumference of a unit circle is 2π , a complete rotation of 360° corresponds exactly to an arc length of 2π radians.
- The conversion between degrees and radians is governed by the fundamental relation:

$$\pi \text{ radians} = 180^\circ.$$

Theorem 3.1.1 (Arc Length and Sector Area). *Let a circle have radius r . If a central angle θ is measured strictly in **radians**, then:*

1. The **arc length** s subtended by the angle is:

$$s = r\theta$$

2. The **area** A of the sector enclosed by the angle is:

$$A = \frac{1}{2}r^2\theta$$

Proof. By definition that $\theta = s$ when $r = 1$. □

Example 3.1.1 (Numerical Test on Conversion). Convert the angle 120° into exact radian measure. Furthermore, calculate the exact arc length subtended by this angle on a circle of radius 6 cm.

Solution. To convert from degrees to radians, we multiply by the conversion factor $\frac{\pi}{180^\circ}$:

$$\theta = 120^\circ \times \frac{\pi}{180^\circ} = \frac{120}{180}\pi = \frac{2}{3}\pi \text{ radians.}$$

Now, using the arc length formula $s = r\theta$ with $r = 6$ cm and $\theta = \frac{2}{3}\pi$:

$$s = 6 \times \left(\frac{2}{3}\pi\right) = 4\pi \text{ cm.}$$

Thus, the radian measure is $\frac{2\pi}{3}$ and the arc length is 4π cm. □

Example 3.1.2 (Algebraic Expression Derivation). Derive the sector area formula $A = \frac{1}{2}r^2\theta$ using the properties of proportional parts of a full circle.

Solution. Let a sector of a circle of radius r have a central angle θ measured in radians.

The area of a full circle is πr^2 , which corresponds to a complete central angle of 2π radians. Because the area of a sector is directly proportional to its central angle, we can set up a ratio:

$$\frac{\text{Area of Sector}}{\text{Area of Full Circle}} = \frac{\text{Central Angle of Sector}}{\text{Central Angle of Full Circle}}.$$

Substitute the known algebraic terms into this proportion:

$$\frac{A}{\pi r^2} = \frac{\theta}{2\pi}.$$

To isolate the sector area A , we multiply both sides of the equation by πr^2 :

$$A = \left(\frac{\theta}{2\pi}\right) \times \pi r^2.$$

Canceling the common factor of π in the numerator and denominator yields the completed derivation:

$$A = \frac{1}{2}r^2\theta.$$

□

Example 3.1.3 (Algebraic Expression Application). A wire of fixed length L is bent to form the boundary of a sector of a circle of radius r and central angle θ (in radians). Express the area A of this sector purely as a quadratic function of r , and determine the value of θ that maximizes the enclosed area.

Solution. The boundary of the sector consists of two straight radii of length r and one curved arc of length $s = r\theta$. Therefore, the total perimeter (wire length) is:

$$L = 2r + r\theta.$$

We can isolate the angle variable θ from this perimeter equation:

$$r\theta = L - 2r \implies \theta = \frac{L - 2r}{r}.$$

Now substitute this expression for θ into the sector area formula $A = \frac{1}{2}r^2\theta$:

$$A = \frac{1}{2}r^2 \left(\frac{L - 2r}{r}\right) = \frac{1}{2}r(L - 2r) = \frac{1}{2}Lr - r^2.$$

This expresses the area as a downward-opening quadratic function of r . The maximum of a quadratic function $f(r) = -r^2 + \frac{L}{2}r$ occurs at its vertex, where:

$$r = -\frac{b}{2a} = -\frac{L/2}{2(-1)} = \frac{L}{4}.$$

Substitute $r = \frac{L}{4}$ back into our expression for θ :

$$\theta = \frac{L - 2(L/4)}{L/4} = \frac{L/2}{L/4} = 2 \text{ radians}.$$

Thus, for any fixed perimeter, the sector area is maximized when the central angle is exactly 2 radians. □

Example 3.1.4 (Abstract Application to Problems). Show abstractly that the radian system is necessary to preserve the classic small-angle approximation $\sin \theta \approx \theta$ when θ is very close to 0.

Solution. Consider a unit circle where a central angle θ is subtended. Let us construct a right-angled triangle inside the unit sector with opposite side $\sin \theta$ and a circular arc of length $s = \theta$ (since $r = 1$ and θ is in radians).

For extremely small values of θ , the curved arc and the straight vertical opposite side become virtually indistinguishable in length, meaning:

$$\sin \theta \approx \theta \quad (\text{for } \theta \approx 0).$$

Now let us see what happens if we use degrees. Let the same angle be ϕ in degrees. Since $\theta = \phi \times \frac{\pi}{180^\circ}$, the approximation becomes:

$$\sin(\phi^\circ) \approx \frac{\pi}{180} \phi \approx 0.01745 \phi.$$

Without radians, the ratio of the limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$ would not equal 1, but rather the messy fraction $\frac{\pi}{180}$. This abstractly demonstrates that the radian is the only angle measure that scales naturally with the linear dimension of distance. $\square$

Example 3.1.5 (Analytic Discussion on Failure of Degrees in Calculus). Discuss the analytical failure of using degrees in calculus by evaluating the derivative of $f(x) = \sin x$ when x is measured in degrees.

Solution. In calculus, the derivative of $\sin x$ is defined using the limit definition:

$$\frac{d}{dx}(\sin x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}.$$

If x and h are measured in radians, we use the fundamental limit $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ to prove that:

$$\frac{d}{dx}(\sin x) = \cos x.$$

However, if the input variable x is measured in degrees, we must apply a scaling factor inside the function to convert it to radians: $f(x) = \sin\left(\frac{\pi}{180}x\right)$.

Using the chain rule to differentiate this function:

$$f'(x) = \frac{d}{dx} \left[\sin \left(\frac{\pi}{180}x \right) \right] = \frac{\pi}{180} \cos \left(\frac{\pi}{180}x \right).$$

In degree notation, this is written as:

$$\frac{d}{dx}(\sin x^\circ) = \frac{\pi}{180} \cos x^\circ.$$

The appearance of the scaling factor $\frac{\pi}{180}$ complicates every derivative and integral in calculus. The simple, elegant rules of calculus completely fail to hold unless we use radians, making radian measure analytically essential. $\square$

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Problems for Section 3.1

1. Convert the following degree angles into exact radian measures: 30° , 45° , 60° , 90° , and 270° .
2. Convert the following radian measures into degrees: $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$, and $\frac{3\pi}{4}$.
3. Find the exact length of an arc of a circle of radius 8 cm that subtends a central angle of $\frac{3\pi}{4}$ radians.
4. Find the area of a sector of a circle of radius 10 cm with a central angle of 2.5 radians.
5. A sector of a circle has a central angle of 60° and a radius of 9 cm. Calculate its area in terms of π .
6. If an arc of length 5π cm subtends a central angle of 150° , find the radius of the circle.
7. Find the central angle (in radians) of a sector of a circle with radius 5 cm if the area of the sector is 20 cm^2 .
8. A pendulum of length 1.2 meters swings through an angle of 0.3 radians. Find the length of the arc swept out by the pendulum bob.
9. Find the perimeter of a sector of a circle with radius 6 cm and central angle $\frac{\pi}{3}$ radians.
10. Solve the equation $\sin x = \frac{\sqrt{3}}{2}$ for the radian interval $0 \leq x \leq 2\pi$.
11. Solve the equation $\cos x = -\frac{\sqrt{2}}{2}$ for the radian interval $-\pi \leq x \leq \pi$.
12. A sector has an area of $12\pi \text{ cm}^2$ and a central angle of $\frac{2\pi}{3}$ radians. Find the radius of the circle.
13. Show that the linear speed v of a point moving along a circle of radius r with constant angular speed ω (in radians per second) is given by $v = r\omega$.
14. An automobile wheel has a diameter of 60 cm. If the wheel rotates at an angular speed of 10 rad/s, calculate the linear speed of the car in meters per second.
15. Simplify the expression $\tan\left(\frac{\pi}{2} - x\right)$ using co-function identities.
16. Find all radian solutions to the equation $\tan(2x) = 1$ in the interval $0 \leq x \leq \pi$.
17. Prove that the area of a segment of a circle (the region bounded by a chord and an arc) with radius r and central angle θ (in radians) is given by $A = \frac{1}{2}r^2(\theta - \sin \theta)$.
18. If $\sin x = -\frac{1}{2}$ and $\pi \leq x \leq \frac{3\pi}{2}$, state the exact value of x in radians.
19. The perimeter of a sector is equal to four times its radius. Show that the central angle of this sector must be exactly 2 radians.
20. Discuss why we do not write the unit symbol “rad” in advanced mathematical contexts, but instead treat radian values purely as real numbers.

3.2 The Algebraic Extension of Trigonometric Ratio

In Subsection 1.4, we defined the trigonometric ratios ($\sin \theta$, $\cos \theta$, $\tan \theta$) using the side lengths of a right-angled triangle. While highly intuitive, this geometric framework is severely limited: it only makes physical sense when the angle θ is strictly acute ($0^\circ < \theta < 90^\circ$). We cannot draw a physical right-angled triangle containing an angle of 120° , 0° , or -45° , which means our ratios have not yet behaved as fully-fledged mathematical functions defined over all real numbers.

To unlock the full potential of trigonometry, we must reformulate our definitions. By using the coordinate geometry established in Section 2, we can extend our view to a **unit circle** (a circle of radius $r = 1$) centered at the origin of a rectangular coordinate system.

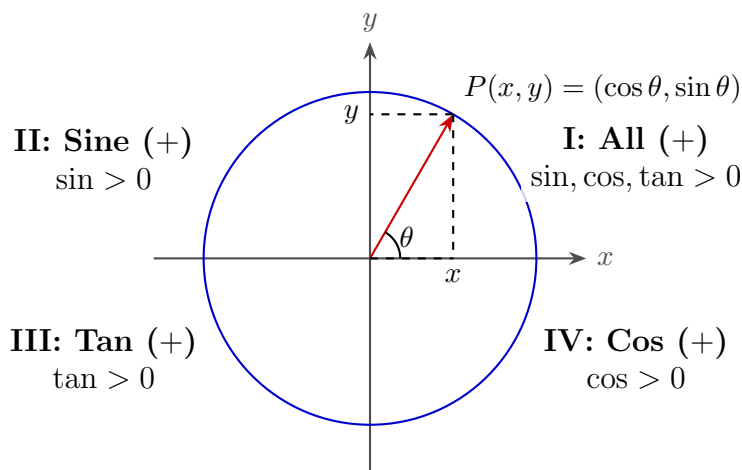


Figure 2: The unit circle definition of trigonometric functions and the ASTC quadrant rule.

By fixing the hypotenuse of our right triangle as the radius of this circle ($r = 1$), any angle θ in standard position sweeps out a terminal ray that intersects the circle at a unique coordinate point (x, y) . The coordinates of this point become our new, generalized definitions of sine and cosine. This acts as a complete mathematical mapping (or function) taking any real angle as an input and returning a coordinate on our unit circle.

Definition 3.2.1 (Unit Circle Definitions of Trigonometric Functions). Let C be the unit circle centered at the origin $O(0, 0)$ in the rectangular coordinate plane, defined by the equation $x^2 + y^2 = 1$. Let an angle θ be placed in standard position (vertex at the origin, initial side lying along the positive x -axis).

- Let $P(x, y)$ be the unique point where the terminal side of the angle θ intersects the unit circle.
- We define the **cosine function** of θ as the x -coordinate of P :

$$\cos \theta = x$$

- We define the **sine function** of θ as the y -coordinate of P :

$$\sin \theta = y$$

- We define the **tangent function** of θ as the ratio of the coordinates (where $x \neq 0$):

$$\tan \theta = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}$$

Remark 3.2.1. This definition beautifully bridges the gap between polar and rectangular coordinates. A polar point $P(1, \theta)$ lying on the unit circle has the exact rectangular coordinates $(x, y) = (\cos \theta, \sin \theta)$.

Theorem 3.2.1 (Sign of Ratios by Quadrants). *Depending on which quadrant the terminal ray of θ lands in, the signs (positive or negative) of the trigonometric values are systematically determined by the signs of the rectangular coordinates x and y :*

Quadrant	Angle Range (θ)	$\sin \theta$ (y)	$\cos \theta$ (x)	$\tan \theta$ (y/x)
Quadrant I	$0^\circ < \theta < 90^\circ$	Positive (+)	Positive (+)	Positive (+)
Quadrant II	$90^\circ < \theta < 180^\circ$	Positive (+)	Negative (−)	Negative (−)
Quadrant III	$180^\circ < \theta < 270^\circ$	Negative (−)	Negative (−)	Positive (+)
Quadrant IV	$270^\circ < \theta < 360^\circ$	Negative (−)	Positive (+)	Negative (−)

This distribution is commonly memorized using the acronym **ASTC** (“All Science Teachers Crazy” or “All Stations To Causeway Bay”), indicating which trigonometric functions are positive in Quadrants I, II, III, and IV respectively.

Proof. Let an angle θ in standard position intersect the unit circle at point $P(x, y)$. By definition, $\cos \theta = x$, $\sin \theta = y$, and $\tan \theta = \frac{y}{x}$. The signs of these functions depend strictly on the quadrant of $P(x, y)$:

1. **Quadrant I** ($0^\circ < \theta < 90^\circ$): Both coordinates are positive ($x > 0, y > 0$). Therefore, $\sin \theta > 0$, $\cos \theta > 0$, and $\tan \theta > 0$. (**All** positive).
2. **Quadrant II** ($90^\circ < \theta < 180^\circ$): Here, $x < 0$ and $y > 0$. Thus, $\sin \theta = y > 0$, while $\cos \theta < 0$ and $\tan \theta = \frac{y}{x} < 0$. (**Sine** positive).
3. **Quadrant III** ($180^\circ < \theta < 270^\circ$): Both coordinates are negative ($x < 0, y < 0$). Thus, $\sin \theta < 0$ and $\cos \theta < 0$, but their ratio $\tan \theta = \frac{-y}{-x} > 0$. (**Tangent** positive).
4. **Quadrant IV** ($270^\circ < \theta < 360^\circ$): Here, $x > 0$ and $y < 0$. Thus, $\cos \theta = x > 0$, while $\sin \theta < 0$ and $\tan \theta < 0$. (**Cosine** positive).

This establishes the mnemonic **ASTC** (All, Sine, Tangent, Cosine). □

Example 3.2.1 (Numerical Test on Coordinates). An angle θ is in standard position, and its terminal side intersects the unit circle at the point $P(-\frac{3}{5}, -\frac{4}{5})$. Find the values of $\sin \theta$, $\cos \theta$, and $\tan \theta$, and identify the quadrant of θ .

Solution. By our unit circle definition, we set:

$$x = -\frac{3}{5}, \quad y = -\frac{4}{5}.$$

Then, we directly evaluate the functions:

$$\begin{aligned} \cos \theta &= x = -\frac{3}{5} \\ \sin \theta &= y = -\frac{4}{5} \\ \tan \theta &= \frac{y}{x} = \frac{-4/5}{-3/5} = \frac{4}{3} \end{aligned}$$

Since both the x and y coordinates are negative, the point lies in the lower-left quadrant. Thus, the angle θ lies in **Quadrant III**. □

Example 3.2.2 (Algebraic Expression Derivation). Using the unit circle equation $x^2 + y^2 = 1$, derive the algebraic relation showing that the values of $\sin \theta$ and $\cos \theta$ must always be bounded between -1 and 1 for any real angle θ .

Solution. The coordinates of any point $P(x, y)$ on the unit circle satisfy:

$$x^2 + y^2 = 1.$$

Since $x^2 \geq 0$ and $y^2 \geq 0$ for any real numbers x and y , we can write:

$$x^2 \leq x^2 + y^2 = 1 \implies x^2 \leq 1$$

$$y^2 \leq x^2 + y^2 = 1 \implies y^2 \leq 1$$

Taking the square root of these inequalities:

$$|x| \leq 1 \implies -1 \leq x \leq 1$$

$$|y| \leq 1 \implies -1 \leq y \leq 1$$

Substituting $x = \cos \theta$ and $y = \sin \theta$ yields the completed derivation:

$$-1 \leq \cos \theta \leq 1 \quad \text{and} \quad -1 \leq \sin \theta \leq 1.$$

□

Example 3.2.3 (Algebraic Expression Application). Let $\sin \theta = -\frac{12}{13}$. If it is known that $\tan \theta > 0$, determine the exact value of $\cos \theta$.

Solution. We are given that $\sin \theta = y = -\frac{12}{13}$, which is negative. This means θ must lie in either Quadrant III or Quadrant IV.

We are also given that $\tan \theta > 0$ (positive). According to our quadrant rules (ASTC), the tangent function is positive only in Quadrants I and III. Therefore, θ must lie in **Quadrant III**.

In Quadrant III, the horizontal coordinate $x = \cos \theta$ must be negative. Using the Pythagoras Theorem:

$$\cos^2 \theta + \sin^2 \theta = 1 \implies \cos^2 \theta + \left(-\frac{12}{13}\right)^2 = 1.$$

$$\cos^2 \theta + \frac{144}{169} = 1 \implies \cos^2 \theta = 1 - \frac{144}{169} = \frac{25}{169}.$$

Taking the negative square root for Quadrant III:

$$\cos \theta = -\frac{5}{13}.$$

□

Example 3.2.4 (Abstract Application to Problems). Let θ be any real angle. Show abstractly that the coordinate distance between $A(\cos \theta, \sin \theta)$ and the origin $O(0, 0)$ is always constant, and state its value.

Solution. Using the 2D Distance Formula from Section 2.1, we find the distance d between $A(\cos \theta, \sin \theta)$ and $O(0, 0)$:

$$d = \sqrt{(\cos \theta - 0)^2 + (\sin \theta - 0)^2} = \sqrt{\cos^2 \theta + \sin^2 \theta}.$$

By the generalized Pythagorean identity, which holds for all real angles on our unit circle, we have $\cos^2 \theta + \sin^2 \theta = 1$. Substituting this value:

$$d = \sqrt{1} = 1.$$

Thus, the distance is always a constant value of 1 , which is consistent with A being a point lying on the unit circle. □

Example 3.2.5 (The Failure of Tangent). Using the coordinate definitions of the unit circle, explain why the function $f(\theta) = \tan \theta$ fails to be defined at $\theta = 90^\circ$ and $\theta = 270^\circ$.

Solution. By our unit circle definition:

$$\tan \theta = \frac{y}{x}.$$

This expression is mathematically undefined whenever the denominator x is equal to 0.

At $\theta = 90^\circ$, the terminal side points straight up along the positive y -axis, intersecting the unit circle at $P(0, 1)$. Here, $x = 0$, so:

$$\tan(90^\circ) = \frac{1}{0} \quad (\text{Undefined}).$$

At $\theta = 270^\circ$, the terminal side points straight down along the negative y -axis, intersecting the unit circle at $P(0, -1)$. Here also, $x = 0$, so:

$$\tan(270^\circ) = \frac{-1}{0} \quad (\text{Undefined}).$$

Thus, the tangent function fails to be defined at these boundary angles where the coordinate x vanishes. □

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Problems for Section 3.2

1. An angle θ intersects the unit circle at $P\left(\frac{5}{13}, -\frac{12}{13}\right)$. Find the exact values of $\sin \theta$ and $\tan \theta$.
2. Determine the quadrant of the angle θ if $\sin \theta < 0$ and $\cos \theta > 0$.
3. If $\cos \theta = -\frac{1}{2}$ and $\sin \theta > 0$, find the exact value of $\tan \theta$.
4. Find the coordinates on the unit circle corresponding to the boundary angles $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270° .
5. Using the boundary coordinates, state the exact value of $\cos(180^\circ) - \sin(270^\circ)$.
6. If $\tan \theta = -\frac{3}{4}$ and $\cos \theta < 0$, find the exact values of $\sin \theta$ and $\cos \theta$.
7. Let $P(\theta) = (\cos \theta, \sin \theta)$ be a point on the unit circle. If $\theta = 120^\circ$, find the exact coordinates of P using special triangles.
8. Prove using our coordinate definitions that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ is true for any quadrant.
9. If $\sin \theta = \frac{2}{3}$, find all possible values of $\cos \theta$.
10. An angle θ is in Quadrant II. If $\sin \theta = \frac{1}{\sqrt{5}}$, find the exact value of $\tan \theta$.
11. If the terminal side of an angle θ passes through the point $(-3, 4)$ (not on the unit circle), show how to scale these coordinates to find $\cos \theta$.
12. Explain using the ASTC rules why the product $\sin \theta \cos \theta$ is positive in Quadrants I and III, and negative in Quadrants II and IV.

13. If θ is in Quadrant IV and $\cos \theta = \frac{8}{17}$, find the value of $\sin \theta - \tan \theta$.
14. Find the coordinates of the point on the unit circle when the angle is -90° .
15. If $\tan \theta = \sqrt{3}$ and θ is in Quadrant III, find the exact values of $\sin \theta$ and $\cos \theta$.
16. Let $\sin \theta = a$. Express $\cos^2 \theta$ and $\tan^2 \theta$ purely in terms of a .
17. Explain why there is no real angle θ such that $\cos \theta = 1.5$.
18. For what angles θ in the range $0^\circ \leq \theta < 360^\circ$ does $\sin \theta$ equal 0?
19. Show that $\tan(135^\circ) = -1$ using coordinates. (Hint: The angle 135° makes a 45° reference angle with the negative x -axis).
20. Discuss why the unit circle definition is considered an "extension" of the right-angled triangle definition rather than an entirely new mathematical concept.

3.3 The Periodicity

In the previous subsection, we extended the definitions of the trigonometric ratios to the unit circle. Because an angle θ in standard position is defined by a terminal ray rotating around the origin, we are no longer restricted to the interval $0^\circ < \theta < 90^\circ$. We can rotate the ray counterclockwise or clockwise indefinitely.

Each complete rotation around the circle covers exactly 360° (or 2π radians). This means that if we add or subtract any integer multiple of 360° to our angle, the terminal ray returns to the exact same position on the unit circle, yielding the exact same coordinates (x, y) . This repeating behavior is called **periodicity**. While this repetitive nature makes trigonometric functions incredibly powerful for modeling cyclic phenomena (like waves, tides, and clocks), it introduces major challenges when solving trigonometric equations.

Definition 3.3.1 (Periodic Function and Periods). A function $f(\theta)$ is said to be **periodic** if there exists a non-zero constant T such that:

$$f(\theta + T) = f(\theta)$$

for all θ in the domain of f . The smallest positive real number T that satisfies this condition is called the **fundamental period** of the function.

Theorem 3.3.1 (Fundamental Periods of Trigonometric Functions). *The fundamental periods of the sine, cosine, and tangent functions are given by:*

1. *The sine and cosine functions have a fundamental period of 360° :*

$$\sin(\theta + 360^\circ) = \sin \theta \quad \text{and} \quad \cos(\theta + 360^\circ) = \cos \theta$$

2. *The tangent function has a fundamental period of 180° :*

$$\tan(\theta + 180^\circ) = \tan \theta$$

Proof. Sine and cosine follows from unit circle, while tangent follows from slope of radius. $\square$

Definition 3.3.2 (Principal Domains and Branch Cuts). A mathematical relation is only a true **function** if every single input maps to exactly one unique output. Because trigonometric functions are periodic, they are many-to-one mapping relations. To define their inverse operations unambiguously, we must restrict their domains to a single, continuous interval called a **principal branch** or perform a **branch cut**:

- For $f(\theta) = \sin \theta$, we restrict the domain to $[-90^\circ, 90^\circ]$.
- For $f(\theta) = \cos \theta$, we restrict the domain to $[0^\circ, 180^\circ]$.
- For $f(\theta) = \tan \theta$, we restrict the domain to $(-90^\circ, 90^\circ)$.

We chose the interval $[-180^\circ, 180^\circ]$ as a standard coordinate convention because it is beautifully anti-symmetric around the origin, keeping track of positive (counterclockwise) and negative (clockwise) directions equally.

Example 3.3.1 (Numerical Test on Periodicity). Find the exact numerical value of $\cos(780^\circ)$ by reducing it to an angle within the standard interval $[0^\circ, 360^\circ)$.

Solution. We can express 780° as a multiple of 360° plus a remainder:

$$780^\circ = 2 \times 360^\circ + 60^\circ = 720^\circ + 60^\circ.$$

By the periodicity of the cosine function, we can subtract the two full rotations:

$$\cos(780^\circ) = \cos(2 \times 360^\circ + 60^\circ) = \cos(60^\circ).$$

From our special angles table:

$$\cos(60^\circ) = \frac{1}{2}.$$

Thus, the exact value of $\cos(780^\circ)$ is $\frac{1}{2}$. □

Example 3.3.2 (Algebraic Expression Derivation). Prove algebraically that $\tan(\theta + 180^\circ) = \tan \theta$ using the unit circle coordinate definition and the fact that a 180° rotation reflects a coordinate point through the origin.

Solution. Let an angle θ intersect the unit circle at $P(x, y)$. By definition:

$$\tan \theta = \frac{y}{x}.$$

If we add 180° to θ , the terminal ray points in the exact opposite direction along the same straight line passing through the origin. This reflects the point $P(x, y)$ through the origin to a new point $P'(-x, -y)$.

Evaluating the tangent function at this new angle:

$$\tan(\theta + 180^\circ) = \frac{-y}{-x}.$$

Since the negative signs cancel out, we obtain:

$$\tan(\theta + 180^\circ) = \frac{y}{x} = \tan \theta.$$

This completes the derivation. □

Example 3.3.3 (Algebraic Expression Application). A student is solving the trigonometric equation $\sin \theta = \frac{1}{2}$ for θ within the range $-180^\circ \leq \theta \leq 180^\circ$. Find all valid algebraic solutions.

Solution. First, find the principal solution using the inverse sine function (restricted to the principal branch $[-90^\circ, 90^\circ]$):

$$\theta_1 = \arcsin\left(\frac{1}{2}\right) = 30^\circ.$$

Because the sine function represents the vertical coordinate y on the unit circle, there is a second angle in Quadrant II where the vertical height is also positive $\frac{1}{2}$. This symmetric angle is:

$$\theta_2 = 180^\circ - \theta_1 = 180^\circ - 30^\circ = 150^\circ.$$

Both 30° and 150° lie within the specified range $[-180^\circ, 180^\circ]$. Checking other periodic rotations (adding or subtracting 360°) yields angles outside this range. Thus, the solutions are:

$$\theta = 30^\circ \quad \text{or} \quad \theta = 150^\circ.$$

□

Example 3.3.4 (Abstract Application to Problems). Let $f(\theta) = \sin(k\theta)$, where $k > 0$ is a positive real constant. Show abstractly that the fundamental period of this modified function is given by $T = \frac{360^\circ}{k}$.

Solution. By the definition of a periodic function, we seek the smallest positive value T such that:

$$f(\theta + T) = f(\theta).$$

Substituting the definition of f :

$$\sin(k(\theta + T)) = \sin(k\theta) \implies \sin(k\theta + kT) = \sin(k\theta).$$

Since the fundamental period of the standard sine function is 360° , this equality is guaranteed to hold for all values of θ if the phase shift term kT is exactly equal to 360° :

$$kT = 360^\circ.$$

Isolating T yields:

$$T = \frac{360^\circ}{k}.$$

Thus, multiplying the input angle by k compresses the period by a factor of k . □

Example 3.3.5 (Failure of Inverse Functions). Explain why the inverse cosine expression $\arccos(\cos(240^\circ))$ does not equal 240° , showing where the direct cancellation fails.

Solution. The inverse trigonometric functions are defined strictly on their principal branches to preserve the single-output rule of mathematical functions. The principal range of the inverse cosine function ($\arccos$) is restricted to the interval $[0^\circ, 180^\circ]$.

Let us evaluate the expressions step-by-step: 1. First, compute the inner value:

$$\cos(240^\circ) = -0.5.$$

2. Next, apply the inverse function to this value, which must return an angle within the branch interval $[0^\circ, 180^\circ]$:

$$\arccos(-0.5) = 120^\circ.$$

Since $120^\circ \neq 240^\circ$, we see that $\arccos(\cos(\theta)) \neq \theta$ for angles outside the principal branch. The direct cancellation fails because the periodicity maps multiple angles to the same coordinate value, and the inverse function is forced to return the unique value from its principal branch. □

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Problems for Section 3.3

1. Find the fundamental period of the function $f(\theta) = \cos(3\theta)$.
2. Find the exact numerical value of $\sin(405^\circ)$.
3. State the exact value of $\tan(-150^\circ)$ using the periodicity and quadrant rules.
4. Find all solutions to the equation $\cos \theta = \frac{\sqrt{2}}{2}$ for the interval $-180^\circ \leq \theta \leq 180^\circ$.
5. Solve $\tan \theta = -1$ for the interval $-180^\circ \leq \theta \leq 180^\circ$.
6. Find the fundamental period of the function $g(\theta) = \tan\left(\frac{\theta}{2}\right)$.
7. Evaluate the exact value of $\sin(1110^\circ)$.
8. Solve $\sin(2\theta) = \frac{\sqrt{3}}{2}$ for the domain $0^\circ \leq \theta \leq 180^\circ$.
9. If $f(\theta)$ has a period of 60° , find the value of $f(150^\circ)$ given that $f(30^\circ) = 5$.
10. Find the value of $\arcsin(\sin(135^\circ))$.
11. Solve $\cos^2 \theta = \frac{1}{4}$ for the interval $-180^\circ \leq \theta \leq 180^\circ$.
12. Why does the function $y = \tan \theta$ have vertical asymptotes? State all asymptote locations in the range $-180^\circ \leq \theta \leq 180^\circ$.
13. Prove that the function $f(\theta) = \cos(4\theta) + \sin(2\theta)$ is periodic, and find its fundamental period.
14. Show that $\sin(\theta - 180^\circ) = -\sin \theta$ for all real angles θ .
15. Solve $\tan(3\theta) = \sqrt{3}$ for the interval $0^\circ \leq \theta \leq 180^\circ$.
16. Find the exact value of $\cos(-420^\circ)$.
17. Explain why the equation $\sin \theta = 2$ has no real solutions, and discuss this in terms of the range of the sine function.
18. If θ is restricted to $[-90^\circ, 90^\circ]$, find the unique solution to $\sin \theta = -0.5$.
19. Find the fundamental period of $h(\theta) = \sin^2 \theta$. (Hint: Think about how squaring values affects negative coordinates).
20. Discuss why solving trigonometric equations often produces infinite solution sets when no domain interval restriction is specified.

3.4 The Algebraic Use

In the previous subsections, we established the theoretical foundations of trigonometry on the unit circle and explored its periodic behavior. However, to excel in both secondary school exams (such as the HKDSE and IGCSE) and advanced mathematical studies, one must master the practical algebraic tools of trigonometry.

Trigonometry is not merely a geometric description of circles; it is an active algebraic language. We can simplify complex trigonometric expressions, prove structural identities, and solve algebraic equations and inequalities. This subsection focuses on the practical application of our core tools, specifically utilizing the fundamental Pythagorean identities to transform equations into manageable polynomial forms.

Theorem 3.4.1 (Fundamental Algebraic Identities). *For all real values of θ where the functions are defined, the following algebraic relationships are universally true:*

1. **Quotient Identity:**

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

2. **Pythagorean Identity:**

$$\sin^2 \theta + \cos^2 \theta = 1$$

3. **Secant-Tangent Identity:**

$$1 + \tan^2 \theta = \sec^2 \theta \quad \left(\text{where } \sec \theta = \frac{1}{\cos \theta} \right)$$

Proof. 1. By slope definition.

2. By the equation of unit circle $x^2 + y^2 = 1$ and that $x = \cos \theta$ and $y = \sin \theta$.

3. From 2, divide both sides by $\cos^2 \theta$ □

Example 3.4.1 (Numerical Test on Identities). If θ is an acute angle and $\sin \theta = 0.6$, calculate the numerical value of the expression $5 \cos \theta + 3 \tan \theta$.

Solution. First, use the Pythagorean identity to find $\cos \theta$:

$$\cos^2 \theta = 1 - \sin^2 \theta = 1 - (0.6)^2 = 1 - 0.36 = 0.64.$$

Since θ is acute, $\cos \theta$ must be positive:

$$\cos \theta = \sqrt{0.64} = 0.8.$$

Next, calculate $\tan \theta$ using the quotient identity:

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{0.6}{0.8} = 0.75 = \frac{3}{4}.$$

Now substitute these values into the given expression:

$$5 \cos \theta + 3 \tan \theta = 5(0.8) + 3 \left(\frac{3}{4} \right) = 4 + 2.25 = 6.25.$$

□

Example 3.4.2 (Algebraic Expression Derivation). Prove the algebraic identity:

$$\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 2 \sec^2 \theta.$$

Solution. We start with the Left-Hand Side (LHS) of the equation and combine the two fractions under a common denominator:

$$\text{LHS} = \frac{(1 + \sin \theta) + (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}.$$

Simplify the numerator and expand the denominator using the difference of squares:

$$\text{LHS} = \frac{2}{1 - \sin^2 \theta}.$$

Using the fundamental Pythagorean identity, we substitute $1 - \sin^2 \theta = \cos^2 \theta$:

$$\text{LHS} = \frac{2}{\cos^2 \theta}.$$

Since $\sec \theta = \frac{1}{\cos \theta}$, this expression simplifies to:

$$\text{LHS} = 2 \sec^2 \theta = \text{RHS}.$$

The identity is proven. □

Example 3.4.3 (Algebraic Expression Application (Quadratic Type)). Solve the trigonometric equation:

$$2 \sin^2 \theta - 3 \cos \theta - 3 = 0$$

for the interval $0^\circ \leq \theta \leq 360^\circ$.

Solution. To solve this equation, we must express all trigonometric terms in terms of a single function. We use the Pythagorean identity to substitute $\sin^2 \theta = 1 - \cos^2 \theta$:

$$2(1 - \cos^2 \theta) - 3 \cos \theta - 3 = 0.$$

Expand the terms:

$$2 - 2 \cos^2 \theta - 3 \cos \theta - 3 = 0 \implies -2 \cos^2 \theta - 3 \cos \theta - 1 = 0.$$

Multiply the entire equation by -1 to make the leading coefficient positive:

$$2 \cos^2 \theta + 3 \cos \theta + 1 = 0.$$

Let $u = \cos \theta$. This yields a standard quadratic equation:

$$2u^2 + 3u + 1 = 0.$$

Factoring this quadratic expression:

$$(2u + 1)(u + 1) = 0 \implies u = -\frac{1}{2} \quad \text{or} \quad u = -1.$$

Now substitute back $u = \cos \theta$ and solve for θ in the range $0^\circ \leq \theta \leq 360^\circ$:

- 1) $\cos \theta = -\frac{1}{2} \implies \theta = 120^\circ$ (Quadrant II) or $\theta = 240^\circ$ (Quadrant III)
- 2) $\cos \theta = -1 \implies \theta = 180^\circ$

Thus, the solutions are $\theta = 120^\circ, 180^\circ$, or 240° . □

Example 3.4.4 (Abstract Application to Problems (Involving Inequalities)). Find all values of θ in the interval $-180^\circ \leq \theta \leq 180^\circ$ that satisfy the trigonometric inequality:

$$2 \sin \theta \geq 1.$$

Solution. First, isolate the sine term by dividing both sides by 2:

$$\sin \theta \geq \frac{1}{2}.$$

We look at the unit circle. The sine function represents the vertical coordinate y . We want to find the set of angles where the vertical coordinate is greater than or equal to $\frac{1}{2}$.

The boundary values occur where $\sin \theta = \frac{1}{2}$. In the interval $[-180^\circ, 180^\circ]$, these boundary angles are:

$$\theta_1 = 30^\circ \quad \text{and} \quad \theta_2 = 150^\circ.$$

The vertical height y is greater than or equal to $\frac{1}{2}$ for all points on the upper arc between these two boundary positions.

Therefore, the solution set is:

$$30^\circ \leq \theta \leq 150^\circ.$$

□

Example 3.4.5 (Failure of Equation Solving by Division). Explain why dividing both sides of the equation $\sin \theta \cos \theta = \sin \theta$ by $\sin \theta$ leads to a failure in finding the complete set of solutions, and show the correct algebraic approach.

Solution. If we divide both sides of the equation directly by $\sin \theta$:

$$\frac{\sin \theta \cos \theta}{\sin \theta} = \frac{\sin \theta}{\sin \theta} \implies \cos \theta = 1.$$

For $0^\circ \leq \theta < 360^\circ$, this yields only the single solution $\theta = 0^\circ$.

This division fails because it assumes that $\sin \theta \neq 0$. In algebra, dividing by a variable term that can equal zero hides valid solutions (analogous to dividing by zero).

The correct approach is to move all terms to one side and factor the expression:

$$\sin \theta \cos \theta - \sin \theta = 0.$$

Factor out the common term $\sin \theta$:

$$\sin \theta (\cos \theta - 1) = 0.$$

This split structure gives two independent cases:

- 1) $\sin \theta = 0 \implies \theta = 0^\circ \quad \text{or} \quad \theta = 180^\circ$
- 2) $\cos \theta - 1 = 0 \implies \cos \theta = 1 \implies \theta = 0^\circ$

Thus, the complete correct set of solutions is $\theta = 0^\circ$ or $\theta = 180^\circ$. Simply dividing by $\sin \theta$ missed the valid solution $\theta = 180^\circ$. $\square$

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Problems for Section 3.4

1. Simplify the algebraic expression $(1 - \cos \theta)(1 + \cos \theta)$ to a single trigonometric term.
2. If $\tan \theta = \frac{4}{3}$ and θ is in Quadrant III, find the exact value of $\sin \theta + \cos \theta$.
3. Show that $\tan \theta + \frac{1}{\tan \theta} = \frac{1}{\sin \theta \cos \theta}$ for all valid angles.
4. Solve the equation $2 \cos \theta - 1 = 0$ for the domain $0^\circ \leq \theta \leq 360^\circ$.
5. Simplify the expression $\frac{\sin^2 \theta}{1 - \cos \theta}$ into a form containing no fractions.
6. Solve the equation $\sin^2 \theta - \sin \theta = 0$ for the domain $-180^\circ \leq \theta \leq 180^\circ$.
7. Prove the identity: $\cos^4 \theta - \sin^4 \theta = \cos^2 \theta - \sin^2 \theta$.
8. Find all solutions to $2 \cos^2 \theta + \sin \theta - 1 = 0$ for the interval $0^\circ \leq \theta \leq 360^\circ$.
9. Solve the trigonometric inequality $2 \cos \theta < -\sqrt{3}$ for $-180^\circ \leq \theta \leq 180^\circ$.
10. Solve the equation $3 \tan^2 \theta - 1 = 0$ for the interval $0^\circ \leq \theta \leq 180^\circ$.
11. Prove that $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = \frac{2}{\sin \theta}$.
12. Solve the equation $2 \sin^2 \theta - \sin \theta - 1 = 0$ for the domain $0^\circ \leq \theta \leq 360^\circ$.

13. If $\sin \theta + \cos \theta = \frac{1}{2}$, find the exact numerical value of the product $\sin \theta \cos \theta$. (Hint: Square both sides).
14. Solve the inequality $\tan \theta > 1$ for the interval $-90^\circ < \theta < 90^\circ$.
15. Simplify the expression $\cos \theta \tan \theta + \sin \theta$ to a single term.
16. Solve the equation $\sqrt{3} \tan \theta \sin \theta - \sin \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$.
17. Prove the identity: $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$.
18. Solve the quadratic-type equation $2 \sin^2 \theta = 3 \cos \theta$ for the domain $-180^\circ \leq \theta \leq 180^\circ$.
19. Find the range of values of k for which the equation $2 \cos \theta = k - 3$ has real solutions for θ .
20. Discuss the differences between solving a standard algebraic polynomial equation (like $x^2 - 4 = 0$) and a trigonometric polynomial equation (like $\sin^2 \theta - 4 = 0$).

3.5 The Special Angle Extend

In the previous subsections, we discovered that the trigonometric functions are periodic and can take any real angle as an input. However, computing the values of these functions for large or negative angles can feel overwhelming if we treat every angle as a completely unique case.

Fortunately, the geometry of the unit circle is highly symmetric. If we reflect a terminal ray across the vertical y -axis, the horizontal x -axis, or the diagonal line $y = x$, the physical distances of the coordinates remain exactly the same—only their signs or their horizontal/vertical roles swap. This symmetry allows us to establish a set of **reduction formulas**. By using these formulas, we can simplify any trigonometric expression containing angles like $\theta \pm 90^\circ$, $\theta \pm 180^\circ$, $\theta \pm 270^\circ$, or $\theta \pm 360^\circ$ to a simple expression involving only the acute angle θ .

Definition 3.5.1 (Reference Angle). Let θ be any angle in standard position. The **reference angle** θ_{ref} is the acute angle ($0^\circ \leq \theta_{\text{ref}} \leq 90^\circ$) formed between the terminal side of θ and the closest portion of the horizontal x -axis.

Theorem 3.5.1 (Symmetry and Reduction Formulas). *For any real angle θ where the functions are defined, the relationships under straight horizontal or vertical reflections are governed by:*

$$\begin{aligned} \sin(-\theta) &= -\sin \theta, & \cos(-\theta) &= \cos \theta, & \tan(-\theta) &= -\tan \theta \\ \sin(180^\circ - \theta) &= \sin \theta, & \cos(180^\circ - \theta) &= -\cos \theta, & \tan(180^\circ - \theta) &= -\tan \theta \\ \sin(180^\circ + \theta) &= -\sin \theta, & \cos(180^\circ + \theta) &= -\cos \theta, & \tan(180^\circ + \theta) &= \tan \theta \\ \sin(360^\circ - \theta) &= -\sin \theta, & \cos(360^\circ - \theta) &= \cos \theta, & \tan(360^\circ - \theta) &= -\tan \theta \end{aligned}$$

Proof. Trivial from coordinates. □

Theorem 3.5.2 (Co-function and Quadrant Shift Formulas). *For any real angle θ where the functions are defined, shifting by odd multiples of 90° swaps the horizontal and vertical coordinate roles, transforming functions into their corresponding co-functions:*

$$\begin{aligned} \sin(90^\circ - \theta) &= \cos \theta, & \cos(90^\circ - \theta) &= \sin \theta, & \tan(90^\circ - \theta) &= \frac{1}{\tan \theta} \\ \sin(90^\circ + \theta) &= \cos \theta, & \cos(90^\circ + \theta) &= -\sin \theta, & \tan(90^\circ + \theta) &= -\frac{1}{\tan \theta} \\ \sin(270^\circ - \theta) &= -\cos \theta, & \cos(270^\circ - \theta) &= -\sin \theta, & \tan(270^\circ - \theta) &= \frac{1}{\tan \theta} \\ \sin(270^\circ + \theta) &= -\cos \theta, & \cos(270^\circ + \theta) &= \sin \theta, & \tan(270^\circ + \theta) &= -\frac{1}{\tan \theta} \end{aligned}$$

Proof. Trivial from coordinates. □

Example 3.5.1 (Numerical Test on Reductions). Using reduction formulas, find the exact numerical value of the expression:

$$\cos(240^\circ) + \sin(-150^\circ) - \tan(315^\circ).$$

Solution. We evaluate each term individually using reference angles and quadrant rules: 1. For $\cos(240^\circ)$: The angle 240° lies in Quadrant III. Its reference angle is $240^\circ - 180^\circ = 60^\circ$. Since cosine is negative in Quadrant III:

$$\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}.$$

2. For $\sin(-150^\circ)$: Using the negative angle identity, $\sin(-\theta) = -\sin \theta$:

$$\sin(-150^\circ) = -\sin(150^\circ).$$

Since 150° lies in Quadrant II, its reference angle is $180^\circ - 150^\circ = 30^\circ$. Sine is positive in Quadrant II:

$$-\sin(150^\circ) = -\sin(30^\circ) = -\frac{1}{2}.$$

3. For $\tan(315^\circ)$: The angle 315° lies in Quadrant IV. Its reference angle is $360^\circ - 315^\circ = 45^\circ$. Since tangent is negative in Quadrant IV:

$$\tan(315^\circ) = -\tan(45^\circ) = -1.$$

Now substitute these evaluated terms back into our expression:

$$\left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right) - (-1) = -1 + 1 = 0.$$

□

Example 3.5.2 (Algebraic Expression Derivation). Using the unit circle definition of coordinates, derive the co-function identity $\sin(90^\circ - \theta) = \cos \theta$ for an acute angle θ .

Solution. Let θ be an acute angle in standard position. Construct a right-angled triangle in Quadrant I where the hypotenuse is $r = 1$. Let the vertex at the origin have angle θ .

The coordinates of the terminal point on the unit circle are $P(x, y) = (\cos \theta, \sin \theta)$. This means the horizontal base of the triangle has length $\cos \theta$ and the vertical height has length $\sin \theta$.

Now consider the angle $90^\circ - \theta$. In our right triangle, the sum of the acute angles is 90° , meaning the second acute angle at the top vertex is exactly $90^\circ - \theta$.

If we place the angle $90^\circ - \theta$ in standard position, the role of the horizontal and vertical sides is swapped. The side opposite to the angle $90^\circ - \theta$ is the horizontal base of our original triangle, which has length $\cos \theta$. By definition:

$$\sin(90^\circ - \theta) = \frac{\text{opposite to } (90^\circ - \theta)}{\text{hypotenuse}} = \frac{\cos \theta}{1} = \cos \theta.$$

This completes the geometric-coordinate derivation. □

Example 3.5.3 (Algebraic Expression Application). Solve the trigonometric equation:

$$\sin(90^\circ - \theta) = \cos(2\theta)$$

for the interval $0^\circ \leq \theta \leq 180^\circ$.

Solution. Using the co-function reduction formula, we substitute $\sin(90^\circ - \theta) = \cos \theta$ into the equation:

$$\cos \theta = \cos(2\theta).$$

For two cosine values to be equal, their angles must either be directly equal (modulo 360°) or opposite in sign (due to $\cos(-\alpha) = \cos \alpha$):

$$\text{Case 1: } 2\theta = \theta + n \cdot 360^\circ \implies \theta = n \cdot 360^\circ$$

$$\text{Case 2: } 2\theta = -\theta + n \cdot 360^\circ \implies 3\theta = n \cdot 360^\circ \implies \theta = n \cdot 120^\circ$$

Now we find all values of θ that fall inside the requested interval $0^\circ \leq \theta \leq 180^\circ$: * From Case 1: $n = 0 \implies \theta = 0^\circ$. * From Case 2: $n = 0 \implies \theta = 0^\circ$; $n = 1 \implies \theta = 120^\circ$. Thus, the valid solutions are $\theta = 0^\circ$ or $\theta = 120^\circ$. $\square$

Example 3.5.4 (Abstract Application to Problems). State the general "Odd-Even Rule" of reduction for any expression of the form $f(n \cdot 90^\circ \pm \theta)$, where n is an integer, and show how this abstracts all co-function and shift formulas.

Solution. The reduction of $f(n \cdot 90^\circ \pm \theta)$ can be simplified using a single abstract rule based on whether n is an odd or even integer:

1. **If n is even** ($2, 4, 6, \dots$, representing shifts by 180° or 360°): The function name **remains the same** (sine stays sine, cosine stays cosine).
2. **If n is odd** ($1, 3, 5, \dots$, representing shifts by 90° or 270°): The function name **swaps to its co-function** (sine becomes cosine, cosine becomes sine, tangent becomes cotangent).
3. **Sign Determination:** The positive or negative sign ($+$ or $-$) of the final reduced expression is determined purely by the sign of the *original* function in the quadrant where $(n \cdot 90^\circ \pm \theta)$ would land, assuming θ is acute.

For example, to reduce $\cos(270^\circ - \theta)$: * $n = 3$ (odd), so cosine changes to its co-function: $\sin \theta$. * The angle $270^\circ - \theta$ lands in Quadrant III, where cosine is negative. Thus:

$$\cos(270^\circ - \theta) = -\sin \theta.$$

$\square$

Example 3.5.5 (Analytic Discussion on Sign Errors). Explain how applying reduction rules blindly without keeping track of the quadrant of the original expression can lead to a sign error, using the reduction of $\cos(90^\circ + \theta)$ as an example.

Solution. A common mistake is to assume that because $\cos(90^\circ - \theta) = \sin \theta$, then $\cos(90^\circ + \theta)$ must also equal $\sin \theta$ because we are simply swapping axes.

However, we must analytically track the quadrant. Let θ be an acute angle ($0^\circ < \theta < 90^\circ$). The angle $90^\circ + \theta$ must therefore lie between 90° and 180° , which is Quadrant II.

In Quadrant II, the horizontal coordinate x is negative, meaning the cosine of any angle in Quadrant II must be a negative value. In contrast, since θ is acute, $\sin \theta$ is positive. Therefore, to preserve mathematical equality, we must introduce a negative sign:

$$\cos(90^\circ + \theta) = -\sin \theta.$$

Failing to track the quadrant of the input angle leads to a sign error of exactly -1 , showing that reduction is a directional mapping. $\square$

—

Problems for Section 3.5

1. Express $\sin(130^\circ)$ as a trigonometric function of an acute angle less than 45° .
2. Find the exact numerical value of $\tan(225^\circ)$.
3. Simplify the expression: $\sin(180^\circ - \theta) \cos(90^\circ - \theta) - \cos(180^\circ - \theta) \sin(90^\circ - \theta)$.
4. Simplify the product $\tan(90^\circ - \theta) \tan \theta$ to a single integer.
5. Find the exact value of $\sin(-240^\circ)$.
6. If $\sin \theta = 0.8$ and θ is acute, find the value of $\cos(90^\circ + \theta)$.
7. Find the reference angle for $\alpha = 310^\circ$.
8. Solve the equation $\cos(90^\circ - \theta) = \frac{1}{2}$ for the interval $0^\circ \leq \theta \leq 360^\circ$.
9. Simplify the ratio $\frac{\sin(360^\circ - \theta)}{\sin(180^\circ + \theta)}$.
10. Find the exact value of $\tan(-135^\circ) + \cos(300^\circ)$.
11. Prove the identity: $\sin(270^\circ - \theta) = -\cos \theta$ using the subtraction formulas.
12. If $\cos \theta = k$, express $\sin(270^\circ + \theta)$ in terms of k .
13. Solve the equation $\tan(180^\circ - \theta) = \sqrt{3}$ for the interval $-180^\circ \leq \theta \leq 180^\circ$.
14. Express $\cos(325^\circ)$ in terms of an acute angle less than 45° .
15. Simplify the expression $\frac{\tan(180^\circ + \theta)}{\tan(90^\circ + \theta)}$.
16. Find the exact value of $\cos(1020^\circ)$.
17. Solve the equation $\sin(180^\circ - \theta) = \cos \theta$ for the domain $0^\circ \leq \theta \leq 180^\circ$.
18. Show that $\sin(180^\circ - \theta) + \sin(180^\circ + \theta) = 0$ for all values of θ .
19. If $\tan(90^\circ - \theta) = \frac{5}{12}$ and θ is acute, find the value of $\sin \theta$.
20. Explain abstractly why the reduction rules for $\theta \pm 360^\circ$ do not involve any changes in the function names (e.g. sine remains sine), whereas rules for $\theta \pm 90^\circ$ do.

4 The Argand Plane

4.1 The Complex Number

In the previous sections, we succeeded in transforming geometry into an algebraic language through coordinate systems. However, we have run into a major algebraic limitation: the real number system $\mathbb{R}$ is not algebraically closed. If we attempt to solve a simple quadratic equation like $x^2 + 1 = 0$, we find that it has no real solutions because the square of any real number is always non-negative ($x^2 \geq 0$).

To resolve this limitation, we must expand our number system. We introduce an imaginary unit, denoted by i , defined by the property that $i^2 = -1$. By combining this unit with real numbers, we construct the set of **complex numbers**, denoted by $\mathbb{C}$.

Crucially, because a complex number is composed of two independent real components—a real part and an imaginary part—it can be mapped directly onto a 2-dimensional flat plane. This plane is called the **Argand Plane**. A complex number is not merely a number; it is a unified algebraic entity that encapsulates both rectangular and polar coordinate systems simultaneously.

Definition 4.1.1 (Complex Numbers and the Argand Plane). A **complex number** z is an expression of the form:

$$z = a + bi,$$

where a and b are real numbers ($a, b \in \mathbb{R}$), and i is the imaginary unit satisfying $i^2 = -1$.

- The real number a is called the **real part** of z , denoted by $\text{Re}(z)$.
- The real number b is called the **imaginary part** of z , denoted by $\text{Im}(z)$.
- The **Argand Plane** is a geometric representation where the horizontal axis (the real axis) represents $\text{Re}(z)$ and the vertical axis (the imaginary axis) represents $\text{Im}(z)$.
- The complex number $z = a + bi$ is plotted as the unique coordinate point (a, b) on this plane.

Definition 4.1.2 (Modulus and Argument). Let $z = a + bi$ be a complex number represented on the Argand plane.

- The **modulus** (or absolute value) of z , denoted by $|z|$, is the straight-line distance from the origin to the point z :

$$|z| = \sqrt{a^2 + b^2}.$$

- The **argument** of z , denoted by $\arg(z)$, is the angle θ (in radians or degrees) swept counterclockwise from the positive real axis to the line segment connecting the origin to z :

$$\tan \theta = \frac{b}{a} \quad (a \neq 0).$$

- To ensure uniqueness, we define the **principal argument**, denoted by $\text{Arg}(z)$ (with a capital A), restricted to the principal interval:

$$-\pi < \text{Arg}(z) \leq \pi \quad (\text{or } -180^\circ < \text{Arg}(z) \leq 180^\circ).$$

Remark 4.1.1. Observe that the modulus $|z|$ naturally corresponds to the polar coordinate radial distance r , while the argument $\arg(z)$ corresponds to the polar angle θ . Complex numbers allow us to perform vector-like 2D coordinate operations using basic scalar algebra!

Theorem 4.1.1 (Properties of Modulus). *For any complex numbers $z, w \in \mathbb{C}$:*

1. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$.
2. $|z \cdot w| = |z| \cdot |w|$.
3. $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ (for $w \neq 0$).
4. **The Triangle Inequality:** $|z + w| \leq |z| + |w|$.

Proof. We prove the Triangle Inequality $|z + w| \leq |z| + |w|$ for any $z, w \in \mathbb{C}$. Consider the square of the modulus $|z + w|^2$. Using the conjugate identity $|u|^2 = u\bar{u}$:

$$|z + w|^2 = (z + w)\overline{(z + w)} = (z + w)(\bar{z} + \bar{w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w}.$$

Recognizing that $z\bar{z} = |z|^2$, $w\bar{w} = |w|^2$, and $w\bar{z} = \overline{z\bar{w}}$:

$$|z + w|^2 = |z|^2 + (z\bar{w} + \overline{z\bar{w}}) + |w|^2.$$

For any complex number u , $u + \bar{u} = 2\operatorname{Re}(u)$. Furthermore, the real part of any number never exceeds its modulus: $\operatorname{Re}(u) \leq |u|$. Therefore:

$$z\bar{w} + \overline{z\bar{w}} = 2\operatorname{Re}(z\bar{w}) \leq 2|z\bar{w}| = 2|z||\bar{w}| = 2|z||w|.$$

Substituting this inequality back into our expression:

$$|z + w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

Taking the principal square root of both non-negative sides yields:

$$|z + w| \leq |z| + |w|.$$

□

Example 4.1.1 (Numerical Test on Modulus and Argument). Let $z = -1 + \sqrt{3}i$. Plot this number on the Argand plane, and find its modulus $|z|$ and its principal argument $\operatorname{Arg}(z)$ in degrees.

Solution. First, identify the real and imaginary parts:

$$a = \operatorname{Re}(z) = -1, \quad b = \operatorname{Im}(z) = \sqrt{3}.$$

Because $a < 0$ and $b > 0$, this point lies in **Quadrant II** of the Argand plane.

Now, calculate the modulus:

$$|z| = \sqrt{a^2 + b^2} = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = \sqrt{4} = 2.$$

Next, find the reference angle θ_{ref} :

$$\tan \theta_{\text{ref}} = \left| \frac{b}{a} \right| = \left| \frac{\sqrt{3}}{-1} \right| = \sqrt{3} \implies \theta_{\text{ref}} = 60^\circ.$$

Since the point is in Quadrant II, the principal argument is:

$$\operatorname{Arg}(z) = 180^\circ - \theta_{\text{ref}} = 180^\circ - 60^\circ = 120^\circ.$$

Thus, $|z| = 2$ and $\operatorname{Arg}(z) = 120^\circ$ (or $\frac{2\pi}{3}$ radians). □

Example 4.1.2 (Algebraic Expression Derivation). Let $z = a + bi$. The complex conjugate of z , denoted by $\bar{z}$, is defined as $\bar{z} = a - bi$. Derive the algebraic identity $z \cdot \bar{z} = |z|^2$.

Solution. Multiply z by its conjugate $\bar{z}$ using standard algebraic expansion:

$$z \cdot \bar{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2i^2.$$

Since $i^2 = -1$, substitute this value into the expression:

$$z \cdot \bar{z} = a^2 - b^2(-1) = a^2 + b^2.$$

From our definition of modulus, we know that $|z| = \sqrt{a^2 + b^2}$, which means:

$$|z|^2 = a^2 + b^2.$$

Comparing the two equations yields the completed derivation:

$$z \cdot \bar{z} = |z|^2.$$

□

Example 4.1.3 (Algebraic Expression Application). Express the complex fraction $w = \frac{2+i}{1-2i}$ in standard rectangular form $a + bi$.

Solution. To simplify a complex fraction, we multiply both the numerator and the denominator by the complex conjugate of the denominator. The conjugate of $1 - 2i$ is $1 + 2i$:

$$w = \frac{(2+i)(1+2i)}{(1-2i)(1+2i)}.$$

Expand the numerator:

$$(2+i)(1+2i) = 2 + 4i + i + 2i^2 = 2 + 5i + 2(-1) = 5i.$$

Expand the denominator using the conjugate identity $z\bar{z} = |z|^2$:

$$(1-2i)(1+2i) = 1^2 + (-2)^2 = 1 + 4 = 5.$$

Combine these results:

$$w = \frac{5i}{5} = i.$$

Thus, the standard form is $w = 0 + 1i$ (or simply i).

□

Example 4.1.4 (Abstract Application to Problems). Let $z = x + yi$. If z satisfies the algebraic equation $|z - 3i| = 2$, show abstractly that the set of all such points forms a circle on the Argand plane, and find its center and radius.

Solution. Substitute $z = x + yi$ into the equation:

$$|x + yi - 3i| = 2 \implies |x + (y - 3)i| = 2.$$

Using the modulus definition:

$$\sqrt{x^2 + (y - 3)^2} = 2.$$

Squaring both sides of the equation:

$$x^2 + (y - 3)^2 = 4.$$

In coordinate geometry, this is the standard equation of a circle:

$$(x - h)^2 + (y - k)^2 = r^2.$$

Thus, the set of points forms a circle on the Argand plane centered at the complex coordinate $3i$ (or $(0, 3)$) with a radius of 2.

□

Example 4.1.5 (Failure of the Concept). Explain why we cannot define a standard inequality relation (such as $z_1 < z_2$) for complex numbers, showing how ordering fails in the complex domain.

Solution. In the real number system $\mathbb{R}$, we have a strict linear ordering: for any two numbers, one is either greater than, less than, or equal to the other. This ordering relies on the property that squares of non-zero elements are always positive.

Suppose we try to extend this ordering to complex numbers $\mathbb{C}$. Since $i \neq 0$, we must have either $i > 0$ or $i < 0$: 1. If we assume $i > 0$: Multiplying both sides of the inequality by the positive number i should preserve the inequality:

$$i \times i > 0 \times i \implies i^2 > 0 \implies -1 > 0.$$

This is a contradiction. 2. If we assume $i < 0$: Multiplying both sides by the negative number i must reverse the inequality:

$$i \times i > 0 \times i \implies i^2 > 0 \implies -1 > 0.$$

This is also a contradiction.

Therefore, we cannot define an ordered field structure on $\mathbb{C}$. While we can compare the *magnitudes* (moduli) of complex numbers because $|z|$ is a real number (e.g. $|z_1| < |z_2|$), the direct comparison $z_1 < z_2$ is mathematically meaningless. $\square$

—

Problems for Section 4.1

1. Simplify the expression: i^3, i^4, i^5 , and i^{102} .
2. Let $z = 3 - 4i$. Calculate the exact values of $|z|$ and $\bar{z}$.
3. Express the complex number $z = -1 - i$ in terms of its modulus and principal argument.
4. Perform the operations and simplify to standard form: $(3+2i) + (5-7i)$ and $(1+i)(2-3i)$.
5. Express the fraction $\frac{5}{2+i}$ in the form $a + bi$.
6. Find the real values of x and y that satisfy the equation $(x + yi)(2 - i) = 5 + 5i$.
7. If $z = 1 - \sqrt{3}i$, find the value of $|z^4|$.
8. Sketch the region on the Argand plane defined by the inequality $|z| \leq 3$.
9. Sketch the locus of points on the Argand plane satisfying $\operatorname{Re}(z) = 2$.
10. Prove that for any complex number z , $z + \bar{z} = 2\operatorname{Re}(z)$ and $z - \bar{z} = 2i\operatorname{Im}(z)$.
11. If $z = 1 + 2i$ and $w = 3 - i$, find the standard form of $\frac{z}{w}$.
12. Prove that the modulus of a product is the product of the moduli, $|zw| = |z||w|$, using $z = a + bi$ and $w = c + di$.
13. Find the principal argument $\operatorname{Arg}(z)$ of $z = -5$ in radians.

14. Find the coordinates of the complex number z on the Argand plane if $|z| = 5$ and $\text{Arg}(z) = -\frac{\pi}{2}$.
15. Solve the quadratic equation $x^2 + 2x + 5 = 0$ over the complex numbers.
16. Find the locus of points on the Argand plane satisfying $|z - 2| = |z - 4|$. Describe its shape geometrically.
17. If $z = 2 + 3i$, find the value of $z^2 - 4z + 13$.
18. Show that $\overline{z + w} = \bar{z} + \bar{w}$ for any complex numbers z and w .
19. Find the complex number z that satisfies the system of equations: $|z| = \sqrt{2}$ and $\text{Arg}(z) = -\frac{3\pi}{4}$.
20. Discuss how the Argand plane is isomorphic (bijectively equivalent) to the $\mathbb{R}^2$ coordinate plane, yet mathematically richer due to the existence of complex multiplication.

4.2 The cis Function

In the previous subsection, we observed that any complex number $z = a + bi$ can be mapped directly to a coordinate point (a, b) on the 2-dimensional Argand plane. By bridging this representation with the polar coordinates (r, θ) established in Section 2, we can rewrite our real components a and b using the unit-circle trigonometric mappings:

$$a = r \cos \theta \quad \text{and} \quad b = r \sin \theta.$$

Substituting these coordinate expressions back into the rectangular form of our complex number, we obtain:

$$z = (r \cos \theta) + (r \sin \theta)i = r(\cos \theta + i \sin \theta).$$

Because the combination term $\cos \theta + i \sin \theta$ appears constantly throughout complex algebra, mathematicians introduce a highly compact shorthand notation: the **cis function**, which stands for “**c**osine plus **i** sine”. This representation serves as an incredibly powerful algebraic tool, converting the complex trigonometric geometry of rotations into simple arithmetic operations.

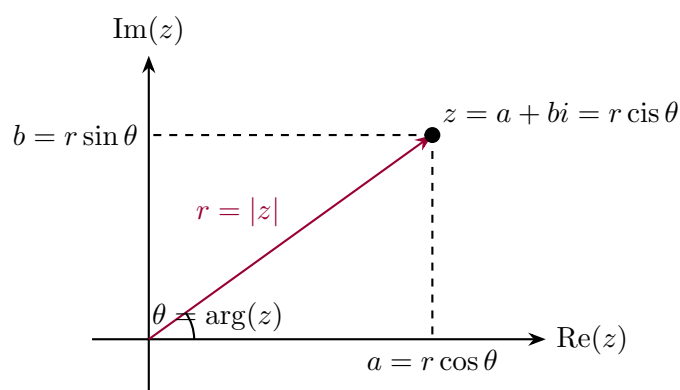


Figure 3: Geometric mapping of a complex number on the Argand plane in rectangular and polar form.

Definition 4.2.1 (The cis Function and Polar Form). For any real angle θ (measured in degrees or radians), the complex-valued function $\text{cis } \theta$ is defined as:

$$\text{cis } \theta = \cos \theta + i \sin \theta.$$

Consequently, any non-zero complex number $z = a + bi$ can be written in its unique **polar form**:

$$z = r \operatorname{cis} \theta,$$

where $r = |z|$ is the non-negative modulus, and $\theta = \arg(z)$ is the argument of z .

Theorem 4.2.1 (Algebraic Properties of cis). *For any real angles α and β , the cis function satisfies the following algebraic operations:*

1. Multiplication Rule:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = \operatorname{cis}(\alpha + \beta)$$

2. Division Rule:

$$\frac{\operatorname{cis} \alpha}{\operatorname{cis} \beta} = \operatorname{cis}(\alpha - \beta)$$

3. Conjugation Rule:

$$\overline{\operatorname{cis} \theta} = \operatorname{cis}(-\theta) = \frac{1}{\operatorname{cis} \theta}$$

Proof. By the rotational property of multiplication. □

Example 4.2.1 (Numerical Test on Polar Form). Express the complex number $z = 4 \operatorname{cis}(150^\circ)$ in standard rectangular form $a + bi$.

Solution. By definition, we expand the polar form:

$$z = 4(\cos(150^\circ) + i \sin(150^\circ)).$$

Using the special angle reduction formulas from Section 3.4, we find the values:

$$\begin{aligned} 1) \quad \cos(150^\circ) &= -\cos(30^\circ) = -\frac{\sqrt{3}}{2} \\ 2) \quad \sin(150^\circ) &= \sin(30^\circ) = \frac{1}{2} \end{aligned}$$

Substitute these values back into the expression:

$$z = 4 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right).$$

Distribute the scalar factor of 4:

$$z = -2\sqrt{3} + 2i.$$

□

Example 4.2.2 (Algebraic Expression Derivation). Prove the multiplication rule:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = \operatorname{cis}(\alpha + \beta)$$

using the compound angle sum identities for sine and cosine.

Solution. Expand the product using standard complex multiplication:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta).$$

Multiply the binomial terms:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = \cos \alpha \cos \beta + i \cos \alpha \sin \beta + i \sin \alpha \cos \beta + i^2 \sin \alpha \sin \beta.$$

Substitute $i^2 = -1$ and group the real and imaginary parts:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta).$$

Now, apply the compound angle sum identities:

$$\begin{aligned} 1) \quad & \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ 2) \quad & \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \end{aligned}$$

Substitute these identities back into the grouped expression:

$$\operatorname{cis} \alpha \cdot \operatorname{cis} \beta = \cos(\alpha + \beta) + i \sin(\alpha + \beta) = \operatorname{cis}(\alpha + \beta).$$

This completes the algebraic derivation. □

Example 4.2.3 (Algebraic Expression Application). Let $z = 6 \operatorname{cis}(75^\circ)$ and $w = 2 \operatorname{cis}(15^\circ)$. Find the exact polar representation of the product $z \cdot w$ and the quotient $\frac{z}{w}$.

Solution. Using the properties of polar multiplication and division: 1. For the product $z \cdot w$:

$$z \cdot w = (6 \cdot 2) \operatorname{cis}(75^\circ + 15^\circ) = 12 \operatorname{cis}(90^\circ).$$

2. For the quotient $\frac{z}{w}$:

$$\frac{z}{w} = \left(\frac{6}{2}\right) \operatorname{cis}(75^\circ - 15^\circ) = 3 \operatorname{cis}(60^\circ).$$

Both results are expressed in their exact, simplified polar representations. □

Example 4.2.4 (Abstract Application to Problems). Show abstractly that the modulus of $z = \operatorname{cis} \theta$ is always exactly 1 for any real angle θ , and use this to prove that the reciprocal of z is equivalent to its complex conjugate $\bar{z}$.

Solution. Let $z = \operatorname{cis} \theta = \cos \theta + i \sin \theta$. Calculating the modulus of z :

$$|z| = \sqrt{\cos^2 \theta + \sin^2 \theta}.$$

By the fundamental Pythagorean identity, $\cos^2 \theta + \sin^2 \theta = 1$ for all real θ . Therefore:

$$|z| = \sqrt{1} = 1.$$

Now, using the conjugate-product identity proven in Subsection 4.1 ($z \cdot \bar{z} = |z|^2$):

$$z \cdot \bar{z} = 1^2 = 1.$$

Dividing both sides of the equation by z (since $z \neq 0$ because $|z| = 1$):

$$\bar{z} = \frac{1}{z}.$$

Thus, for any complex number lying on the unit circle, its reciprocal and complex conjugate are identical. □

Example 4.2.5 (Failure of the Concept). Explain why the polar equation $z = r \operatorname{cis} \theta$ fails to be unique if we do not restrict the range of r and the branch of θ .

Solution. The polar representation fails to be unique for two primary reasons: 1. **Periodicity of Angles:** Because $\cos \theta$ and $\sin \theta$ have a fundamental period of 360° , we have:

$$\operatorname{cis} \theta = \operatorname{cis}(\theta + n \cdot 360^\circ)$$

for any integer n . For example, $2 \operatorname{cis}(30^\circ)$ and $2 \operatorname{cis}(390^\circ)$ represent the exact same complex number. 2. **Negative Modulus Ambiguity:** If we allow the modulus r to be negative, then a negative scalar can be absorbed by rotating the angle by 180° :

$$-r \operatorname{cis} \theta = r \operatorname{cis}(\theta + 180^\circ) \quad (\text{since } \operatorname{cis}(180^\circ) = -1).$$

Therefore, to prevent algebraic ambiguity and maintain a unique mathematical description, we must enforce the constraints $r \geq 0$ and $-\pi < \theta \leq \pi$. $\square$

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Problems for Section 4.2

- Convert the complex number $z = 2\sqrt{2} \operatorname{cis}(45^\circ)$ into rectangular form.
- Convert $w = 5 \operatorname{cis}(-90^\circ)$ into rectangular form.
- Let $z_1 = 3 \operatorname{cis}(40^\circ)$ and $z_2 = 4 \operatorname{cis}(80^\circ)$. Find the exact polar form of $z_1 \cdot z_2$.
- Let $z_1 = 8 \operatorname{cis}(100^\circ)$ and $z_2 = 2 \operatorname{cis}(40^\circ)$. Find the exact polar form of $\frac{z_1}{z_2}$.
- Find the polar form of the complex conjugate of $z = 7 \operatorname{cis}(120^\circ)$.
- Find the exact polar form of the reciprocal of $z = 3 \operatorname{cis}(30^\circ)$.
- If $z = \operatorname{cis} \theta$, show that $z + \bar{z} = 2 \cos \theta$ and $z - \bar{z} = 2i \sin \theta$.
- Convert the rectangular number $z = 1 + i$ into polar form $r \operatorname{cis} \theta$, choosing $0 \leq \theta < 2\pi$.
- Convert $w = -3 + 3\sqrt{3}i$ into polar form, choosing the principal argument $-\pi < \theta \leq \pi$.
- Simplify the expression: $\frac{4 \operatorname{cis}(250^\circ) \cdot 3 \operatorname{cis}(120^\circ)}{6 \operatorname{cis}(200^\circ)}$ into a single polar form term.
- If $z = \operatorname{cis}\left(\frac{\pi}{6}\right)$, evaluate the rectangular value of z^2 using polar multiplication.
- Find the polar representation of the imaginary unit i .
- Find the polar representation of the real number -1 .
- Solve the equation $z \cdot \operatorname{cis}(60^\circ) = 2 \operatorname{cis}(15^\circ)$ for z .
- Show that $\operatorname{cis}(-\theta) = \cos \theta - i \sin \theta$.
- If $z = \operatorname{cis} \theta$, show that $\frac{z^2 - 1}{z^2 + 1} = i \tan \theta$.
- Let $z = r_1 \operatorname{cis} \theta_1$ and $w = r_2 \operatorname{cis} \theta_2$. Prove that their quotients satisfy $\left| \frac{z}{w} \right| = \frac{r_1}{r_2}$ and $\arg\left(\frac{z}{w}\right) = \theta_1 - \theta_2$.
- Evaluate the exact rectangular value of $\operatorname{cis}(45^\circ) \cdot \operatorname{cis}(15^\circ)$.
- Find the set of all real angles $\theta \in [0, 2\pi)$ such that $\operatorname{cis} \theta$ is purely imaginary.
- Discuss why the multiplication of complex numbers in polar form ($r_1 \operatorname{cis} \theta_1 \cdot r_2 \operatorname{cis} \theta_2$) is computationally simpler than in rectangular form ($(a + bi)$).

4.3 The Geometry of i

In our introductory discussion of complex numbers, we defined the imaginary unit i from a purely algebraic perspective: it is the unique mathematical entity whose square is equal to -1 ($i^2 = -1$). While this algebraic definition successfully expands the domain of solvable polynomial equations, it does not immediately explain why complex numbers are so deeply connected to the geometry of the real plane.

The beauty of the Argand plane is that algebraic operations correspond perfectly to geometric transformations. When we add two complex numbers, we are performing vector addition (translation). But what happens when we multiply by the imaginary unit i ?

If we look at the algebra:

$$(a + bi) \cdot i = ai + bi^2 = -b + ai.$$

On the coordinate grid, this operation maps the point (a, b) to the new point $(-b, a)$. If you plot this transformation on paper, you will notice something extraordinary: the position vector has not changed in length, but it has been rotated counterclockwise by exactly 90° (a right angle) around the origin. This reveals that i is not merely an algebraic symbol; it is a geometric operator representing a quarter-turn rotation.

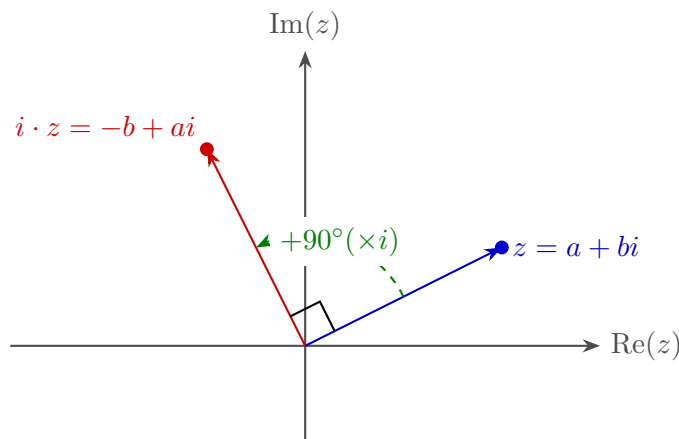


Figure 4: Multiplication by i acts as a 90° counterclockwise rotation operator on the Argand plane.

Definition 4.3.1 (The Rotation Operator i). On the Argand plane, multiplication by the imaginary unit i acts as a linear transformation that rotates any complex number z counterclockwise around the origin by exactly 90° ($\frac{\pi}{2}$ radians) without changing its modulus.

Theorem 4.3.1 (Geometric Rotation by i). Let $z = r \operatorname{cis} \theta$ be a complex number in polar form. Multiplying z by i results in:

$$i \cdot z = r \operatorname{cis} (\theta + 90^\circ).$$

Proof. Trivial from definition. □

Theorem 4.3.2 (General Rotational Multiplier). To rotate any complex number z counterclockwise around the origin by an arbitrary angle ϕ without changing its magnitude, we multiply z by the complex number $\operatorname{cis} \phi$:

$$R_\phi(z) = z \cdot \operatorname{cis} \phi.$$

Proof. Note that when $z = 1$, the cis function represent exactly how much rotates. As long as multiplication is commutative in complex number, the theorem follows. □

Example 4.3.1 (Numerical Test on Rotation). Let $z = 3 + 4i$. Calculate the numerical coordinates of $w = i \cdot z$, and verify using the Pythagorean theorem that the distance from the origin remains unchanged.

Solution. Multiply z by i :

$$w = i(3 + 4i) = 3i + 4i^2.$$

Since $i^2 = -1$:

$$w = -4 + 3i.$$

The original coordinates are $(3, 4)$ and the transformed coordinates are $(-4, 3)$.

Now, calculate the moduli of both numbers:

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

$$|w| = \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5.$$

Since $|z| = |w| = 5$, the modulus (distance from the origin) remains completely unchanged under multiplication by i . $\square$

Example 4.3.2 (Algebraic Expression Derivation). Using the transformation of coordinates $(x, y) \rightarrow (-y, x)$ under a 90° counterclockwise rotation, derive the algebraic relation of the slopes of the lines connecting the origin to $z = x + yi$ and $w = i \cdot z$.

Solution. Let the line connecting the origin to $z = x + yi$ have slope m_1 . By coordinate definitions:

$$m_1 = \frac{y}{x}.$$

Multiplying z by i gives the rotated point $w = -y + xi$. Let the line connecting the origin to w have slope m_2 :

$$m_2 = \frac{x}{-y} = -\frac{x}{y}.$$

Now, calculate the product of these two slopes:

$$m_1 \cdot m_2 = \left(\frac{y}{x}\right) \cdot \left(-\frac{x}{y}\right) = -1.$$

In coordinate geometry, two lines with non-zero slopes are perpendicular if and only if the product of their slopes is -1 . This algebraic derivation proves that multiplication by i rotates the position vector by exactly 90° , making it perpendicular to the original vector. $\square$

Example 4.3.3 (Algebraic Expression Application). You are given a complex number $z = 2 \operatorname{cis}(40^\circ)$. Find the exact rectangular coordinates of the point obtained by rotating z counterclockwise by 15° and then doubling its distance from the origin.

Solution. First, set up the rotation multiplier for 15° :

$$R_{15^\circ} = \operatorname{cis}(15^\circ).$$

To rotate z and double its magnitude, we multiply z by the scalar 2 and the rotation operator $\operatorname{cis}(15^\circ)$:

$$w = 2 \cdot z \cdot \operatorname{cis}(15^\circ).$$

Substitute $z = 2 \operatorname{cis}(40^\circ)$:

$$w = 2 \cdot [2 \operatorname{cis}(40^\circ)] \cdot \operatorname{cis}(15^\circ).$$

Multiply the moduli and add the angles:

$$w = 4 \operatorname{cis}(40^\circ + 15^\circ) = 4 \operatorname{cis}(55^\circ).$$

To express this in rectangular coordinates, expand the polar form:

$$w = 4 \cos(55^\circ) + 4i \sin(55^\circ).$$

The exact coordinates are $(4 \cos(55^\circ), 4 \sin(55^\circ))$. □

Example 4.3.4 (Abstract Application to Problems). Let z_1, z_2, z_3 represent the vertices of an equilateral triangle ordered counterclockwise on the Argand plane. Show abstractly that the vertices must satisfy the relation:

$$z_3 - z_1 = (z_2 - z_1) \operatorname{cis}(60^\circ).$$

Solution. An equilateral triangle has three equal interior angles of 60° ($\frac{\pi}{3}$ radians) and three sides of equal length.

Consider the side vectors originating from vertex z_1 : * The vector pointing from z_1 to z_2 is represented by the complex number $z_2 - z_1$. * The vector pointing from z_1 to z_3 is represented by the complex number $z_3 - z_1$.

Because the side lengths are equal, the magnitude of these two vectors is identical:

$$|z_3 - z_1| = |z_2 - z_1|.$$

Furthermore, because the interior angle at z_1 is 60° and the vertices are ordered counterclockwise, we can obtain the vector $z_3 - z_1$ by rotating the vector $z_2 - z_1$ counterclockwise by exactly 60° .

Applying our general rotational multiplier theorem with $\phi = 60^\circ$:

$$z_3 - z_1 = (z_2 - z_1) \operatorname{cis}(60^\circ).$$

This abstract relation holds for any equilateral triangle in the plane. □

Example 4.3.5 (Failure of the Concept). Explain how the geometric interpretation of i as a 90° rotation fails to make physical sense if we attempt to apply it to 1-dimensional real numbers on a standard number line without expanding to a 2D plane.

Solution. If we are restricted to a 1-dimensional real number line, we can only move in two directions: forward (positive) and backward (negative).

When we multiply a real number like 3 by -1 , we get -3 . Geometrically, this can be viewed as a “reflection” across the origin, or a rotation of 180° .

If we define i such that $i \times i = -1$, then multiplying by i twice must produce a 180° rotation. This suggests that multiplying by i once should represent a half-way step: a 90° rotation.

However, on a 1-dimensional line, there is no “upward” or “downward” direction to rotate *into*. There is no space to turn 90° . Therefore, the geometric rotation of i cannot exist in one dimension; it requires the introduction of a second vertical dimension (the imaginary axis) to serve as the space for the rotation to occur. □

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Problems for Section 4.3

1. Let $z = 5 + 2i$. Find the standard rectangular coordinates of the point obtained by rotating z counterclockwise around the origin by 90° .
2. Rotate the complex number $w = -4 + i$ clockwise around the origin by 90° . State the result in rectangular form.
3. What is the geometric effect of multiplying a complex number z by $-i$? Explain in terms of rotation angle and direction.
4. Let $z = 2 \operatorname{cis}(30^\circ)$. Find the polar form of the complex number obtained by rotating z counterclockwise by 45° and scaling its modulus by 3.
5. If you multiply a complex number z by i^3 , by how many degrees (and in which direction) have you rotated it?
6. Show that multiplying a complex number $z = x + yi$ by i^2 is geometrically equivalent to reflecting the point through the origin.
7. A square on the Argand plane has its center at the origin. If one vertex is at $z_1 = 1 + 2i$, find the other three vertices in order of counterclockwise rotation.
8. The vector pointing from the origin to z is rotated counterclockwise by 120° . If $z = 4 \operatorname{cis}(15^\circ)$, write the exact polar and rectangular coordinates of the rotated point.
9. Show that for any complex number $z \neq 0$, the ratio $\frac{iz}{z}$ is always equal to i . Explain this geometrically.
10. Let $z_1 = 2 + i$ and $z_2 = -1 + 2i$. Show that the line segments from the origin to these points are perpendicular to each other.
11. A developer is coding a 2D video game and needs to rotate a velocity vector (u, v) by 90° counterclockwise. Express this rotation using complex multiplication.
12. If $z = r \operatorname{cis} \theta$, prove that $\operatorname{Re}(i \cdot z) = -\operatorname{Im}(z)$ and $\operatorname{Im}(i \cdot z) = \operatorname{Re}(z)$.
13. Let $ABCD$ be a square ordered counterclockwise on the Argand plane. If vertex A is represented by the complex number a and vertex B is represented by b , find the complex expressions for vertices C and D in terms of a and b .
14. Rotate the complex number $z = 6 \operatorname{cis}(200^\circ)$ clockwise by 110° . What is the argument of the resulting number?
15. Explain why rotating a complex number by 360° is algebraically represented by multiplying it by i^4 .
16. Find the complex number that acts as a rotation operator for a counterclockwise rotation of 45° with no change in scale.
17. A line segment connects $z_1 = 1 + i$ to $z_2 = 4 + 3i$. If this entire segment is rotated 90° counterclockwise around z_1 (which acts as the pivot), find the coordinates of the new endpoint of the rotated segment.
18. Show that multiplying a complex number z by the scalar $\frac{1}{2}$ and the imaginary unit i yields a point whose distance from the origin is halved, and whose direction is perpendicular to z .

19. Let $z = \text{cis } \theta$. If we rotate z by $-\theta$, show that the result is the real number 1.
20. Discuss how the geometric rotation of i provides an elegant, algebraic way of representing coordinate transformations in a 2D plane without using transformation matrices.

4.4 Multiplication of Complex Numbers

In Subsection 4.2, we established the polar form of complex numbers $z = r \text{cis } \theta$ and proved the remarkable multiplicative rule:

$$(r_1 \text{cis } \alpha) \cdot (r_2 \text{cis } \beta) = (r_1 r_2) \text{cis}(\alpha + \beta).$$

If we analyze this rule closely, we observe a fascinating algebraic property: while the moduli r_1 and r_2 are multiplied, the arguments α and β are **added**. This behavior of converting a product of two functions into a sum of their inputs is the defining characteristic of an **exponential function**. That is, if $f(x) = b^x$, then $f(x) \cdot f(y) = b^{x+y}$.

This observation strongly suggests that the cis function is indeed a disguised exponential function where the exponent is imaginary. To prove this connection rigorously and move beyond a mere analogy, we must look at the power series expansions of these functions. By expanding the real exponential function e^x , the sine function, and the cosine function as infinite polynomials, we can prove the celebrated **Euler's Formula**, which bridges algebra, calculus, and geometry.

Theorem 4.4.1 (Taylor Series Expansions). *The functions e^x , $\cos x$, and $\sin x$ can be expressed as infinite power series for any real number x :*

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \end{aligned}$$

Remark 4.4.1. Take the series as definition is enough for Secondary education, and just leave the discussion to Calculus part.

Proof. Using Taylor series, behind the scope. □

Theorem 4.4.2 (Euler's Formula). *For any real number θ (measured in radians):*

$$e^{i\theta} = \cos \theta + i \sin \theta = \text{cis } \theta.$$

Proof. We prove $e^{i\theta} = \cos \theta + i \sin \theta$ by defining the auxiliary function:

$$f(\theta) = \frac{e^{i\theta}}{\cos \theta + i \sin \theta} = e^{i\theta}(\cos \theta - i \sin \theta).$$

We differentiate $f(\theta)$ with respect to θ using the product rule:

$$f'(\theta) = \left[\frac{d}{d\theta}(e^{i\theta}) \right] (\cos \theta - i \sin \theta) + e^{i\theta} \left[\frac{d}{d\theta}(\cos \theta - i \sin \theta) \right].$$

Since $\frac{d}{d\theta}(e^{i\theta}) = ie^{i\theta}$, we substitute the derivatives:

$$f'(\theta) = ie^{i\theta}(\cos \theta - i \sin \theta) + e^{i\theta}(-\sin \theta - i \cos \theta).$$

Distributing the terms inside the parentheses:

$$f'(\theta) = e^{i\theta}(i \cos \theta - i^2 \sin \theta - \sin \theta - i \cos \theta).$$

Since $i^2 = -1$, $-i^2 \sin \theta = +\sin \theta$. All terms inside the parentheses cancel out:

$$f'(\theta) = e^{i\theta}(0) = 0.$$

Because the derivative $f'(\theta)$ is identically zero for all real θ , $f(\theta)$ must be a constant function ($f(\theta) = C$). To find C , evaluate $f(0)$:

$$f(0) = e^{i(0)}(\cos 0 - i \sin 0) = 1(1 - 0) = 1.$$

Thus, $f(\theta) = 1$ for all θ , proving that $e^{i\theta} = \cos \theta + i \sin \theta$. □

Example 4.4.1 (Geometric Exponentiation via Euler's Formula). Using Euler's Formula, evaluate $e^{i\pi/2}$ and $e^{i\pi}$, and explain their geometric meaning on the Argand plane.

Solution. 1. For $\theta = \frac{\pi}{2}$ (90°):

$$e^{i\pi/2} = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) = 0 + i(1) = i.$$

Geometrically, multiplying by $e^{i\pi/2}$ rotates a number by 90° counterclockwise, which aligns with our discovery that i is a quarter-turn rotation operator! 2. For $\theta = \pi$ (180°):

$$e^{i\pi} = \cos(\pi) + i \sin(\pi) = -1 + i(0) = -1.$$

Geometrically, a 180° rotation reverses a vector's direction, mapping 1 to -1 . Adding 1 to both sides yields Euler's Identity: $e^{i\pi} + 1 = 0$. □

Theorem 4.4.3 (Euler's Identity). *By substituting the specific angle $\theta = \pi$ into Euler's Formula, we obtain the famous relation combining five of the most fundamental constants in mathematics:*

$$e^{i\pi} + 1 = 0.$$

Proof. Trivial by taking $\theta = \pi$. □

Example 4.4.2 (Numerical Test on Euler's Formula). Using Euler's Formula, express the complex number $z = 2e^{i\pi/3}$ in standard rectangular coordinates $a + bi$.

Solution. By Euler's Formula, we can write:

$$z = 2 \operatorname{cis}\left(\frac{\pi}{3}\right) = 2 \left[\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right].$$

Converting the radian measure $\frac{\pi}{3}$ to degrees gives 60° . Substituting the special values $\cos(60^\circ) = \frac{1}{2}$ and $\sin(60^\circ) = \frac{\sqrt{3}}{2}$:

$$z = 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right).$$

Distributing the factor of 2:

$$z = 1 + \sqrt{3}i.$$

The standard form is $1 + \sqrt{3}i$. □

Example 4.4.3 (Algebraic Expression Derivation). Using the series expansions of e^x , $\cos x$, and $\sin x$, derive Euler's Formula $e^{i\theta} = \cos \theta + i \sin \theta$.

Solution. Substitute $x = i\theta$ directly into the Taylor series expansion for the exponential function:

$$e^{i\theta} = 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots$$

Evaluate the powers of the imaginary unit i using the cyclic relations $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$:

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \dots$$

Group the real terms and the imaginary terms together:

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right).$$

We recognize the first grouped term as the Taylor series for $\cos \theta$, and the second grouped term as the Taylor series for $\sin \theta$. Substituting these series back into our equation yields:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

This completes the algebraic power-series derivation. □

Example 4.4.4 (Algebraic Expression Application). Using Euler's Formula, express the complex product $z_1 \cdot z_2$ as a single exponential term, where $z_1 = 3e^{i\pi/4}$ and $z_2 = 4e^{-i\pi/12}$.

Solution. Using the laws of exponents (which are fully valid for complex exponents due to Euler's Formula):

$$z_1 \cdot z_2 = \left(3e^{i\pi/4}\right) \cdot \left(4e^{-i\pi/12}\right) = (3 \cdot 4)e^{i\pi/4 - i\pi/12}.$$

Find a common denominator to subtract the fractions in the exponent:

$$\frac{\pi}{4} - \frac{\pi}{12} = \frac{3\pi - \pi}{12} = \frac{2\pi}{12} = \frac{\pi}{6}.$$

Thus:

$$z_1 \cdot z_2 = 12e^{i\pi/6}.$$

□

Example 4.4.5 (Abstract Application to Problems). Show abstractly that the sine and cosine functions can be expressed purely in terms of complex exponential functions as:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Solution. By Euler's Formula, we have:

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (\text{Equation 1})$$

Now, evaluate Euler's Formula for the negative angle $-\theta$:

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta).$$

Since cosine is an even function ($\cos(-\theta) = \cos \theta$) and sine is an odd function ($\sin(-\theta) = -\sin \theta$), we can write:

$$e^{-i\theta} = \cos \theta - i \sin \theta. \quad (\text{Equation 2})$$

1. To isolate the cosine term, we add Equation 1 and Equation 2:

$$e^{i\theta} + e^{-i\theta} = (\cos \theta + i \sin \theta) + (\cos \theta - i \sin \theta) = 2 \cos \theta.$$

Dividing by 2 yields:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}.$$

2. To isolate the sine term, we subtract Equation 2 from Equation 1:

$$e^{i\theta} - e^{-i\theta} = (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) = 2i \sin \theta.$$

Dividing by $2i$ yields:

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These relations are known as the exponential definitions of the circular functions. $\square$

Example 4.4.6 (Failure of the Concept). Explain why the real exponential function e^x is strictly one-to-one, whereas the complex exponential function e^z is periodic (many-to-one), showing how the real properties fail to transfer to the complex domain.

Solution. For real numbers $x \in \mathbb{R}$, the function $f(x) = e^x$ is strictly increasing, meaning if $e^{x_1} = e^{x_2}$, then $x_1 = x_2$.

However, in the complex domain, let $z = x + yi$. Using Euler's Formula:

$$e^z = e^{x+yi} = e^x \cdot e^{yi} = e^x (\cos y + i \sin y).$$

Since the cosine and sine functions are periodic with a period of 2π , adding any integer multiple of $2\pi i$ to the exponent yields:

$$e^{z+2\pi i} = e^z \cdot e^{2\pi i} = e^z (\cos(2\pi) + i \sin(2\pi)) = e^z (1 + 0) = e^z.$$

Thus, e^z is periodic with an imaginary period of $2\pi i$. Therefore, the one-to-one property of the real exponential function fails completely in the complex domain. $\square$

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Problems for Section 4.4

1. Convert the complex exponential number $z = 5e^{i\pi}$ into standard rectangular form.
2. Convert $w = \sqrt{2}e^{-i\pi/4}$ into rectangular coordinates.
3. Let $z_1 = 2e^{i\pi/3}$ and $z_2 = 3e^{i\pi/6}$. Express their product $z_1 \cdot z_2$ as a single complex exponential.
4. Express the quotient $\frac{8e^{i\pi/2}}{2e^{i\pi/4}}$ as a single complex exponential.
5. Convert the rectangular complex number $z = 1 - \sqrt{3}i$ into its exact exponential form $re^{i\theta}$.
6. Express the complex conjugate of $z = re^{i\theta}$ in exponential form.
7. Show that $e^{i\theta} \cdot e^{-i\theta} = 1$ using Euler's Formula.
8. Find the exact real and imaginary parts of the complex number $z = e^{2+i\pi/4}$.
9. Use Euler's Formula to evaluate the exact rectangular coordinates of $e^{i\pi/2} + e^{-i\pi/2}$.
10. Evaluate the infinite sum $1 - \frac{\pi^2}{2!} + \frac{\pi^4}{4!} - \frac{\pi^6}{6!} + \dots$ by recognizing its Taylor series.

11. Show that $|e^{i\theta}| = 1$ for any real number θ .
12. Solve the equation $e^{i\theta} = -1$ for the interval $-\pi < \theta \leq \pi$.
13. Convert the imaginary unit i into its exact exponential form $re^{i\theta}$.
14. Find the exact exponential representation of the complex number $z = -3i$.
15. Using the identity $i = e^{i\pi/2}$, evaluate the value of i^i . Show that this value is a purely real number!
16. If $z = e^{i\theta}$, show that the reciprocal $\frac{1}{z}$ is equal to $e^{-i\theta}$.
17. Prove that the derivative of $f(\theta) = e^{i\theta}$ with respect to θ satisfies $f'(\theta) = ie^{i\theta}$ by differentiating the rectangular components.
18. Show that $e^{i(\theta+2\pi)} = e^{i\theta}$ for all real θ .
19. Use the series definition of e^x to write the first four non-zero terms of e^{2i} .
20. Discuss how Euler's identity $e^{i\pi} + 1 = 0$ acts as a bridge uniting the five branches of mathematics: arithmetic (0, 1), algebra (i), geometry (π), and analysis (e).

4.5 Taking Power on Complex Numbers

In the previous subsections, we proved that complex numbers can be represented in the incredibly compact polar and exponential forms:

$$z = r \operatorname{cis} \theta = re^{i\theta}.$$

We also discovered that when we multiply two complex numbers, we multiply their moduli and add their arguments.

What happens if we want to multiply a complex number by itself repeatedly? In other words, how do we evaluate a power like z^n ? If we were to use the standard rectangular form $(a+bi)^n$, we would be forced to perform a massive binomial expansion, resulting in a complicated algebraic mess. However, by leveraging our polar representation and the exponential rules, taking powers becomes incredibly elegant. This leads us to one of the most powerful theorems in complex algebra: **De Moivre's Theorem**. This theorem not only simplifies powers but also allows us to reverse the process to find the fractional roots of complex numbers, laying the foundation for solving advanced polynomial equations.

Theorem 4.5.1 (De Moivre's Theorem). *For any real number θ and any integer $n \in \mathbb{Z}$:*

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

In our compact notation, this is written as:

$$(\operatorname{cis} \theta)^n = \operatorname{cis}(n\theta).$$

Proof. Trivially follows from addition □

Theorem 4.5.2 (Powers of General Complex Numbers). *Let $z = r \operatorname{cis} \theta$ be a complex number. For any integer $n \in \mathbb{Z}$, the n -th power of z is given by:*

$$z^n = r^n \operatorname{cis}(n\theta).$$

Proof. Trivially follows from Euler's formula. □

Theorem 4.5.3 (The n -th Roots of Unity). *The solutions to the polynomial equation $z^n = 1$, where n is a positive integer, are called the **n -th roots of unity**. There are exactly n distinct solutions, which are symmetrically distributed on the unit circle on the Argand plane:*

$$z_k = \text{cis} \left(\frac{2k\pi}{n} \right) \quad \text{for } k = 0, 1, 2, \dots, n-1.$$

Proof. By Fundamental Theorem of Arithmetic, there are n roots to the equation $z^n = 1$. Then, by the generator $e^{2i\pi/n}$, we are done. $\square$

Example 4.5.1 (Numerical Test on Powers). Evaluate the exact value of $(1+i)^{10}$ in standard rectangular form.

Solution. First, convert the base $z = 1+i$ into polar form. Calculate the modulus:

$$r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Calculate the argument (since z is in Quadrant I):

$$\theta = \arctan \left(\frac{1}{1} \right) = 45^\circ \quad \left(\text{or } \frac{\pi}{4} \text{ radians} \right).$$

Thus, the polar form of the base is:

$$z = \sqrt{2} \text{cis}(45^\circ).$$

Now, apply De Moivre's Theorem to evaluate the 10-th power:

$$z^{10} = (\sqrt{2})^{10} \text{cis}(10 \times 45^\circ).$$

Simplify the modulus term:

$$(\sqrt{2})^{10} = (2^{1/2})^{10} = 2^5 = 32.$$

Simplify the angle:

$$10 \times 45^\circ = 450^\circ.$$

Using periodicity, reduce the angle 450° to its coterminal equivalent in $[0^\circ, 360^\circ)$:

$$450^\circ - 360^\circ = 90^\circ.$$

Now evaluate the polar form:

$$z^{10} = 32 \text{cis}(90^\circ) = 32(\cos(90^\circ) + i \sin(90^\circ)).$$

Since $\cos(90^\circ) = 0$ and $\sin(90^\circ) = 1$:

$$z^{10} = 32(0 + 1i) = 32i.$$

The exact value is $32i$. $\square$

Example 4.5.2 (Algebraic Expression Derivation). Using De Moivre's Theorem and the binomial expansion theorem, derive the triple angle formula for cosine:

$$\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta.$$

Solution. By De Moivre's Theorem with $n = 3$, we have:

$$\cos(3\theta) + i \sin(3\theta) = (\cos \theta + i \sin \theta)^3.$$

Expand the right-hand side of the equation using the binomial formula $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$:

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3.$$

Evaluate the powers of i (since $i^2 = -1$ and $i^3 = -i$):

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta.$$

Group the real terms and imaginary terms:

$$(\cos \theta + i \sin \theta)^3 = (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta).$$

For two complex numbers to be equal, their real parts must be identical. Setting the real parts of both sides equal:

$$\cos(3\theta) = \cos^3 \theta - 3 \cos \theta \sin^2 \theta.$$

To express this purely in terms of cosine, substitute the Pythagorean identity $\sin^2 \theta = 1 - \cos^2 \theta$:

$$\cos(3\theta) = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) = \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta.$$

Combining like terms yields the completed derivation:

$$\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta.$$

□

Example 4.5.3 (Algebraic Expression Application). Find the three distinct complex solutions to the equation $z^3 = 8$ in exact polar form, and describe their arrangement on the Argand plane.

Solution. First, write the complex number 8 in polar form:

$$8 = 8 \operatorname{cis}(0^\circ).$$

Let our unknown solutions be represented as $z = r \operatorname{cis} \theta$. Substituting this into the equation:

$$(r \operatorname{cis} \theta)^3 = 8 \operatorname{cis}(0^\circ) \implies r^3 \operatorname{cis}(3\theta) = 8 \operatorname{cis}(0^\circ + k \cdot 360^\circ),$$

where k is any integer. By setting the moduli and arguments equal:

$$\begin{aligned} 1) \quad r^3 &= 8 \implies r = 2 \\ 2) \quad 3\theta &= k \cdot 360^\circ \implies \theta_k = k \cdot 120^\circ \end{aligned}$$

To find the three distinct solutions, we evaluate this for $k = 0, 1, 2$: * For $k = 0$: $z_0 = 2 \operatorname{cis}(0^\circ)$. * For $k = 1$: $z_1 = 2 \operatorname{cis}(120^\circ)$. * For $k = 2$: $z_2 = 2 \operatorname{cis}(240^\circ)$. Geometrically, these three solutions represent points on a circle of radius 2 centered at the origin. They form the vertices of a perfect equilateral triangle, separated by equal angular intervals of exactly 120° . □

Example 4.5.4 (Abstract Application to Problems). Show abstractly that if ω is a non-real cube root of unity (satisfying $\omega^3 = 1$ and $\omega \neq 1$), then it must satisfy the algebraic relation:

$$1 + \omega + \omega^2 = 0.$$

Solution. Consider the polynomial equation:

$$x^3 - 1 = 0.$$

Using the algebraic difference of cubes factorization, we can rewrite this as:

$$(x - 1)(x^2 + x + 1) = 0.$$

Since ω is a cube root of unity, it is a root of the equation $x^3 - 1 = 0$. Therefore:

$$(\omega - 1)(\omega^2 + \omega + 1) = 0.$$

We are given that ω is non-real, which means $\omega \neq 1$. Since the factor $(\omega - 1) \neq 0$, we can divide both sides of the equation by $(\omega - 1)$ to obtain:

$$\omega^2 + \omega + 1 = 0.$$

Rearranging terms yields the requested relation:

$$1 + \omega + \omega^2 = 0.$$

This abstract property is central to simplifying complex systems of roots. □

Example 4.5.5 (Failure of the Theorem). Explain why De Moivre's Theorem cannot be applied directly to evaluate the power of an expression of the form $(\cos \theta - i \sin \theta)^n$ without first converting it into the standard form $(\cos \phi + i \sin \phi)$.

Solution. De Moivre's Theorem states strictly that:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

This formula relies on having a positive addition sign (+) separating the real cosine and imaginary sine terms.

If we have the negative expression $(\cos \theta - i \sin \theta)^n$, we cannot blindly distribute the exponent as $\cos(n\theta) - i \sin(n\theta)$ without mathematical verification.

To resolve this and apply the theorem correctly, we must use the negative angle symmetry identities (cosine is even, sine is odd) to rewrite the base expression in standard form:

$$\cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta).$$

Now, the base is in standard position with angle $(-\theta)$. We can now safely apply De Moivre's Theorem:

$$[\cos(-\theta) + i \sin(-\theta)]^n = \cos(-n\theta) + i \sin(-n\theta) = \cos(n\theta) - i \sin(n\theta).$$

While the final result matches, attempting to apply the formula without this step is a logical failure because the negative term does not match the theorem's defined base. □

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Problems for Section 4.5

1. Evaluate $(1 + i)^4$ in standard rectangular form.
2. Use De Moivre's Theorem to find the exact value of $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{12}$.
3. Simplify $(2 \operatorname{cis}(18^\circ))^5$ into rectangular form.
4. Find the exact rectangular coordinates of z^8 given $z = -1 + \sqrt{3}i$.
5. Using De Moivre's Theorem to derive the double angle formula for sine: $\sin(2\theta) = 2 \sin \theta \cos \theta$.
6. Find the four distinct roots of the equation $z^4 = 1$ in exact polar form.
7. Sketch the locations of the four roots of $z^4 = 1$ on the Argand plane.
8. Solve $z^3 = -27i$ for z in polar form.
9. Let ω be a non-real cube root of unity. Evaluate the exact numerical value of the expression $(1 - \omega)(1 - \omega^2)$.
10. Simplify the expression: $\frac{(\operatorname{cis} \theta)^5}{(\operatorname{cis} \theta)^3}$ using De Moivre's Theorem.
11. If $z = \operatorname{cis}\left(\frac{2\pi}{5}\right)$, find the value of $1 + z + z^2 + z^3 + z^4$.
12. Find all complex solutions to $z^5 = -32$ in polar form.
13. Express $\cos(4\theta)$ purely in terms of powers of $\cos \theta$.
14. If $z + \frac{1}{z} = 2 \cos \theta$, use De Moivre's Theorem to prove that $z^n + \frac{1}{z^n} = 2 \cos(n\theta)$.
15. Find the value of $(1 - i)^{-6}$ in standard rectangular form.
16. Solve $z^2 = 2i$ in polar form.
17. Let $w = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$. Calculate the exact value of w^6 .
18. If $z = re^{i\theta}$, write the exponential form of z^{-3} .
19. Find the three roots of $z^3 = 1$ and show that their sum is exactly 0.
20. Discuss the geometric significance of how taking powers of a complex number with a modulus greater than 1 (e.g., $|z| = 1.5$) creates a spiral pattern expanding away from the origin.

5 The Compound Angle Formula

5.1 The Compound Angle Formula of Sine

In the previous sections, we succeeded in extending trigonometry to coordinate planes and the complex domain. We observed that complex numbers represent rotations, and that the multiplication of complex exponentials converts multiplication of operators into the addition of their angles:

$$e^{iA} \cdot e^{iB} = e^{i(A+B)}.$$

Now, we will harvest the algebraic fruits of this relationship. In standard secondary school geometry, proving compound angle formulas (such as $\sin(A + B)$) requires complicated and tedious geometric constructions involving auxiliary lines.

By using Euler's Formula, we can bypass these geometric steps entirely. The algebraic properties of complex exponentials automatically handle the geometry for us. In this subsection, we will derive the compound angle formulas of sine for two, three, and in general n angles. We will also examine the specialized cases of double angle, triple angle, and general n -angle formulas.

Theorem 5.1.1 (Compound Angle Formula of Sine for Two Angles). *For any real angles A and B :*

$$\begin{aligned}\sin(A + B) &= \sin A \cos B + \cos A \sin B \\ \sin(A - B) &= \sin A \cos B - \cos A \sin B\end{aligned}$$

Proof. We prove both compound angle formulas simultaneously using Euler's Formula and complex exponentiation. For any real angles A and B , Euler's Formula states:

$$e^{i(A+B)} = \cos(A + B) + i \sin(A + B).$$

By the laws of exponents, we can separate the exponential term into a product:

$$e^{i(A+B)} = e^{iA} \cdot e^{iB} = (\cos A + i \sin A)(\cos B + i \sin B).$$

Expanding the product of these two complex binomials:

$$e^{iA} \cdot e^{iB} = \cos A \cos B + i \cos A \sin B + i \sin A \cos B + i^2 \sin A \sin B.$$

Substituting $i^2 = -1$ and grouping the real and imaginary components:

$$e^{i(A+B)} = (\cos A \cos B - \sin A \sin B) + i(\sin A \cos B + \cos A \sin B).$$

Two complex expressions are equal if and only if their real parts are equal and their imaginary parts are equal. Equating the real parts yields:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

Equating the imaginary parts yields:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

Replacing B with $-B$ and applying even/odd identities ($\cos(-B) = \cos B$, $\sin(-B) = -\sin B$) yields the subtraction formulas. $\square$

Theorem 5.1.2 (Double and Triple Angle Formulas of Sine). *By setting the angles equal, we obtain the double angle and triple angle formulas of sine:*

1. Double Angle Formula:

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

2. Triple Angle Formula:

$$\sin(3\theta) = 3 \sin \theta - 4 \sin^3 \theta$$

Proof. From the previous theorem, set $A = B$ and the rest is trivial. $\square$

Theorem 5.1.3 (Compound Angle Formula of Sine for Three Angles). *For any real angles A , B , and C :*

$$\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$$

Proof. Trivial. $\square$

Theorem 5.1.4 (General n -Angle Formula of Sine). *For any real angle θ and any positive integer $n \in \mathbb{Z}^+$:*

$$\sin(n\theta) = \sum_{k \text{ odd}, k=1}^n \binom{n}{k} (-1)^{\frac{k-1}{2}} \cos^{n-k} \theta \sin^k \theta$$

Proof. Trivial. $\square$

Example 5.1.1 (Numerical Test on Compound Angles). Find the exact numerical value of $\sin(75^\circ)$ using the compound angle sum formula.

Solution. We can express the non-special angle 75° as the sum of two special angles from our table in Subsection 1.4:

$$75^\circ = 45^\circ + 30^\circ.$$

Apply the compound angle sum formula for sine with $A = 45^\circ$ and $B = 30^\circ$:

$$\sin(75^\circ) = \sin(45^\circ + 30^\circ) = \sin(45^\circ) \cos(30^\circ) + \cos(45^\circ) \sin(30^\circ).$$

Substitute the exact values of these special ratios:

$$\sin(75^\circ) = \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) + \left(\frac{\sqrt{2}}{2} \right) \left(\frac{1}{2} \right).$$

Multiply the fractions:

$$\sin(75^\circ) = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$\square$

Example 5.1.2 (Algebraic Expression Derivation). Using Euler's Formula, derive the compound angle formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$.

Solution. By Euler's Formula, we have the identity:

$$e^{i(A+B)} = \cos(A + B) + i \sin(A + B).$$

Using the laws of exponents, we can also expand this term as a product:

$$e^{i(A+B)} = e^{iA} \cdot e^{iB}.$$

Apply Euler's Formula to each of the product terms on the right-hand side:

$$e^{iA} \cdot e^{iB} = (\cos A + i \sin A)(\cos B + i \sin B).$$

Expand the product of these two binomials:

$$e^{iA} \cdot e^{iB} = \cos A \cos B + i \cos A \sin B + i \sin A \cos B + i^2 \sin A \sin B.$$

Substitute $i^2 = -1$ and group the real and imaginary parts:

$$e^{iA} \cdot e^{iB} = (\cos A \cos B - \sin A \sin B) + i(\sin A \cos B + \cos A \sin B).$$

Now, equate the imaginary part of this expanded product with the imaginary part of our first identity:

$$\operatorname{Im} \left(e^{i(A+B)} \right) = \sin(A+B).$$

$$\operatorname{Im} (e^{iA} \cdot e^{iB}) = \sin A \cos B + \cos A \sin B.$$

This algebraic equivalence completes the derivation:

$$\sin(A+B) = \sin A \cos B + \cos A \sin B.$$

□

Example 5.1.3 (Algebraic Expression Application). Using the double angle formula, show how to express $\sin(2\theta)$ purely in terms of $\tan \theta$.

Solution. Start with the double angle formula:

$$\sin(2\theta) = 2 \sin \theta \cos \theta.$$

We can divide this term by 1 without changing its value, and then substitute the Pythagorean identity $1 = \cos^2 \theta + \sin^2 \theta$ into the denominator:

$$\sin(2\theta) = \frac{2 \sin \theta \cos \theta}{1} = \frac{2 \sin \theta \cos \theta}{\cos^2 \theta + \sin^2 \theta}.$$

To express this in terms of tangent, divide both the numerator and the denominator by $\cos^2 \theta$ (assuming $\cos \theta \neq 0$):

$$\sin(2\theta) = \frac{\frac{2 \sin \theta \cos \theta}{\cos^2 \theta}}{\frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta}}.$$

Simplify each term:

$$\sin(2\theta) = \frac{2 \left(\frac{\sin \theta}{\cos \theta} \right)}{1 + \frac{\sin^2 \theta}{\cos^2 \theta}} = \frac{2 \tan \theta}{1 + \tan^2 \theta}.$$

This completed identity is highly useful in calculus for substitutions. □

Example 5.1.4 (Abstract Application to Problems). Let θ be an angle in Quadrant II such that $\sin \theta = \frac{3}{5}$. Use the triple angle formula to find the exact value of $\sin(3\theta)$.

Solution. The triple angle formula of sine states:

$$\sin(3\theta) = 3 \sin \theta - 4 \sin^3 \theta.$$

This formula depends *only* on the value of $\sin \theta$ itself; the quadrant of θ and the value of $\cos \theta$ do not affect the calculation.

Substitute the given value $\sin \theta = \frac{3}{5}$ into the formula:

$$\sin(3\theta) = 3 \left(\frac{3}{5} \right) - 4 \left(\frac{3}{5} \right)^3.$$

Calculate the powers:

$$\sin(3\theta) = \frac{9}{5} - 4 \left(\frac{27}{125} \right) = \frac{9}{5} - \frac{108}{125}.$$

Find a common denominator:

$$\sin(3\theta) = \frac{225}{125} - \frac{108}{125} = \frac{117}{125}.$$

Thus, the exact value is $\frac{117}{125}$. □

Example 5.1.5 (The Failure of Addition). Explain why the direct addition statement $\sin(A + B) = \sin A + \sin B$ is a failure of basic algebra, and provide a numerical counterexample.

Solution. The notation $\sin(A + B)$ represents a function evaluated at the single combined input $(A + B)$. It is not a multiplication of $\sin$ by $(A + B)$. Therefore, the distributive property of algebra:

$$a(b + c) = ab + ac$$

does not apply to functions.

To prove this statement fails, let us choose a simple numerical counterexample using special angles: $A = 30^\circ$ and $B = 60^\circ$. 1. Evaluate the left-hand side:

$$\sin(A + B) = \sin(30^\circ + 60^\circ) = \sin(90^\circ) = 1.$$

2. Evaluate the right-hand side:

$$\sin A + \sin B = \sin(30^\circ) + \sin(60^\circ) = \frac{1}{2} + \frac{\sqrt{3}}{2} \approx 0.5 + 0.866 = 1.366.$$

Since $1 \neq 1.366$, the direct addition assumption fails completely. $\square$

—

Problems for Section 5.1

- Find the exact value of $\sin(15^\circ)$ using the compound angle subtraction formula.
- If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, where A and B are acute angles, calculate the exact value of $\sin(A + B)$.
- Find the exact value of $\sin(105^\circ)$ by expressing it as the sum of two special angles.
- Simplify the expression: $\sin(x + 45^\circ) + \sin(x - 45^\circ)$.
- If $\tan \theta = 3$ and θ is acute, find the exact value of $\sin(2\theta)$.
- Use the triple angle formula of sine to find the exact value of $\sin(3\theta)$ when $\sin \theta = \frac{1}{3}$.
- If $\sin(x - 30^\circ) = \cos x$, solve for x in the interval $0^\circ \leq x \leq 180^\circ$.
- Prove the identity: $\frac{\sin(2\theta)}{\sin \theta} - \frac{\cos(2\theta)}{\cos \theta} = \frac{1}{\cos \theta}$ for all valid angles.
- Express $\sin(4\theta)$ as an expression containing only $\sin \theta$ and $\cos \theta$ terms.
- If $\sin(A + B) = 3 \sin(A - B)$, show that $\tan A = 2 \tan B$.
- Find the exact value of the product $\sin(15^\circ) \cos(15^\circ)$ using the double angle formula in reverse.
- Solve the equation $\sin(2x) - \sin x = 0$ for the domain $0^\circ \leq x \leq 360^\circ$.
- Simplify the three-angle term $\sin(x + y + z)$ when $y = 90^\circ$ and $z = 180^\circ$.
- If $\cos \theta = -\frac{4}{5}$ and θ is in Quadrant III, find the exact value of $\sin(2\theta)$.
- Show that $\sin(3\theta) = \sin \theta (2 \cos(2\theta) + 1)$.

16. Solve the equation $2 \sin x \cos x = \frac{1}{2}$ for the interval $0^\circ \leq x \leq 180^\circ$.
17. If $A + B + C = 180^\circ$, prove that $\sin A \cos B + \cos A \sin B = \sin C$.
18. Let $\sin \theta = t$. Express $\sin(3\theta)$ purely in terms of t .
19. Prove that $\sin(5\theta) = 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta$ using the general n -angle formula.
20. Discuss how the compound angle formulas act as a critical bridge in physics for analyzing the interference of overlapping sound waves.

5.2 The Compound Angle Formula of Cosine

In the previous subsection, we used the algebraic properties of complex exponentials to elegantly derive the compound angle formulas for the sine function. We now turn our attention to the companion of sine: the cosine function.

Just as with sine, deriving the compound angle formula for cosine via classical Euclidean geometry requires intricate constructions of right-angled triangles packed within one another. By maintaining our modern perspective on the Argand plane and using Euler's Formula, we can extract the cosine formulas from the exact same complex product:

$$e^{i(A+B)} = e^{iA} \cdot e^{iB}.$$

While the sine formulas are captured by the imaginary parts of this product, the cosine formulas are encoded within the real parts. In this subsection, we will derive these formulas for two, three, and general n angles, and examine the highly important double and triple angle identities.

Theorem 5.2.1 (Compound Angle Formula of Cosine for Two Angles). *For any real angles A and B :*

$$\begin{aligned}\cos(A + B) &= \cos A \cos B - \sin A \sin B \\ \cos(A - B) &= \cos A \cos B + \sin A \sin B\end{aligned}$$

Proof. Similar to sine. □

Theorem 5.2.2 (Double and Triple Angle Formulas of Cosine). *By setting the angles equal, we obtain the double angle and triple angle formulas of cosine:*

1. Double Angle Formulas:

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

2. Triple Angle Formula:

$$\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta$$

Proof. We prove both compound angle formulas simultaneously using Euler's Formula and complex exponentiation. For any real angles A and B , Euler's Formula states:

$$e^{i(A+B)} = \cos(A+B) + i \sin(A+B).$$

By the laws of exponents, we can separate the exponential term into a product:

$$e^{i(A+B)} = e^{iA} \cdot e^{iB} = (\cos A + i \sin A)(\cos B + i \sin B).$$

Expanding the product of these two complex binomials:

$$e^{iA} \cdot e^{iB} = \cos A \cos B + i \cos A \sin B + i \sin A \cos B + i^2 \sin A \sin B.$$

Substituting $i^2 = -1$ and grouping the real and imaginary components:

$$e^{i(A+B)} = (\cos A \cos B - \sin A \sin B) + i(\sin A \cos B + \cos A \sin B).$$

Two complex expressions are equal if and only if their real parts are equal and their imaginary parts are equal. Equating the real parts yields:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

Equating the imaginary parts yields:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

Replacing B with $-B$ and applying even/odd identities ($\cos(-B) = \cos B$, $\sin(-B) = -\sin B$) yields the subtraction formulas. $\square$

Theorem 5.2.3 (Compound Angle Formula of Cosine for Three Angles). *For any real angles A , B , and C :*

$$\cos(A + B + C) = \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C$$

Proof. Trivial. $\square$

Theorem 5.2.4 (General n -Angle Formula of Cosine). *For any real angle θ and any positive integer $n \in \mathbb{Z}^+$:*

$$\cos(n\theta) = \sum_{k \text{ even}, k=0}^n \binom{n}{k} (-1)^{\frac{k}{2}} \cos^{n-k} \theta \sin^k \theta$$

Proof. Trivial. $\square$

Example 5.2.1 (Numerical Test on Compound Angles). Find the exact numerical value of $\cos(15^\circ)$ using the compound angle subtraction formula.

Solution. We can express the acute angle 15° as the difference of two special angles:

$$15^\circ = 45^\circ - 30^\circ.$$

Apply the compound angle subtraction formula for cosine with $A = 45^\circ$ and $B = 30^\circ$:

$$\cos(15^\circ) = \cos(45^\circ - 30^\circ) = \cos(45^\circ) \cos(30^\circ) + \sin(45^\circ) \sin(30^\circ).$$

Substitute the exact values of these special ratios:

$$\cos(15^\circ) = \left(\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right).$$

Multiply the fractions:

$$\cos(15^\circ) = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Note that this exact value is identical to $\sin(75^\circ)$, which is consistent with the co-function identity $\cos(15^\circ) = \sin(90^\circ - 15^\circ) = \sin(75^\circ)$. $\square$

Example 5.2.2 (Algebraic Expression Derivation). Using Euler's Formula, derive the compound angle formula $\cos(A + B) = \cos A \cos B - \sin A \sin B$.

Solution. By Euler's Formula, we have the identity:

$$e^{i(A+B)} = \cos(A+B) + i \sin(A+B).$$

We can expand this exponential term as a product:

$$e^{i(A+B)} = e^{iA} \cdot e^{iB} = (\cos A + i \sin A)(\cos B + i \sin B).$$

Multiply the two complex binomial expressions:

$$e^{iA} \cdot e^{iB} = \cos A \cos B + i \cos A \sin B + i \sin A \cos B + i^2 \sin A \sin B.$$

Substitute $i^2 = -1$ and group the real and imaginary parts:

$$e^{iA} \cdot e^{iB} = (\cos A \cos B - \sin A \sin B) + i(\sin A \cos B + \cos A \sin B).$$

For two complex expressions to be equal, their real parts must be identical. Equating the real parts of both sides of our identities:

$$\operatorname{Re} \left(e^{i(A+B)} \right) = \cos(A+B).$$

$$\operatorname{Re} (e^{iA} \cdot e^{iB}) = \cos A \cos B - \sin A \sin B.$$

Setting these equal yields the completed derivation:

$$\cos(A+B) = \cos A \cos B - \sin A \sin B.$$

□

Example 5.2.3 (Algebraic Expression Application). Using the double angle formulas of cosine, derive the linear power-reducing identities for $\cos^2 \theta$ and $\sin^2 \theta$ that are widely used to simplify calculations in calculus.

Solution. 1. To isolate $\cos^2 \theta$, start with the double angle formula containing only cosine terms:

$$\cos(2\theta) = 2 \cos^2 \theta - 1.$$

Add 1 to both sides of the equation:

$$1 + \cos(2\theta) = 2 \cos^2 \theta.$$

Divide both sides by 2:

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}.$$

2. To isolate $\sin^2 \theta$, start with the double angle formula containing only sine terms:

$$\cos(2\theta) = 1 - 2 \sin^2 \theta.$$

Rearrange the terms to isolate the sine squared term:

$$2 \sin^2 \theta = 1 - \cos(2\theta).$$

Divide both sides by 2:

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}.$$

These two algebraic identities allow us to reduce quadratic powers of trigonometric functions into linear expressions. □

Example 5.2.4 (Abstract Application to Problems). Let θ be an angle in Quadrant IV such that $\cos \theta = \frac{4}{5}$. Calculate the exact value of $\cos(2\theta)$.

Solution. The double angle formula of cosine has three equivalent forms. Since we are given the value of $\cos \theta$ directly, we can choose the form that depends only on cosine:

$$\cos(2\theta) = 2 \cos^2 \theta - 1.$$

This choice of formula means we do not need to calculate the sign or value of $\sin \theta$, making our calculation elegant and minimizing potential sign errors in Quadrant IV.

Substitute $\cos \theta = \frac{4}{5}$ into the formula:

$$\cos(2\theta) = 2 \left(\frac{4}{5} \right)^2 - 1 = 2 \left(\frac{16}{25} \right) - 1.$$

Multiply the terms:

$$\cos(2\theta) = \frac{32}{25} - \frac{25}{25} = \frac{7}{25}.$$

Thus, the exact value is $\frac{7}{25}$. □

Example 5.2.5 (The Failure of Sign Rules). Explain why the compound angle sum formula for cosine contains a subtraction sign ($\cos(A + B) = \cos A \cos B - \sin A \sin B$), whereas the sine formula contains an addition sign ($\sin(A + B) = \sin A \cos B + \cos A \sin B$).

Solution. This difference is not a random algebraic rule, but a direct consequence of complex number multiplication.

When we multiply the imaginary components of two complex numbers:

$$(i \sin A)(i \sin B) = i^2 \sin A \sin B.$$

Because i^2 is defined to be exactly -1 , this imaginary product undergoes a sign change and is projected onto the negative real axis.

Because the real part of the complex product $e^{iA}e^{iB}$ defines the cosine sum formula, this -1 factor from i^2 is directly responsible for the negative sign in $\cos(A + B)$. Thus, the subtraction sign reflects the algebraic rotation properties of the 2D plane, showing why a simple additive sign rule fails to carry over from sine. □

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Problems for Section 5.2

1. Find the exact numerical value of $\cos(75^\circ)$ using the compound angle sum formula.
2. If $\sin A = \frac{5}{13}$ and $\sin B = \frac{3}{5}$, where A and B are acute angles, calculate the exact value of $\cos(A + B)$.
3. Find the exact value of $\cos(105^\circ)$ by expressing it as the sum of two special angles.
4. Simplify the algebraic expression: $\cos(x - 60^\circ) - \cos(x + 60^\circ)$.
5. If $\cos \theta = -\frac{1}{3}$ and θ is in Quadrant II, find the exact value of $\cos(2\theta)$.
6. Use the triple angle formula of cosine to find the exact value of $\cos(3\theta)$ when $\cos \theta = \frac{1}{4}$.

7. Solve the equation $\cos(2x) + \cos x = 0$ for the domain $0^\circ \leq x \leq 360^\circ$.
8. Prove the identity: $\cos(2\theta) = \frac{1-\tan^2\theta}{1+\tan^2\theta}$ for all valid angles.
9. Express $\cos(4\theta)$ purely in terms of $\cos \theta$.
10. If $\cos(A - B) = 2\cos(A + B)$, show that $\tan A \tan B = \frac{1}{3}$.
11. Simplify the expression $\cos^2(15^\circ) - \sin^2(15^\circ)$ using a double angle formula in reverse.
12. Find all solutions to the equation $2\cos^2 x - 1 = \cos x$ for the interval $0^\circ \leq x \leq 360^\circ$.
13. Simplify the three-angle term $\cos(x + y + z)$ when $y = 90^\circ$ and $z = 90^\circ$.
14. If $\sin \theta = -\frac{4}{5}$ and θ is in Quadrant III, find the exact value of $\cos(2\theta)$.
15. Show that $\cos(3\theta) = \cos \theta(2\cos(2\theta) - 1)$.
16. Solve the equation $\cos(2x) + 3\sin x - 2 = 0$ for the interval $0^\circ \leq x \leq 360^\circ$.
17. If $A + B + C = 180^\circ$, prove that $\cos(A + B) + \cos C = 0$.
18. Let $\cos \theta = u$. Express $\cos(3\theta)$ purely in terms of u .
19. Prove that $\cos(5\theta) = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$ using the general n -angle formula.
20. Discuss how the compound angle formulas of cosine are applied in acoustics to analyze the destructive interference (phase-cancellation) used in active noise-canceling headphones.

5.3 The Compound Angle Formula of Tangent

In the previous two subsections, we derived the compound angle formulas for sine and cosine. We now turn our attention to the third fundamental trigonometric function: the tangent function.

By our definition, the tangent function is the ratio of the sine and cosine functions:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

Rather than performing new geometric constructions or complex exponentials from scratch, we can elegantly construct the compound angle formulas for tangent by directly dividing the compound angle formulas of sine and cosine. In this subsection, we will derive these formulas for two, three, and general n angles, and examine the resulting double and triple angle identities.

Theorem 5.3.1 (Compound Angle Formula of Tangent for Two Angles). *For any real angles A and B where the functions are defined:*

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Proof. □

Theorem 5.3.2 (Double and Triple Angle Formulas of Tangent). *By setting the angles equal, we obtain the double angle and triple angle formulas of tangent:*

1. **Double Angle Formula:**

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

2. **Triple Angle Formula:**

$$\tan(3\theta) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

Proof. Trivial by taking $3\theta = \theta + 2\theta$ □

Theorem 5.3.3 (Compound Angle Formula of Tangent for Three Angles). *For any real angles A , B , and C where the functions are defined:*

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

Proof. Trivial, by recursively applying compound angle formula. □

Theorem 5.3.4 (General n -Angle Formula of Tangent). *For any real angle θ and any positive integer $n \in \mathbb{Z}^+$ where the functions are defined:*

$$\tan(n\theta) = \frac{\sum_{k \text{ odd}} \binom{n}{k} (-1)^{\frac{k-1}{2}} \tan^k \theta}{\sum_{k \text{ even}} \binom{n}{k} (-1)^{\frac{k}{2}} \tan^k \theta}$$

Proof. Trivial, by De Moivre's Theorem. □

Example 5.3.1 (Numerical Test on Compound Angles). Find the exact numerical value of $\tan(75^\circ)$ using the compound angle sum formula.

Solution. We can express the angle 75° as the sum of two special angles:

$$75^\circ = 45^\circ + 30^\circ.$$

Apply the compound angle sum formula for tangent with $A = 45^\circ$ and $B = 30^\circ$:

$$\tan(75^\circ) = \tan(45^\circ + 30^\circ) = \frac{\tan(45^\circ) + \tan(30^\circ)}{1 - \tan(45^\circ) \tan(30^\circ)}.$$

Substitute the exact values of these special ratios ($\tan(45^\circ) = 1$ and $\tan(30^\circ) = \frac{\sqrt{3}}{3}$):

$$\tan(75^\circ) = \frac{1 + \frac{\sqrt{3}}{3}}{1 - (1) \left(\frac{\sqrt{3}}{3} \right)} = \frac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}.$$

To rationalize the denominator, multiply the numerator and the denominator by the conjugate $(3 + \sqrt{3})$:

$$\tan(75^\circ) = \frac{(3 + \sqrt{3})^2}{(3 - \sqrt{3})(3 + \sqrt{3})} = \frac{9 + 6\sqrt{3} + 3}{9 - 3} = \frac{12 + 6\sqrt{3}}{6} = 2 + \sqrt{3}.$$

□

Example 5.3.2 (Algebraic Expression Derivation). Derive the compound angle formula for $\tan(A + B)$ using the compound angle formulas of sine and cosine.

Solution. By definition:

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)}.$$

Substitute the compound angle formulas for sine and cosine:

$$\tan(A + B) = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}.$$

To express this quotient entirely in terms of tangent, we divide every term in both the numerator and the denominator by the product $\cos A \cos B$ (assuming $\cos A \cos B \neq 0$):

$$\tan(A + B) = \frac{\frac{\sin A \cos B}{\cos A \cos B} + \frac{\cos A \sin B}{\cos A \cos B}}{\frac{\cos A \cos B}{\cos A \cos B} - \frac{\sin A \sin B}{\cos A \cos B}}.$$

Simplify each individual fraction:

$$\tan(A + B) = \frac{\left(\frac{\sin A}{\cos A}\right)(1) + (1)\left(\frac{\sin B}{\cos B}\right)}{1 - \left(\frac{\sin A}{\cos A}\right)\left(\frac{\sin B}{\cos B}\right)}.$$

Substituting the definition of tangent yields the completed derivation:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

□

Example 5.3.3 (Algebraic Expression Application). Using the double angle formula of tangent, show how to solve the quadratic-type trigonometric equation:

$$\tan(2\theta) = 3 \tan \theta$$

for the interval $0^\circ \leq \theta \leq 180^\circ$ (excluding $\theta = 90^\circ$).

Solution. Substitute the double angle formula of tangent into the equation:

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = 3 \tan \theta.$$

To solve this equation, we avoid dividing by $\tan \theta$ directly (which would hide valid solutions). Instead, rearrange the terms to one side:

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} - 3 \tan \theta = 0.$$

Factor out the common term $\tan \theta$:

$$\tan \theta \left(\frac{2}{1 - \tan^2 \theta} - 3 \right) = 0.$$

This split structure yields two independent cases:

$$\text{Case 1: } \tan \theta = 0 \implies \theta = 0^\circ \text{ or } \theta = 180^\circ$$

$$\text{Case 2: } \frac{2}{1 - \tan^2 \theta} - 3 = 0 \implies 2 = 3(1 - \tan^2 \theta) \implies 2 = 3 - 3 \tan^2 \theta$$

Isolate the tangent squared term in Case 2:

$$3 \tan^2 \theta = 1 \implies \tan^2 \theta = \frac{1}{3} \implies \tan \theta = \pm \frac{\sqrt{3}}{3}.$$

Now, solve for θ in the range $0^\circ \leq \theta \leq 180^\circ$:

$$\tan \theta = \frac{\sqrt{3}}{3} \implies \theta = 30^\circ \text{ (Quadrant I)}$$

$$\tan \theta = -\frac{\sqrt{3}}{3} \implies \theta = 150^\circ \text{ (Quadrant II)}$$

Combining both cases, the valid solutions are $\theta = 0^\circ, 30^\circ, 150^\circ$, or 180° .

□

Example 5.3.4 (Abstract Application to Problems). Let α and β be acute angles such that $\tan \alpha = \frac{1}{2}$ and $\tan \beta = \frac{1}{3}$. Prove abstractly that the sum of these two angles must be exactly 45° ($\frac{\pi}{4}$ radians).

Solution. Let us calculate the tangent of their sum, $\tan(\alpha + \beta)$, using the compound angle sum formula:

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}.$$

Substitute the given values into the formula:

$$\tan(\alpha + \beta) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}.$$

Evaluate the numerator and the denominator:

$$\text{Numerator} = \frac{3 + 2}{6} = \frac{5}{6}$$

$$\text{Denominator} = 1 - \frac{1}{6} = \frac{5}{6}.$$

Thus:

$$\tan(\alpha + \beta) = \frac{5/6}{5/6} = 1.$$

For acute angles, if $\tan(\alpha + \beta) = 1$, then the angle must be exactly:

$$\alpha + \beta = 45^\circ \quad \left(\text{or } \frac{\pi}{4}\right).$$

This completes the abstract proof. □

Example 5.3.5 (Failure of the Tangent Sum). Explain why the compound angle formula for tangent fails to be defined when either $\tan A$, $\tan B$, or the denominator term $(1 - \tan A \tan B)$ is undefined, and show a specific angular configuration where this occurs.

Solution. The compound angle formula of tangent is a rational expression, meaning it fails to be defined whenever a division by zero occurs. This happens in two ways: 1. ****Co-domain Boundaries:**** If $A = 90^\circ$ or $B = 90^\circ$, the individual terms $\tan A$ or $\tan B$ are themselves mathematically undefined, preventing their substitution into the formula. 2. ****Asymptote Shifts:**** If the sum angle $A + B = 90^\circ$, then $\tan(A + B)$ is undefined.

Let us see how the formula's algebraic denominator reflects this. If $A + B = 90^\circ$, then $B = 90^\circ - A$. Applying the co-function identity:

$$\tan B = \tan(90^\circ - A) = \frac{1}{\tan A}.$$

Now, evaluate the denominator of the compound sum formula:

$$1 - \tan A \tan B = 1 - \tan A \left(\frac{1}{\tan A} \right) = 1 - 1 = 0.$$

Because the denominator collapses to zero, the formula yields a division by zero, perfectly reflecting the physical asymptote of the tangent function. Thus, the formula fails for any angular configuration where $A + B = 90^\circ + n \cdot 180^\circ$. □

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Problems for Section 5.3

1. Find the exact numerical value of $\tan(15^\circ)$ using the compound angle subtraction formula.
2. If $\tan A = \frac{3}{4}$ and $\tan B = \frac{5}{12}$, calculate the exact value of $\tan(A + B)$.
3. Find the exact value of $\tan(105^\circ)$ by expressing it as the sum of two special angles.
4. Simplify the algebraic expression: $\frac{\tan x + \tan(45^\circ)}{1 - \tan x \tan(45^\circ)}$.
5. If $\sin \theta = \frac{3}{5}$ and θ is acute, find the exact value of $\tan(2\theta)$.
6. Use the triple angle formula of tangent to find the exact value of $\tan(3\theta)$ when $\tan \theta = \frac{1}{2}$.
7. Solve the equation $\tan(2x) = \tan x$ for the domain $0^\circ \leq x \leq 180^\circ$ (excluding $x = 90^\circ$).
8. Prove the identity: $\tan(45^\circ + \theta) \tan(45^\circ - \theta) = 1$ for all valid angles.
9. Express $\tan(4\theta)$ as an algebraic fraction containing only $\tan \theta$ terms.
10. If $A + B = 45^\circ$, prove that $(1 + \tan A)(1 + \tan B) = 2$.
11. Simplify the expression $\frac{2 \tan(15^\circ)}{1 - \tan^2(15^\circ)}$ using a double angle formula in reverse.
12. Find all solutions to the equation $\tan(2x) + 2 \tan x = 0$ for the interval $0^\circ \leq x \leq 180^\circ$.
13. Simplify the three-angle term $\tan(A + B + C)$ when $A + B + C = 180^\circ$ (the interior angles of a triangle).
14. If $\cos \theta = -\frac{12}{13}$ and θ is in Quadrant II, find the exact value of $\tan(2\theta)$.
15. Show that $\tan(3\theta) - \tan(2\theta) - \tan \theta = \tan(3\theta) \tan(2\theta) \tan \theta$.
16. Solve the equation $\tan^2 x + (1 - \sqrt{3}) \tan x - \sqrt{3} = 0$ for the interval $0^\circ \leq x \leq 180^\circ$.
17. If $\tan(A - B) = \frac{1}{3}$ and $\tan B = \frac{2}{5}$, find the exact value of $\tan A$.
18. Let $\tan \theta = y$. Express $\tan(3\theta)$ purely in terms of y .
19. Prove that $\tan(5\theta) = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$ using the general n -angle formula.
20. Discuss how the compound angle formula of tangent is utilized in surveying to find the slope of a hill combined from two diagonal transit measurements.

5.4 The Half Angle Formula

In the previous subsections, we derived the compound angle and double angle formulas. These equations allow us to scale up: if we know the trigonometric values for an angle θ , we can calculate the values for 2θ , 3θ , or $n\theta$.

Now, we will reverse this direction. We want to scale down: if we know the trigonometric properties of a given angle θ , how can we evaluate the properties of its half, $\frac{\theta}{2}$? This is particularly useful for finding the exact values of non-standard acute angles (such as 22.5° or 15°) without relying on decimal approximations.

By taking the double-angle power-reducing formulas of cosine that we derived in Subsection 5.2 and performing an algebraic substitution, we can elegantly establish the **half-angle formulas** for sine, cosine, and tangent.

Theorem 5.4.1 (Half Angle Formulas of Sine and Cosine). *For any real angle θ :*

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 + \cos\theta}{2}}$$

where the choice of the positive (+) or negative (−) sign depends strictly on the quadrant in which the half-angle $\frac{\theta}{2}$ lies.

Proof. By taking $\theta = 2(\theta/2)$. □

Theorem 5.4.2 (Half Angle Formulas of Tangent). *For any real angle θ where the functions are defined:*

$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} = \frac{\sin\theta}{1 + \cos\theta} = \frac{1 - \cos\theta}{\sin\theta}$$

Proof. Follow directly from Theorem 5.4.1. □

Example 5.4.1 (Numerical Test on Half Angles). Find the exact numerical value of $\sin(22.5^\circ)$ using the half-angle formula.

Solution. We recognize that 22.5° is exactly half of the special angle 45° :

$$22.5^\circ = \frac{45^\circ}{2}.$$

Apply the half-angle formula for sine with $\theta = 45^\circ$:

$$\sin(22.5^\circ) = \sin\left(\frac{45^\circ}{2}\right) = \pm\sqrt{\frac{1 - \cos(45^\circ)}{2}}.$$

Since 22.5° lies in Quadrant I, its sine value must be positive. We choose the positive sign:

$$\sin(22.5^\circ) = \sqrt{\frac{1 - \frac{\sqrt{2}}{2}}{2}}.$$

Simplify the fraction inside the square root:

$$\sin(22.5^\circ) = \sqrt{\frac{\frac{2 - \sqrt{2}}{2}}{2}} = \sqrt{\frac{2 - \sqrt{2}}{4}} = \frac{\sqrt{2 - \sqrt{2}}}{2}.$$

□

Example 5.4.2 (Algebraic Expression Derivation). Derive the rational, sign-free half-angle formula $\tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1 + \cos\theta}$ using the double angle formulas of sine and cosine.

Solution. Let us start with the right-hand side of the identity:

$$\text{RHS} = \frac{\sin\theta}{1 + \cos\theta}.$$

We treat θ as double the angle $\frac{\theta}{2}$. Apply the double-angle formulas:

$$\begin{aligned} 1) \quad \sin\theta &= 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) \\ 2) \quad \cos\theta &= 2\cos^2\left(\frac{\theta}{2}\right) - 1 \end{aligned}$$

Substitute these expressions into our fraction:

$$\text{RHS} = \frac{2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)}{1 + \left(2 \cos^2\left(\frac{\theta}{2}\right) - 1\right)}.$$

Simplify the denominator:

$$\text{RHS} = \frac{2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)}{2 \cos^2\left(\frac{\theta}{2}\right)}.$$

Cancel the common terms ($2 \cos\left(\frac{\theta}{2}\right)$) in the numerator and denominator (assuming $\cos\left(\frac{\theta}{2}\right) \neq 0$):

$$\text{RHS} = \frac{\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)} = \tan\left(\frac{\theta}{2}\right) = \text{LHS}.$$

This completes the algebraic derivation, yielding an elegant formula that contains no square roots or sign ambiguities. $\square$

Example 5.4.3 (Algebraic Expression Application). Let $\cos \theta = -\frac{7}{25}$, where θ lies in Quadrant III ($180^\circ < \theta < 270^\circ$). Calculate the exact value of $\cos\left(\frac{\theta}{2}\right)$.

Solution. First, determine the quadrant of the half-angle $\frac{\theta}{2}$ by dividing the interval bounds by 2:

$$\frac{180^\circ}{2} < \frac{\theta}{2} < \frac{270^\circ}{2} \implies 90^\circ < \frac{\theta}{2} < 135^\circ.$$

This range indicates that $\frac{\theta}{2}$ lies in **Quadrant II**. In Quadrant II, the cosine function is negative. Therefore, we must choose the negative sign for our half-angle calculation:

$$\cos\left(\frac{\theta}{2}\right) = -\sqrt{\frac{1 + \cos \theta}{2}}.$$

Substitute the given value $\cos \theta = -\frac{7}{25}$ into the formula:

$$\cos\left(\frac{\theta}{2}\right) = -\sqrt{\frac{1 + \left(-\frac{7}{25}\right)}{2}} = -\sqrt{\frac{\frac{18}{25}}{2}}.$$

Simplify the fraction:

$$\cos\left(\frac{\theta}{2}\right) = -\sqrt{\frac{18}{50}} = -\sqrt{\frac{9}{25}} = -\frac{3}{5}.$$

$\square$

Example 5.4.4 (Abstract Application to Problems). Use the half-angle tangent formula to find the exact value of $\tan(15^\circ)$ using $\theta = 30^\circ$, and show it matches our result from Subsection 5.3.

Solution. Apply the sign-free half-angle formula for tangent with $\theta = 30^\circ$:

$$\tan(15^\circ) = \tan\left(\frac{30^\circ}{2}\right) = \frac{1 - \cos(30^\circ)}{\sin(30^\circ)}.$$

Substitute the special values $\cos(30^\circ) = \frac{\sqrt{3}}{2}$ and $\sin(30^\circ) = \frac{1}{2}$:

$$\tan(15^\circ) = \frac{1 - \frac{\sqrt{3}}{2}}{\frac{1}{2}}.$$

Multiply the numerator and the denominator by 2 to clear the fractions:

$$\tan(15^\circ) = 2 \left(1 - \frac{\sqrt{3}}{2} \right) = 2 - \sqrt{3}.$$

This exact value is identical to our compound angle derivation, demonstrating the absolute consistency of our trigonometric system. $\square$

Example 5.4.5 (The Failure of Sign Selection). Explain why a standard calculator fails to return the correct sign for a half-angle formula like $\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$ if only given the value of $\cos\theta$, and show how to resolve this.

Solution. The algebraic expression inside the radical, $\frac{1-\cos\theta}{2}$, is always non-negative because $-1 \leq \cos\theta \leq 1$. Therefore, taking the principal square root always returns a positive real number.

However, the physical angle $\frac{\theta}{2}$ can lie in any quadrant depending on the size of the original angle θ . For example, if we are only told that $\cos\theta = 0.5$, θ could be:

- 1) $\theta = 60^\circ \implies \frac{\theta}{2} = 30^\circ$ (Quadrant I) $\implies \sin(30^\circ) = 0.5$
- 2) $\theta = 300^\circ \implies \frac{\theta}{2} = 150^\circ$ (Quadrant II) $\implies \sin(150^\circ) = 0.5$
- 3) $\theta = -60^\circ \implies \frac{\theta}{2} = -30^\circ$ (Quadrant IV) $\implies \sin(-30^\circ) = -0.5$

Because multiple coterminal angles have the exact same cosine value but lead to different half-angle quadrants, the algebraic formula alone cannot determine the correct sign. We must manually resolve this failure by specifying the original domain of θ to locate the exact quadrant of $\frac{\theta}{2}$. $\square$

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Problems for Section 5.4

1. Find the exact numerical value of $\cos(22.5^\circ)$ using the half-angle formula.
2. Find the exact numerical value of $\tan(22.5^\circ)$ using a rational half-angle formula.
3. If $\cos\theta = \frac{3}{5}$ and θ is an acute angle, find the exact value of $\sin\left(\frac{\theta}{2}\right)$.
4. If $\cos\theta = \frac{12}{13}$ and θ is acute, find the exact value of $\cos\left(\frac{\theta}{2}\right)$.
5. Let $\cos\theta = -\frac{3}{5}$, where $90^\circ < \theta < 180^\circ$. Find the exact value of $\tan\left(\frac{\theta}{2}\right)$.
6. Find the exact value of $\sin(15^\circ)$ using the half-angle formula with $\theta = 30^\circ$.
7. Prove the identity: $\tan\left(\frac{\theta}{2}\right) + \frac{1}{\tan\left(\frac{\theta}{2}\right)} = \frac{2}{\sin\theta}$ for all valid angles.
8. Solve the equation $\cos^2\left(\frac{x}{2}\right) = \cos x$ for the domain $0^\circ \leq x \leq 180^\circ$.
9. If $\tan\theta = \frac{4}{3}$ and θ is in Quadrant III, find the exact value of $\cos\left(\frac{\theta}{2}\right)$.
10. Express $\sin^2\left(\frac{\theta}{2}\right)$ purely in terms of $\cos\theta$.

11. Simplify the expression $\frac{1-\cos(2x)}{\sin(2x)}$ using half-angle/double-angle relations.
12. Solve the equation $\sin\left(\frac{x}{2}\right) - \cos x = 0$ for the domain $0^\circ \leq x \leq 360^\circ$.
13. Prove the identity: $\tan\left(\frac{\theta}{2}\right) = \csc \theta - \cot \theta$.
14. If $\sin \theta = \frac{8}{17}$ and θ is in Quadrant II, find the exact value of $\tan\left(\frac{\theta}{2}\right)$.
15. Find the exact value of $\cos(11.25^\circ)$ in nested radical form. (Hint: Apply the half-angle formula twice starting from 45°).
16. Solve the equation $2\sin^2\left(\frac{x}{2}\right) + \cos x = 1$ for the interval $0^\circ \leq x \leq 360^\circ$.
17. Show that the area of an isosceles triangle with equal sides s and vertex angle θ can be written as $\mathcal{A} = s^2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)$.
18. If $t = \tan\left(\frac{\theta}{2}\right)$, show that $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$. (These are the famous t -substitution formulas of calculus).
19. Using the t -substitution formulas, solve the equation $\sin \theta + \cos \theta = 1$ for $0^\circ \leq \theta \leq 180^\circ$.
20. Discuss why the half-angle formulas are highly valued in pure geometry for constructing regular polygons with 2^n sides using only a straightedge and compass.

5.5 Enrichment: Why The One-Third Cut Fails

In the previous subsection, we established the half-angle formulas, which allow us to algebraically and geometrically bisect any angle. Geometrically, bisecting an angle is incredibly simple: using only a standard compass and a straightedge, any student can construct the perfect half of a given angle in a few steps.

Naturally, this raises a historical question: can we perform a **one-third cut**? That is, can we trisect an arbitrary angle? For thousands of years, ancient Greek mathematicians and their successors attempted to find a general method to divide any given angle into three equal parts using only an unmarked straightedge and a compass.

The answer to this quest is a surprising and profound **no**. While we can easily trisect a few specific angles (such as 90° to get 30°), it is mathematically impossible to trisect an arbitrary angle (such as 60° to get 20°) using classical tools. Proving this impossibility requires translating the geometric problem into modern algebra, linking the trisection of angles to the theory of cubic equations, field extensions, and ultimately, Galois Theory.

Theorem 5.5.1 (The Angle Trisection Cubic Equation). *Let θ be a given angle, and let $x = \cos\left(\frac{\theta}{3}\right)$ be the value we wish to find. The algebraic relation to trisect the angle is governed by the cubic polynomial equation:*

$$4x^3 - 3x - \cos \theta = 0.$$

Proof. Directly from Triple angle formula of cosine □

Theorem 5.5.2 (Constructible Numbers and Field Extensions). *A geometric length is constructible using a straightedge and compass if and only if the corresponding real number lies in a sequence of quadratic field extensions over the rational numbers $\mathbb{Q}$. Consequently, the minimal polynomial of any constructible number must have a degree that is a power of 2 (i.e., $1, 2, 4, 8, \dots$).*

Proof. Far beyond the scope. □

Example 5.5.1 (Numerical Test on Trisection Equations). Suppose we wish to trisect the angle $\theta = 60^\circ$ to find the value of $x = \cos(20^\circ)$. Write the explicit cubic equation that x must satisfy, and evaluate the equation for $\cos(20^\circ) \approx 0.9397$ to show it holds numerically.

Solution. The trisection equation is:

$$4x^3 - 3x - \cos(60^\circ) = 0.$$

We know that $\cos(60^\circ) = 0.5$. Substituting this value yields the cubic equation:

$$4x^3 - 3x - 0.5 = 0 \implies 8x^3 - 6x - 1 = 0.$$

Let us test the numerical value $x = \cos(20^\circ) \approx 0.9397$:

$$8(0.9397)^3 - 6(0.9397) - 1 \approx 8(0.8301) - 5.6382 - 1 = 6.6408 - 5.6382 - 1 = 0.0026.$$

The result is extremely close to 0, with the minor deviation due entirely to rounding error. This confirms the validity of the cubic trisection equation. $\square$

Example 5.5.2 (Algebraic Expression Derivation). Derive the cubic trisection relation $4x^3 - 3x - \cos \theta = 0$ using the triple angle formula of cosine.

Solution. Let $\phi = \frac{\theta}{3}$ represent our target trisected angle, meaning $3\phi = \theta$.

Recall the triple angle formula of cosine derived in Subsection 5.2:

$$\cos(3\phi) = 4\cos^3 \phi - 3\cos \phi.$$

Substitute $3\phi = \theta$ and $\cos \phi = x$ into this identity:

$$\cos \theta = 4x^3 - 3x.$$

Rearranging all the algebraic terms to one side of the equation yields:

$$4x^3 - 3x - \cos \theta = 0.$$

This completes the derivation, showing that trisecting an angle is algebraically equivalent to finding the roots of a cubic polynomial. $\square$

Example 5.5.3 (Algebraic Expression Application (Trisecting 90°)). Show that when we trisect a right angle ($\theta = 90^\circ$), the cubic equation has a rational root, making $\cos(30^\circ)$ constructible.

Solution. For a right angle, $\theta = 90^\circ$. The cubic trisection equation becomes:

$$4x^3 - 3x - \cos(90^\circ) = 0.$$

Since $\cos(90^\circ) = 0$, the equation simplifies to:

$$4x^3 - 3x = 0.$$

Factor out the variable x :

$$x(4x^2 - 3) = 0.$$

This gives three possible algebraic solutions:

$$x = 0 \quad \text{or} \quad x = \pm \frac{\sqrt{3}}{2}.$$

Since our trisected angle $\phi = 30^\circ$ lies in Quadrant I, the cosine value must be positive. Therefore:

$$x = \frac{\sqrt{3}}{2}.$$

Because the minimal polynomial of $x = \frac{\sqrt{3}}{2}$ over $\mathbb{Q}$ is $4x^2 - 3 = 0$ (which is of degree $2 = 2^1$), this value is geometrically constructible. This explains why we can easily construct a 30° angle using a compass and straightedge. $\square$

Example 5.5.4 (Abstract Application to Problems (Irreducibility)). Using the Rational Root Theorem, prove abstractly that the cubic polynomial $8x^3 - 6x - 1 = 0$ (for trisecting 60°) has no rational roots.

Solution. The Rational Root Theorem states that any rational root of a polynomial with integer coefficients, written in lowest terms as $\frac{p}{q}$, must have p as a factor of the constant term (-1) and q as a factor of the leading coefficient (8). * The possible values for p are: ± 1 . * The possible values for q are: $\pm 1, \pm 2, \pm 4, \pm 8$. Therefore, the only possible rational roots are:

$$x \in \left\{ \pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8} \right\}.$$

Let us test these candidates in our polynomial $P(x) = 8x^3 - 6x - 1$:

$$P(1) = 8(1) - 6(1) - 1 = 1 \neq 0$$

$$P(-1) = 8(-1) - 6(-1) - 1 = -3 \neq 0$$

$$P(1/2) = 8(1/8) - 6(1/2) - 1 = 1 - 3 - 1 = -3 \neq 0$$

$$P(-1/2) = 8(-1/8) - 6(-1/2) - 1 = -1 + 3 - 1 = 1 \neq 0$$

$$P(1/4) = 8(1/64) - 6(1/4) - 1 = \frac{1}{8} - \frac{3}{2} - 1 = -\frac{19}{8} \neq 0$$

$$P(-1/4) = 8(-1/64) - 6(-1/4) - 1 = -\frac{1}{8} + \frac{3}{2} - 1 = \frac{3}{8} \neq 0$$

Testing the remaining values $\pm \frac{1}{8}$ also yields non-zero results.

Since none of the rational candidates are roots, the polynomial $8x^3 - 6x - 1$ is irreducible over the rational numbers $\mathbb{Q}$. Its roots lie in a cubic field extension of degree 3. Because 3 is not a power of 2, we conclude abstractly that the roots are not constructible. Thus, trisecting a 60° angle is geometrically impossible. $\square$

Example 5.5.5 (Failure of the Trisection Analogy). Explain why the algebraic bisection of an angle always succeeds under classical construction, whereas the trisection fails, highlighting the core structural difference.

Solution. The failure to construct a trisection lies in a deep structural difference in the degrees of their corresponding polynomial equations:

1. ****Bisection:**** Bisecting an angle θ to find $y = \cos\left(\frac{\theta}{2}\right)$ relies on the half-angle formula:

$$2y^2 - 1 = \cos \theta \implies y = \sqrt{\frac{1 + \cos \theta}{2}}.$$

This is a quadratic equation (degree 2). Because compass-and-straightedge operations can physically draw circles (which intersect to form quadratic relations), we can always construct square roots. Thus, bisection always succeeds.

2. ****Trisection:**** Trisecting an angle θ to find $x = \cos\left(\frac{\theta}{3}\right)$ relies on a cubic equation (degree 3). Because intersecting straight lines and circles can only generate systems of degree 1 or 2, classical tools can never solve an irreducible cubic equation.

Therefore, the geometric analogy of dividing angles fails to carry over from bisection to trisection because the algebraic degree of the operation jumps from a constructible power of 2 to a non-constructible prime 3. $\square$

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Problems for Section 5.5

1. Write the trisection cubic equation for $\theta = 180^\circ$.
2. Solve the trisection cubic equation for $\theta = 180^\circ$ to find the exact value of $\cos(60^\circ)$. Show that this root is rational and constructible.
3. Explain why we can construct an angle of 45° but cannot construct an angle of 15° starting from a 45° angle.
4. What is the minimal degree of a field extension required to construct the roots of an irreducible cubic equation over $\mathbb{Q}$?
5. Show that $\cos(40^\circ)$ is not constructible by writing its trisection equation starting from $\theta = 120^\circ$.
6. Use the Rational Root Theorem to show that $4x^3 - 3x - \frac{\sqrt{2}}{2} = 0$ (for trisecting 45°) has no rational roots.
7. Can an angle of 90° be divided into four equal parts using only a straightedge and compass? Explain why or why not using power-of-two degree arguments.
8. Let θ be an angle such that $\cos \theta$ is a rational number. Under what algebraic condition on the cubic polynomial $4x^3 - 3x - \cos \theta$ is the angle θ trisectible?
9. Explain the role of a "Neusis construction" (which uses a marked ruler) in allowing ancient mathematicians to bypass the trisection impossibility.
10. Show that if we could trisect any angle, we could construct a regular heptadecagon (17-sided polygon) using only a straightedge and compass.
11. Prove that the cubic equation $4x^3 - 3x - 1 = 0$ has a rational root, and identify which angle trisection this corresponds to.
12. Find the three real roots of the cubic equation $4x^3 - 3x = 0$ and state the three corresponding trisected angles.
13. Discuss why the impossibility of angle trisection was not rigorously proven until the 19th century, despite being suspected for thousands of years.
14. Let $K = \mathbb{Q}(\cos \theta)$. If $[K(x) : K] = 3$, explain what this degree tells us about the constructibility of $\cos(\frac{\theta}{3})$.
15. Show that if an angle θ is constructible, then the angle 3θ is always constructible. (Contrast this with the reverse operation).
16. If we are given a trisected angle $\phi = 20^\circ$, explain how we can easily construct the original angle 60° .
17. Why does the existence of a cubic formula (Cardano's Formula) for solving cubic equations not help us construct the roots using only a compass and straightedge?
18. Show that the equation $4x^3 - 3x + 0.5 = 0$ corresponds to trisecting 240° to find $\cos(80^\circ)$.
19. Prove that we can always trisect an angle of the form $\theta = \frac{360^\circ}{2^n}$ under classical construction.
20. Discuss how the bridge between geometric constructions and abstract field theory exemplifies the power of unifying different branches of mathematics.

5.6 The Application

In the previous subsections of Section 5, we derived a massive library of algebraic tools: compound angle formulas, double angle formulas, half-angle formulas, and even analyzed the cubic nature of the one-third cut. Now, we must put these algebraic engines to work.

A primary focus of secondary and matriculation exams (like the HKDSE and HKALE) is the resolution of advanced trigonometric equations. In particular, we often encounter equations where sine and cosine are combined linearly, such as $a \sin x + b \cos x = k$. Solving this directly using standard algebra is incredibly difficult because sine and cosine represent orthogonal directions on our unit circle.

To solve this, we introduce the highly elegant **Subsidiary Angle Method** (sometimes called the R -formula). This method combines a linear sum of sine and cosine into a single, unified sine or cosine wave. By doing so, we can instantly read off the wave's maximum and minimum values and solve complex equations with ease.

Theorem 5.6.1 (The Subsidiary Angle Formula). *For any real numbers a and b (not both zero), the linear combination $a \sin \theta + b \cos \theta$ can be expressed as a single sine function:*

$$a \sin \theta + b \cos \theta = R \sin(\theta + \alpha),$$

where the amplitude R is given by:

$$R = \sqrt{a^2 + b^2},$$

and the auxiliary (or subsidiary) angle α satisfies:

$$\cos \alpha = \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}, \quad \text{and} \quad \tan \alpha = \frac{b}{a}.$$

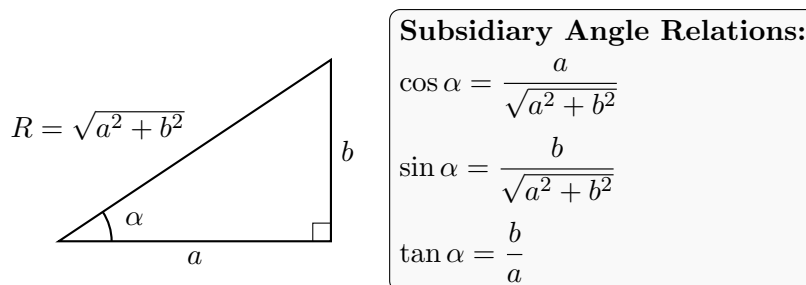


Figure 5: The auxiliary right-angled triangle used to transform $a \sin \theta + b \cos \theta$ into $R \sin(\theta + \alpha)$.

Proof. Let $a, b \in \mathbb{R}$ not both zero. We wish to express $a \sin \theta + b \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$. Expanding the target expression using the compound angle sum formula for sine:

$$R \sin(\theta + \alpha) = R(\sin \theta \cos \alpha + \cos \theta \sin \alpha) = (R \cos \alpha) \sin \theta + (R \sin \alpha) \cos \theta.$$

Equating the coefficients of $\sin \theta$ and $\cos \theta$ with our original expression $a \sin \theta + b \cos \theta$ gives the system of equations:

$$R \cos \alpha = a \quad (1)$$

$$R \sin \alpha = b \quad (2)$$

To solve for R , square both equations and add them together:

$$(R \cos \alpha)^2 + (R \sin \alpha)^2 = a^2 + b^2 \implies R^2(\cos^2 \alpha + \sin^2 \alpha) = a^2 + b^2.$$

Since $\cos^2 \alpha + \sin^2 \alpha = 1$ and we require $R > 0$:

$$R = \sqrt{a^2 + b^2}.$$

To solve for α , divide equation (2) by equation (1) (assuming $a \neq 0$):

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{b}{a} \implies \tan \alpha = \frac{b}{a}.$$

Since $\cos \alpha = \frac{a}{R}$ and $\sin \alpha = \frac{b}{R}$, the unique quadrant of α is determined precisely by the signs of a and b . $\square$

Example 5.6.1 (Numerical Test on Subsidiary Angles). Express the linear expression $y = \sqrt{3} \sin \theta + \cos \theta$ in the unified form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

Solution. Identify the coefficients from the expression:

$$a = \sqrt{3}, \quad b = 1.$$

Calculate the amplitude R :

$$R = \sqrt{a^2 + b^2} = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4} = 2.$$

Next, find the auxiliary angle α using the tangent relation:

$$\tan \alpha = \frac{b}{a} = \frac{1}{\sqrt{3}}.$$

From our special angles table in Subsection 1.4, we know that the acute angle satisfying this is:

$$\alpha = 30^\circ.$$

Thus, we can write the unified expression as:

$$\sqrt{3} \sin \theta + \cos \theta = 2 \sin(\theta + 30^\circ).$$

$\square$

Example 5.6.2 (Algebraic Expression Derivation). Derive the subsidiary angle relations $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$ using the compound angle sum formula of sine.

Solution. We start by assuming that the linear combination can be equated to a shifted sine wave:

$$a \sin \theta + b \cos \theta = R \sin(\theta + \alpha).$$

Expand the right-hand side using the compound angle sum formula for sine:

$$a \sin \theta + b \cos \theta = R(\sin \theta \cos \alpha + \cos \theta \sin \alpha).$$

Distribute the scalar R :

$$a \sin \theta + b \cos \theta = (R \cos \alpha) \sin \theta + (R \sin \alpha) \cos \theta.$$

For this equation to hold as an identity for all values of θ , the coefficients of $\sin \theta$ and $\cos \theta$ on both sides must be equal:

- 1) $R \cos \alpha = a$
- 2) $R \sin \alpha = b$

To find R , square both equations and add them together:

$$(R \cos \alpha)^2 + (R \sin \alpha)^2 = a^2 + b^2 \implies R^2(\cos^2 \alpha + \sin^2 \alpha) = a^2 + b^2.$$

Using the Pythagorean identity $\cos^2 \alpha + \sin^2 \alpha = 1$:

$$R^2 = a^2 + b^2 \implies R = \sqrt{a^2 + b^2}.$$

To find the angle α , divide the second coefficient equation by the first:

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{b}{a} \implies \tan \alpha = \frac{b}{a}.$$

This completes the algebraic derivation of the R -formula. □

Example 5.6.3 (Algebraic Expression Application). Solve the trigonometric equation:

$$\sin \theta + \cos \theta = 1$$

for the interval $0^\circ \leq \theta \leq 360^\circ$ using the subsidiary angle method.

Solution. First, express the left-hand side in the unified form $R \sin(\theta + \alpha)$. Here, $a = 1$ and $b = 1$:

$$R = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

$$\tan \alpha = \frac{1}{1} = 1 \implies \alpha = 45^\circ.$$

Thus, we rewrite the equation as:

$$\sqrt{2} \sin(\theta + 45^\circ) = 1.$$

Divide both sides by $\sqrt{2}$:

$$\sin(\theta + 45^\circ) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Let $u = \theta + 45^\circ$. Since $0^\circ \leq \theta \leq 360^\circ$, the range for u is:

$$45^\circ \leq u \leq 405^\circ.$$

Now, find all solutions for $\sin u = \frac{\sqrt{2}}{2}$ in this range:

- 1) $u_1 = 45^\circ$
- 2) $u_2 = 180^\circ - 45^\circ = 135^\circ$
- 3) $u_3 = 45^\circ + 360^\circ = 405^\circ$

Now, solve for θ by subtracting 45° from each u value:

$$\begin{aligned}\theta_1 &= 45^\circ - 45^\circ = 0^\circ \\ \theta_2 &= 135^\circ - 45^\circ = 90^\circ \\ \theta_3 &= 405^\circ - 45^\circ = 360^\circ\end{aligned}$$

Thus, the solutions are $\theta = 0^\circ, 90^\circ$, or 360° . □

Example 5.6.4 (Abstract Application to Problems). Find the exact maximum and minimum values of the algebraic function:

$$f(\theta) = 3 \sin \theta - 4 \cos \theta + 5.$$

Solution. First, focus on the variable linear combination part: $3 \sin \theta - 4 \cos \theta$. Here, $a = 3$ and $b = -4$:

$$R = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

Thus, we can express this part as $5 \sin(\theta + \alpha)$ for some angle α . The function becomes:

$$f(\theta) = 5 \sin(\theta + \alpha) + 5.$$

Because the standard sine function is strictly bounded between -1 and 1 :

$$-1 \leq \sin(\theta + \alpha) \leq 1.$$

Multiply the inequality by the amplitude 5:

$$-5 \leq 5 \sin(\theta + \alpha) \leq 5.$$

Add 5 to all parts of the inequality:

$$-5 + 5 \leq 5 \sin(\theta + \alpha) + 5 \leq 5 + 5 \implies 0 \leq f(\theta) \leq 10.$$

Thus, the exact minimum value of the function is 0, and the exact maximum value is 10. $\square$

Example 5.6.5 (Failure of the Application). Explain why the subsidiary angle formula cannot be applied to unify the expression $f(\theta) = \sin \theta + \cos(2\theta)$ into a single sine term, and state the correct algebraic alternative.

Solution. The subsidiary angle theorem relies strictly on the condition that both the sine and cosine terms share the exact same frequency (the same input angle θ).

In the expression $f(\theta) = \sin \theta + \cos(2\theta)$, the input angles are different (θ and 2θ). Because their periods are different, they do not oscillate in harmony, meaning their linear combination cannot be mapped to a single shifted sine wave of a constant frequency. Attempting to apply the R -formula directly here is an algebraic failure.

The correct alternative is to use the double-angle formulas of cosine to match the frequencies. We substitute $\cos(2\theta) = 1 - 2 \sin^2 \theta$:

$$f(\theta) = \sin \theta + (1 - 2 \sin^2 \theta) = -2 \sin^2 \theta + \sin \theta + 1.$$

This transforms the expression into a quadratic polynomial in terms of $\sin \theta$, which can then be analyzed using standard quadratic factoring or the quadratic formula. $\square$

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Problems for Section 5.6

1. Express $5 \sin \theta + 12 \cos \theta$ in the form $R \sin(\theta + \alpha)$ where $R > 0$ and α is acute.
2. Express $\sin \theta - \cos \theta$ in the form $R \sin(\theta - \alpha)$.
3. Find the exact maximum value of the expression $2 \sin \theta + 3 \cos \theta$.
4. Solve the equation $\sqrt{3} \sin x + \cos x = \sqrt{2}$ for $0^\circ \leq x \leq 360^\circ$.
5. Solve the equation $\sin x - \sqrt{3} \cos x = 1$ for $-180^\circ \leq x \leq 180^\circ$.

6. Solve the equation $3 \sin \theta + 4 \cos \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$.
7. Find the range of values of k for which the equation $5 \sin x + 2 \cos x = k$ has real solutions.
8. Find the minimum value of the function $g(\theta) = 12 \sin \theta - 5 \cos \theta - 7$.
9. Express $\sin(2\theta) + \sqrt{3} \cos(2\theta)$ as a single sine term.
10. Solve the equation $\sin(2x) + \cos(2x) = 0$ for the domain $0^\circ \leq x \leq 180^\circ$.
11. If $f(x) = \frac{1}{3 \sin x + 4 \cos x + 10}$, find the maximum and minimum values of $f(x)$.
12. Solve the equation $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ using quadratic factoring for $0^\circ \leq \theta \leq 360^\circ$.
13. Solve the equation $\cos(2\theta) - \cos \theta = 0$ for the domain $-180^\circ \leq \theta \leq 180^\circ$.
14. Express the linear combination $a \sin \theta - b \cos \theta$ (where $a, b > 0$) in the form $R \sin(\theta - \alpha)$.
15. Show that $\cos \theta + \sin \theta = \sqrt{2} \cos(\theta - 45^\circ)$ using the compound angle subtraction formula of cosine.
16. Solve the equation $3 \sin x + 3 \cos x = 3\sqrt{2}$ for the interval $0^\circ \leq x \leq 360^\circ$.
17. Find all real solutions to $\sin x + \cos x = 2$. Explain your answer geometrically.
18. Solve the equation $\tan x \sin x - \sin x = 0$ for $0^\circ \leq x \leq 360^\circ$.
19. If $x = R \cos(\theta - \alpha)$ is used to represent $2 \cos \theta + 5 \sin \theta$, find the exact values of R and $\tan \alpha$.
20. Discuss how the subsidiary angle formula is utilized in mechanical engineering to calculate the combined peak stress on a beam subjected to both vertical bending and horizontal twisting forces.

6 Extending Trigonometric Functions from Real to Complex Domain

6.1 The Complex Angle Discussion

In Section 3.5, we established a rigorous, natural definition of an angle using **radian measure**. By mapping our rotations to the unit circle, we defined an angle purely as the arc length s subtended on the circle's boundary. Because the boundary of a circle is a continuous line, these arc lengths—and therefore our angles—have always been represented by real numbers ($\theta \in \mathbb{R}$).

However, in Section 4, we discovered the magic of the complex number system ($\mathbb{C}$), where we solved algebraic equations that were impossible in the real domain by introducing the imaginary unit i . This raises a profound and mind-bending question:

If a real angle represents a physical arc length traversed along a circular curve in flat space, what does a **complex angle** ($\theta = x + yi$) represent?

To answer this, we must step outside of our flat, two-dimensional drawing board. In a flat plane—which we call **Euclidean space**—the shortest path between two points is a straight line, and the geometry of our circles is perfectly uniform. But when we allow angles to have imaginary components, the geometry deforms.

A complex angle represents a transition from flat planes to curved, warped surfaces—known as **non-Euclidean space**. On these warped surfaces, straight lines are no longer the shortest paths. Instead, the shortest paths are determined by the physical curvature of the space, traveling along twisted, contorted paths called **geodesics**. In this subsection, we discuss the transition from real angles to complex angles and the geometric shift from Euclidean to non-Euclidean spaces.

Definition 6.1.1 (Euclidean and Non-Euclidean Space). A space is said to be **Euclidean** ($\mathbb{R}^n$) if it satisfies classical planar axioms, particularly Euclid's parallel postulate, meaning the shortest distance between any two points is a unique straight line, and the interior angles of any triangle sum to exactly 180° .

A space is **non-Euclidean** if the space is curved. On these surfaces, the shortest path connecting two points is a curved path called a **geodesic**, and triangle interior angles do not sum to 180° .

Definition 6.1.2 (The Complex Angle). A **complex angle** z is an angle of the form:

$$z = x + yi,$$

where x represents the real rotational component (arc length on a circle of real curvature) and y represents the imaginary component (movement along a hyperbolic trajectory, corresponding to negative or pseudospherical curvature).

Theorem 6.1.1 (Euler's Extension to Complex Angles). *The definition of sine and cosine can be extended to any complex angle $z = x + yi$ using their infinite power series or complex exponential definitions:*

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Proof. Direct verification from Euler's formula. □

Example 6.1.1 (Numerical Test on Complex Inputs). Evaluate the exact complex value of $\cos(i)$ in terms of the real mathematical constant e .

Solution. By the complex exponential definition of the cosine function:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

Substitute the purely imaginary angle $z = i$ into this formula:

$$\cos(i) = \frac{e^{i(i)} + e^{-i(i)}}{2}.$$

Since $i^2 = -1$ and $-i^2 = 1$, simplify the exponents:

$$\cos(i) = \frac{e^{-1} + e^1}{2} = \frac{e + \frac{1}{e}}{2}.$$

Using the decimal approximation $e \approx 2.718$, we can estimate the value:

$$\cos(i) \approx \frac{2.718 + 0.368}{2} = 1.543.$$

Remarkably, because the input angle is imaginary, the output value of the cosine function is a real number strictly greater than 1. This is impossible for any real angle input! $\square$

Example 6.1.2 (Algebraic Expression Derivation). Prove algebraically that the fundamental Pythagorean identity $\cos^2 z + \sin^2 z = 1$ remains universally true even when the angle z is a complex number.

Solution. We use the complex exponential definitions of cosine and sine:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Square both expressions:

$$\cos^2 z = \left(\frac{e^{iz} + e^{-iz}}{2} \right)^2 = \frac{e^{2iz} + 2e^{iz}e^{-iz} + e^{-2iz}}{4} = \frac{e^{2iz} + 2 + e^{-2iz}}{4}.$$

$$\sin^2 z = \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 = \frac{e^{2iz} - 2e^{iz}e^{-iz} + e^{-2iz}}{4i^2}.$$

Since $i^2 = -1$, substitute this into the denominator of the sine squared term:

$$\sin^2 z = \frac{e^{2iz} - 2 + e^{-2iz}}{-4} = \frac{-e^{2iz} + 2 - e^{-2iz}}{4}.$$

Now, add the two squared terms together:

$$\cos^2 z + \sin^2 z = \frac{(e^{2iz} + 2 + e^{-2iz}) + (-e^{2iz} + 2 - e^{-2iz})}{4}.$$

Combine like terms in the numerator:

$$\cos^2 z + \sin^2 z = \frac{2 + 2}{4} = \frac{4}{4} = 1.$$

This algebraic derivation proves that the unit-circle relation is preserved in the complex domain. $\square$

Example 6.1.3 (Algebraic Expression Application). Show that if a complex angle $z = yi$ is purely imaginary, the sine function returns a purely imaginary output.

Solution. Substitute the purely imaginary angle $z = yi$ into the complex sine formula:

$$\sin(yi) = \frac{e^{i(yi)} - e^{-i(yi)}}{2i} = \frac{e^{-y} - e^y}{2i}.$$

Multiply the numerator and the denominator by i to move the imaginary unit out of the denominator:

$$\sin(yi) = \frac{i(e^{-y} - e^y)}{2i^2} = \frac{i(e^{-y} - e^y)}{-2}.$$

Distribute the negative sign in the numerator:

$$\sin(yi) = i \left(\frac{e^y - e^{-y}}{2} \right).$$

Because the term inside the parentheses is a real number for any real y , the output is indeed a purely imaginary number of the form bi , where $b = \frac{e^y - e^{-y}}{2}$. $\square$

Example 6.1.4 (Abstract Application to Problems). Explain abstractly why the concept of a triangle with interior angles summing to 270° can exist on the curved surface of a sphere (non-Euclidean), but is a complete failure in a standard Euclidean plane.

Solution. In a standard Euclidean plane, our straight lines are flat. By Theorem 1.3.1, the interior angle sum of any flat triangle is locked at exactly 180° .

However, consider a sphere (like the Earth). Let us draw three geodesic paths (the “straightest” possible paths on a sphere): 1. Start at the North Pole. Travel straight south along the prime meridian (0° longitude) down to the Equator. 2. Turn left at the Equator by exactly 90° , and travel east along the Equator until you reach 90°E longitude. 3. Turn left by exactly 90° again, and travel straight north back to the North Pole.

At the North Pole, the angle of intersection between the 0° and 90°E meridians is exactly 90° . This forms a spherical triangle with three vertices: the North Pole, the intersection at $(0^\circ, 0^\circ)$, and the intersection at $(0^\circ, 90^\circ\text{E})$. The interior angles of this triangle are:

$$\text{Sum} = 90^\circ + 90^\circ + 90^\circ = 270^\circ.$$

This occurs because the positive curvature of the sphere warps the paths outward, showing how Euclidean parallel rules fail in curved, non-Euclidean spaces. $\square$

Example 6.1.5 (The Failure of Real Bounds). Explain why the real trigonometric bounding constraint $-1 \leq \sin \theta \leq 1$ fails to hold when we extend the domain to complex angles, and find a complex angle z such that $\sin z = 2$.

Solution. The boundary $-1 \leq \sin \theta \leq 1$ is derived strictly from the coordinates of the real unit circle, where $y^2 \leq 1$. However, in the complex domain, we can solve $\sin z = k$ for any real k , even if $|k| > 1$.

Let us find a purely imaginary angle $z = yi$ such that:

$$\sin(yi) = 2.$$

Using our third example, we substitute the imaginary sine expression:

$$i \left(\frac{e^y - e^{-y}}{2} \right) = 2.$$

This equation contains an imaginary unit on the left but a real number on the right, which cannot balance directly. To resolve this, we must use a complex angle with both real and imaginary components: $z = \frac{\pi}{2} + yi$.

Substitute this complex angle into the sine formula:

$$\sin\left(\frac{\pi}{2} + yi\right) = \sin\left(\frac{\pi}{2}\right)\cos(yi) + \cos\left(\frac{\pi}{2}\right)\sin(yi).$$

Since $\sin(\frac{\pi}{2}) = 1$ and $\cos(\frac{\pi}{2}) = 0$:

$$\sin\left(\frac{\pi}{2} + yi\right) = \cos(yi) = \frac{e^{-y} + e^y}{2} = 2.$$

Now, solve for e^y :

$$e^y + e^{-y} = 4.$$

Let $u = e^y$. This yields a quadratic equation:

$$u + \frac{1}{u} = 4 \implies u^2 - 4u + 1 = 0.$$

Using the quadratic formula:

$$u = \frac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}.$$

Thus, $y = \ln(2 + \sqrt{3})$ (taking the positive root). This proves that the complex angle:

$$z = \frac{\pi}{2} + i\ln(2 + \sqrt{3})$$

satisfies $\sin z = 2$, showing that real bounds fail in the complex domain. □

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Problems for Section 6.1

1. Evaluate the exact value of $\cos(2i)$ in terms of e .
2. Find the exact value of $\sin(i)$ in terms of e .
3. State whether the output of $\cos(3i)$ is real, imaginary, or complex.
4. Show that $\sin(-z) = -\sin z$ holds true for any complex angle z .
5. Show that $\cos(-z) = \cos z$ holds true for any complex angle z .
6. Calculate the exact real value of $\cos\left(\frac{\pi}{2} - i\right)$.
7. Explain the difference between a geodesic path and a standard Euclidean straight line.
8. Prove that the compound angle formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$ remains valid when A and B are complex angles.
9. Solve the equation $\cos z = 1$ for complex angles z .
10. Let $z = x + yi$. Find the real and imaginary parts of $\cos z$ as functions of x and y .
11. Find the real and imaginary parts of $\sin z$ as functions of x and y .
12. Show that $\cos(i \ln 2) = 1.25$.

13. Solve the equation $\sin z = 0$ over the complex numbers. Show that all solutions are purely real.
14. Under what condition on the complex angle z is the value of $\tan z$ undefined?
15. Explain why the sum of the interior angles of a triangle drawn on a saddle-shaped surface (hyperbolic plane) is strictly less than 180° .
16. Solve the complex equation $\cos z = 2$ for z .
17. Express $\cos(2z)$ purely in terms of $\cos z$ for a complex variable z .
18. If $z = yi$, find the limit of $\cos(yi)$ as $y \rightarrow \infty$.
19. Find the value of $\sin(x + yi)$ when $y \rightarrow 0$. Show that it collapses to the real sine function.
20. Discuss how extending the domain of a function to complex numbers (analytic continuation) is analogous to adding a vertical axis to a 1D map to understand the true terrain.

6.2 The Computation: Hyperbolic Functions

In the previous subsection, we discussed how extending our domain to complex angles deforms our geometry. We discovered that a purely imaginary angle input, such as $\cos(yi)$, rotates our trigonometric calculations off the standard circular tracks of the flat Euclidean plane and projects them onto hyperbolic curves.

To handle these warped projections without carrying around the imaginary unit i in every calculation, mathematicians defined a new family of functions: the **hyperbolic functions**. Just as the standard circular functions $(\cos \theta, \sin \theta)$ trace out a unit circle defined by $x^2 + y^2 = 1$, the hyperbolic functions $(\cosh x, \sinh x)$ trace out a unit hyperbola defined by $x^2 - y^2 = 1$. These functions serve as the primary computational tools in non-Euclidean geometry, relativity, and engineering.

Definition 6.2.1 (Hyperbolic Functions). For any real number x :

- The **hyperbolic sine** function, denoted by $\sinh x$ (pronounced “sinch”), is defined as:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

- The **hyperbolic cosine** function, denoted by $\cosh x$ (pronounced “kosh”), is defined as:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

- The **hyperbolic tangent** function, denoted by $\tanh x$ (pronounced “tansh”), is defined as:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Theorem 6.2.1 (Circular-Hyperbolic Conversion Relations). *The circular trigonometric functions and the hyperbolic functions are deeply unified across the complex plane through the imaginary angle relations:*

$$\cos(ix) = \cosh x \quad \text{and} \quad \sin(ix) = i \sinh x$$

$$\cosh(ix) = \cos x \quad \text{and} \quad \sinh(ix) = i \sin x$$

Proof. Direct verification from definition. □

Theorem 6.2.2 (The Hyperbolic Pythagorean Identity). *For any real number x , the coordinates $(\cosh x, \sinh x)$ satisfy the equation of the unit hyperbola:*

$$\cosh^2 x - \sinh^2 x = 1$$

Proof. Direct verification from definition. □

Example 6.2.1 (Numerical Test on Hyperbolic Evaluation). Calculate the exact numerical value of $\cosh(\ln 3) + \sinh(\ln 3)$.

Solution. By our definitions:

$$\cosh(\ln 3) = \frac{e^{\ln 3} + e^{-\ln 3}}{2} \quad \text{and} \quad \sinh(\ln 3) = \frac{e^{\ln 3} - e^{-\ln 3}}{2}.$$

Recall the properties of logarithms and exponentials: $e^{\ln 3} = 3$ and $e^{-\ln 3} = e^{\ln(1/3)} = \frac{1}{3}$. Substitute these values into the equations:

$$\cosh(\ln 3) = \frac{3 + \frac{1}{3}}{2} = \frac{\frac{10}{3}}{2} = \frac{5}{3}.$$

$$\sinh(\ln 3) = \frac{3 - \frac{1}{3}}{2} = \frac{\frac{8}{3}}{2} = \frac{4}{3}.$$

Now, add the two evaluated terms together:

$$\cosh(\ln 3) + \sinh(\ln 3) = \frac{5}{3} + \frac{4}{3} = \frac{9}{3} = 3.$$

□

Example 6.2.2 (Algebraic Expression Derivation). Derive the hyperbolic Pythagorean identity $\cosh^2 x - \sinh^2 x = 1$ directly from their exponential definitions.

Solution. Substitute the definitions of hyperbolic cosine and hyperbolic sine into the expression:

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2.$$

Expand the squared terms:

$$\begin{aligned} \cosh^2 x &= \frac{e^{2x} + 2e^x e^{-x} + e^{-2x}}{4} = \frac{e^{2x} + 2 + e^{-2x}}{4}. \\ \sinh^2 x &= \frac{e^{2x} - 2e^x e^{-x} + e^{-2x}}{4} = \frac{e^{2x} - 2 + e^{-2x}}{4}. \end{aligned}$$

Subtract the two expressions:

$$\cosh^2 x - \sinh^2 x = \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{4}.$$

Simplify the numerator by canceling common terms:

$$\cosh^2 x - \sinh^2 x = \frac{2 - (-2)}{4} = \frac{4}{4} = 1.$$

This completes the algebraic derivation. □

Example 6.2.3 (Algebraic Expression Application). Solve the hyperbolic equation:

$$2 \cosh x + \sinh x = 2$$

for $x \in \mathbb{R}$.

Solution. Substitute the exponential definitions into the equation:

$$2 \left(\frac{e^x + e^{-x}}{2} \right) + \left(\frac{e^x - e^{-x}}{2} \right) = 2.$$

Simplify the terms:

$$(e^x + e^{-x}) + \frac{e^x - e^{-x}}{2} = 2.$$

Multiply the entire equation by 2 to clear the fraction:

$$2(e^x + e^{-x}) + (e^x - e^{-x}) = 4.$$

Distribute and combine like terms:

$$2e^x + 2e^{-x} + e^x - e^{-x} = 4 \implies 3e^x + e^{-x} = 4.$$

Multiply the entire equation by e^x (since $e^x \neq 0$):

$$3(e^x)^2 + 1 = 4e^x.$$

Let $u = e^x$. This yields a standard quadratic equation:

$$3u^2 - 4u + 1 = 0.$$

Factoring the quadratic expression:

$$(3u - 1)(u - 1) = 0 \implies u = \frac{1}{3} \quad \text{or} \quad u = 1.$$

Substitute back $u = e^x$ and solve for the real variable x :

$$\begin{aligned} 1) \quad e^x = 1 &\implies x = \ln(1) = 0 \\ 2) \quad e^x = \frac{1}{3} &\implies x = \ln\left(\frac{1}{3}\right) = -\ln 3 \end{aligned}$$

Thus, the valid solutions are $x = 0$ or $x = -\ln 3$. □

Example 6.2.4 (Abstract Application to Problems). Show abstractly that the addition formula for hyperbolic sine satisfies the relation:

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y.$$

Solution. We can prove this elegant relation by using the imaginary conversion relations and the circular compound angle sum formula.

Start with the circular compound angle formula for sine from Subsection 5.1, evaluated for the imaginary angles ix and iy :

$$\sin(i(x + y)) = \sin(ix + iy) = \sin(ix) \cos(iy) + \cos(ix) \sin(iy).$$

Apply the Circular-Hyperbolic conversion relations to every term:

$$\begin{aligned} \sin(i(x + y)) &= i \sinh(x + y) \\ \sin(ix) &= i \sinh x \\ \cos(iy) &= \cosh y \\ \cos(ix) &= \cosh x \\ \sin(iy) &= i \sinh y \end{aligned}$$

Substitute these back into the compound equation:

$$i \sinh(x + y) = (i \sinh x)(\cosh y) + (\cosh x)(i \sinh y).$$

Factor out the common imaginary unit i on the right-hand side:

$$i \sinh(x + y) = i (\sinh x \cosh y + \cosh x \sinh y).$$

Since $i \neq 0$, we divide both sides by i to complete the abstract proof:

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y.$$

□

Example 6.2.5 (Failure of the Real Periodicity). Explain why the hyperbolic functions $\sinh x$ and $\cosh x$ do not have a real period of 2π (unlike their circular counterparts), showing what happens instead.

Solution. The circular functions are periodic because as we rotate along the unit circle, we return to our starting point every 360° (2π radians).

However, a hyperbola is an open, non-closed curve. As the parameter x increases, the coordinates $(\cosh x, \sinh x)$ expand infinitely along the branches of the hyperbola, never returning to their starting values. Therefore, for real inputs $x \in \mathbb{R}$:

$$\sinh(x + 2\pi) \neq \sinh x.$$

Instead, their periodicity exists strictly in the **imaginary direction**. Using the complex exponential definitions:

$$\sinh(x + 2\pi i) = \frac{e^{x+2\pi i} - e^{-(x+2\pi i)}}{2}.$$

Recall that $e^{2\pi i} = \cos(2\pi) + i \sin(2\pi) = 1$, which means:

$$e^{x+2\pi i} = e^x \cdot e^{2\pi i} = e^x \quad \text{and} \quad e^{-(x+2\pi i)} = e^{-x} \cdot e^{-2\pi i} = e^{-x}.$$

Thus:

$$\sinh(x + 2\pi i) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

Therefore, the circular functions fail to preserve real-periodicity when converted to hyperbolic functions; instead, their real period 2π is rotated into an imaginary period $2\pi i$. $\square$

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Problems for Section 6.2

- Find the exact numerical value of $\sinh(0)$.
- Find the exact numerical value of $\cosh(0)$.
- Evaluate the exact value of $\tanh(\ln 2)$.
- Show that $\cosh(-x) = \cosh x$ is true for all real x .
- Show that $\sinh(-x) = -\sinh x$ is true for all real x .
- Prove the identity: $1 - \tanh^2 x = \frac{1}{\cosh^2 x}$ (often written as $1 - \tanh^2 x = \operatorname{sech}^2 x$).
- Solve the equation $\sinh x = 0$ for $x \in \mathbb{R}$.
- Solve the equation $\cosh x = 1$ for $x \in \mathbb{R}$.
- Solve the equation $e^x - 3e^{-x} = 2$ by rewriting it as a hyperbolic equation.
- If $\sinh x = \frac{3}{4}$, find the exact positive value of $\cosh x$.
- If $\cosh x = \frac{13}{12}$ and $x > 0$, find the exact value of $\tanh x$.
- Solve the equation $5\sinh x - 3\cosh x = 3$ for $x \in \mathbb{R}$.
- Prove the addition formula: $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$.
- Prove the double angle formula: $\sinh(2x) = 2\sinh x \cosh x$.
- Prove the double angle formula: $\cosh(2x) = \cosh^2 x + \sinh^2 x$.

16. Show that $\cosh(2x) = 2 \cosh^2 x - 1$ and $\cosh(2x) = 1 + 2 \sinh^2 x$.
17. Solve the equation $\cosh(2x) - 3 \cosh x + 2 = 0$ for $x \in \mathbb{R}$.
18. Express $\tanh(2x)$ purely in terms of $\tanh x$.
19. Find the range of the function $f(x) = \cosh x$ for $x \in \mathbb{R}$.
20. Discuss why the physical shape of a heavy hanging cable (such as a power line suspended between two poles) is modeled by a hyperbolic cosine curve, $y = a \cosh(x/a)$, rather than a standard parabola.

7 Spherical Geometry

7.1 The 3-Dimensional Conversion

In Section 2, we mastered the coordinate systems of a 2-dimensional plane, exploring the global consistency of rectangular coordinates (x, y) and the local relevance of polar coordinates (r, θ) . We then bridged them using basic trigonometry.

Now, we will scale our coordinate systems up to model the 3-dimensional universe we physically inhabit. To describe any position in 3D space, we can easily extend our rectangular grid by adding a third vertical height axis, giving us the triplet (x, y, z) .

However, when modeling spherical bodies—such as the Earth, the path of a satellite, or the field of view of a telescope—rectangular coordinates become highly inefficient. Instead, we require a 3-dimensional generalization of polar coordinates. This is known as the **Spherical Coordinate System**. Rather than measuring along flat perpendicular grids, we describe a point using its direct distance from the center of the world, along with two distinct angular directions.

Definition 7.1.1 (Spherical Coordinate System). In 3-dimensional space, the position of any point P can be uniquely represented by the ordered triple:

$$(R, \theta, \phi) \in [0, \infty) \times [0, 2\pi) \times [0, \pi),$$

where:

- R (or ρ) is the **radial distance**, representing the straight-line distance from the origin $O(0, 0, 0)$ to the point P .
- θ is the **azimuthal angle**, representing the counterclockwise angle of rotation in the horizontal xy -plane, measured from the positive x -axis (exactly like the 2D polar angle).
- ϕ is the **polar angle** (or co-latitude), representing the vertical angle of inclination measured downward from the positive vertical z -axis.

Remark 7.1.1. In geography, the azimuthal angle θ corresponds directly to **longitude** (East / West positioning), while the polar angle ϕ corresponds to the angular offset from the North Pole, which is directly related to **latitude** (North/South positioning).

Theorem 7.1.1 (Spherical-Rectangular Conversion Equations). *Let a point have rectangular coordinates (x, y, z) and spherical coordinates (R, θ, ϕ) . The mathematical conversions between these systems are given by:*

$$\begin{aligned} \text{Spherical to Rectangular: } & x = R \sin \phi \cos \theta, \quad y = R \sin \phi \sin \theta, \quad z = R \cos \phi \\ \text{Rectangular to Spherical: } & R = \sqrt{x^2 + y^2 + z^2}, \quad \cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad \tan \theta = \frac{y}{x} \end{aligned}$$

Proof. Left to Readers. □

Example 7.1.1 (Numerical Test on 3D Conversion). A point P has spherical coordinates given by $(R, \theta, \phi) = (4, \frac{\pi}{3}, \frac{\pi}{6})$. Convert these coordinates into standard rectangular coordinates (x, y, z) .

Solution. We are given the spherical parameters: $R = 4$, $\theta = \frac{\pi}{3}$ (equivalent to 60°), and $\phi = \frac{\pi}{6}$ (equivalent to 30°).

Using the conversion formulas:

$$x = R \sin \phi \cos \theta = 4 \sin \left(\frac{\pi}{6} \right) \cos \left(\frac{\pi}{3} \right)$$

$$y = R \sin \phi \sin \theta = 4 \sin \left(\frac{\pi}{6} \right) \sin \left(\frac{\pi}{3} \right)$$

$$z = R \cos \phi = 4 \cos \left(\frac{\pi}{6} \right)$$

Substitute the exact values of these special angles:

$$\sin \left(\frac{\pi}{6} \right) = \frac{1}{2}, \quad \cos \left(\frac{\pi}{6} \right) = \frac{\sqrt{3}}{2}, \quad \cos \left(\frac{\pi}{3} \right) = \frac{1}{2}, \quad \sin \left(\frac{\pi}{3} \right) = \frac{\sqrt{3}}{2}.$$

Compute each component:

$$x = 4 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 1$$

$$y = 4 \left(\frac{1}{2} \right) \left(\frac{\sqrt{3}}{2} \right) = \sqrt{3}$$

$$z = 4 \left(\frac{\sqrt{3}}{2} \right) = 2\sqrt{3}$$

Thus, the rectangular coordinates are $(1, \sqrt{3}, 2\sqrt{3})$. □

Example 7.1.2 (Algebraic Expression Derivation). Derive the relation $x^2 + y^2 = R^2 \sin^2 \phi$ using the spherical conversion formulas, and explain its geometric interpretation in the horizontal plane.

Solution. We write the sum of the squares of the horizontal components x and y :

$$x^2 + y^2 = (R \sin \phi \cos \theta)^2 + (R \sin \phi \sin \theta)^2.$$

Distribute the exponents over the multiplied terms:

$$x^2 + y^2 = R^2 \sin^2 \phi \cos^2 \theta + R^2 \sin^2 \phi \sin^2 \theta.$$

Factor out the common term $R^2 \sin^2 \phi$:

$$x^2 + y^2 = R^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta).$$

Using the fundamental 2D Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$, we obtain:

$$x^2 + y^2 = R^2 \sin^2 \phi.$$

****Geometric Interpretation:**** The term $\sqrt{x^2 + y^2}$ represents the perpendicular distance r from the point P to the vertical z -axis (the radius of the horizontal cylinder passing through P). This derivation proves that this horizontal radius r is equal to the projection of the 3D radius R onto the horizontal plane, scaled by the sine of the vertical inclination angle ϕ :

$$r = R \sin \phi.$$

□

Example 7.1.3 (Algebraic Expression Application). Convert the rectangular coordinates of a point $A(1, 1, \sqrt{2})$ into its exact spherical representation (R, θ, ϕ) .

Solution. First, calculate the radial distance R :

$$R = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = \sqrt{1 + 1 + 2} = \sqrt{4} = 2.$$

Next, find the azimuthal angle θ in the horizontal plane using $x = 1$ and $y = 1$:

$$\tan \theta = \frac{y}{x} = \frac{1}{1} = 1.$$

Since both x and y are positive, θ lies in Quadrant I:

$$\theta = \frac{\pi}{4} \quad (45^\circ).$$

Now, find the vertical polar angle ϕ using $z = \sqrt{2}$ and $R = 2$:

$$\cos \phi = \frac{z}{R} = \frac{\sqrt{2}}{2}.$$

Since the range of ϕ is strictly $[0, \pi]$, we take the unique angle:

$$\phi = \frac{\pi}{4} \quad (45^\circ).$$

Thus, the spherical coordinates are $(2, \frac{\pi}{4}, \frac{\pi}{4})$. □

Example 7.1.4 (Abstract Discussion: Why the Polar Angle is Restricted to $[0, \pi)$). Explain abstractly why the polar angle ϕ is restricted to the interval $[0, \pi)$ ($0^\circ \leq \phi < 180^\circ$) instead of the full circle range $[0, 2\pi)$ used for the azimuthal angle θ .

Solution. To map out all of 3-dimensional space uniquely, we need our coordinate variables to cover the entire globe without overlapping.

Suppose we let both θ and ϕ rotate through a full circle $[0, 2\pi)$: 1. The azimuthal angle θ sweeps out a full horizontal circle of 360° around the vertical z -axis, covering all longitudes. 2. If we also let the vertical inclination angle ϕ sweep out 360° from the North Pole down to the South Pole and back up the other side, we would end up mapping every point in space twice.

For example, starting at the North Pole and swinging down by 180° along the front of the sphere ($\phi = \pi$) reaches the South Pole. If we continue swinging past 180° to 270° ($\phi = \frac{3\pi}{2}$), we are now moving up the back of the sphere.

However, this back-of-the-sphere region is already covered by keeping the vertical angle restricted to $\phi \leq 180^\circ$ and simply rotating the horizontal longitude angle θ by 180° (a half-turn).

Therefore, to prevent this duplicate coordinate mapping and maintain a one-to-one correspondence between physical locations and mathematical coordinates, we restrict ϕ to $[0, \pi]$. This range allows us to travel from the North Pole (0) down to the South Pole (π), while the full rotation of θ through $[0, 2\pi)$ sweeps this vertical arc around the entire globe to cover all remaining directions. □

Example 7.1.5 (Failure of the Coordinate System (Coordinate Singularity)). Explain why the spherical coordinate system fails to assign a unique azimuthal angle θ to any point lying directly on the vertical z -axis (including the origin), and state the mathematical name for this phenomenon.

Solution. Let a point P lie directly on the positive z -axis, meaning its rectangular coordinates are $(0, 0, z)$ with $z > 0$. Let us attempt to convert this into spherical coordinates: 1. The radial

distance is $R = z$. 2. The polar angle is $\cos \phi = \frac{z}{R} = \frac{z}{z} = 1 \implies \phi = 0$. 3. Now, let us find the azimuthal angle θ :

$$\tan \theta = \frac{y}{x} = \frac{0}{0} \quad (\text{Undefined}).$$

Because both x and y are exactly zero, the projection of the point onto the horizontal plane is a single point at the origin of that plane. Since a point has zero dimensions, it has no horizontal direction. Therefore, the angle θ can take *any* value from 0 to 2π while still representing the exact same physical point on the z -axis.

This failure of uniqueness is called a **coordinate singularity** (specifically, a **gimbal lock** or pole singularity). It is the 3D equivalent of the polar coordinate failure at the origin $r = 0$ discussed in Section 2.2, showing that mapping curved coordinates to flat grids always introduces singularities at the boundaries. $\square$

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Problems for Section 7.1

1. Convert the spherical coordinates $(6, 0, \frac{\pi}{2})$ into rectangular coordinates.
2. Convert the spherical coordinates $(2, \pi, \frac{\pi}{4})$ into rectangular coordinates.
3. Convert the rectangular coordinates $(0, 3, 4)$ into spherical coordinates, keeping angles in radians.
4. Find the exact radial distance R of the point $A(3, -4, 12)$ from the origin.
5. Convert the rectangular coordinates $(-1, 1, -\sqrt{2})$ into exact spherical coordinates.
6. A satellite is orbiting at a constant altitude of $R = 7000$ km. If its angular coordinates are $\theta = \frac{\pi}{3}$ and $\phi = \frac{\pi}{2}$, find its rectangular coordinates relative to the Earth's center.
7. Prove using conversion equations that $x^2 + y^2 + z^2 = R^2$.
8. Show that the spherical equation $R = a$ (where $a > 0$) represents a sphere centered at the origin.
9. Explain the geometric shape represented by the spherical coordinate equation $\phi = \frac{\pi}{4}$.
10. Explain the geometric shape represented by the spherical coordinate equation $\theta = \frac{\pi}{6}$.
11. Convert the rectangular plane equation $z = 5$ into its equivalent spherical equation, expressing R as a function of ϕ .
12. Convert the spherical equation $R = 2 \cos \phi$ into its rectangular form, and identify the shape.
13. Find the distance between two points whose spherical coordinates are $A(1, 0, 0)$ and $B(1, 0, \pi)$.
14. If a point has spherical coordinates $(5, \theta, \frac{\pi}{2})$, describe its location relative to the horizontal xy -plane.
15. Let $\phi = \frac{\pi}{2}$. Show that the 3D spherical coordinates (R, θ, ϕ) collapse to standard 2D polar coordinates.

16. A laser is mounted at the origin. To hit a target at rectangular coordinates $(1, \sqrt{3}, 2)$, what azimuthal angle θ and inclination angle ϕ must the laser be steered to?
17. Find the spherical coordinates of the South Pole on a unit sphere centered at the origin.
18. Convert the rectangular cylinder equation $x^2 + y^2 = 9$ into its equivalent spherical equation.
19. Prove that the volume element of a cuboid $dV = dx dy dz$ converts to $dV = R^2 \sin \phi dR d\theta d\phi$ in spherical coordinates. (Hint: Think about the physical dimensions of the curved spherical wedge).
20. Discuss how the concept of latitude and longitude on the Earth's surface represents a real-world, physical implementation of the spherical coordinate system with a fixed radius R .

7.2 Projective Geometry: Stereographic Projection

In the previous subsection, we expanded our geometric workspace to three dimensions and established the conversion formulas between rectangular coordinates (x, y, z) and spherical coordinates (R, θ, ϕ) . Now, let us examine a fascinating constraint: what happens if we lock the radial distance variable, fixing $R = 1$?

By restricting our coordinates to $R = 1$, we are no longer free to move throughout the entirety of 3D space. Instead, we are confined entirely to the 2-dimensional boundary surface of a unit sphere:

$$x^2 + y^2 + z^2 = 1.$$

Even though this surface exists within a 3-dimensional world, it only has two degrees of freedom (the angles θ and ϕ). This means the surface of a sphere is fundamentally a 2-dimensional space.

To analyze this 2D spherical space using our powerful complex variables from Section 4, we need a way to flatten the sphere onto a flat sheet of paper—the complex Argand plane. The mathematical bridge that performs this mapping is called **Stereographic Projection**. By drawing straight projection lines from the North Pole of our sphere through any point on its surface down to a flat plane resting at the equator, we establish a perfect, one-to-one mapping between the curved sphere and the flat complex plane. This framework is known as the **Riemann Sphere**, and it serves as the foundation of projective geometry and complex analysis.

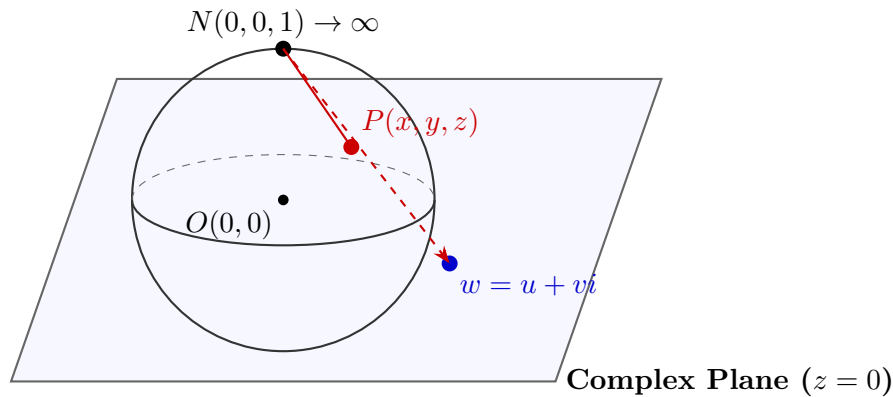


Figure 6: Stereographic projection mapping a point P on the Riemann Sphere from the North Pole N onto the complex plane.

Definition 7.2.1 (Stereographic Projection). Let S^2 be the unit sphere in 3D space defined by the equation $x^2 + y^2 + z^2 = 1$. Let the point $N(0, 0, 1)$ represent the **North Pole** of the sphere.

- Let the horizontal plane $z = 0$ represent the flat complex Argand plane, where any point is written as $w = u + vi$.
- For any point $P(x, y, z)$ on the sphere (excluding the North Pole N), we draw a unique straight line connecting N and P .
- The point where this straight line intersects the equatorial plane $z = 0$ defines the **stereographic projection** of P onto the complex plane, denoted by $w = u + vi$.

Theorem 7.2.1 (Stereographic Projection Equations). *The algebraic mapping between a point $P(x, y, z)$ on the unit sphere and its projected complex coordinate $w = u + vi$ is given by:*

$$\begin{aligned} \text{Sphere to Complex Plane: } u &= \frac{x}{1-z}, \quad v = \frac{y}{1-z} \implies w = \frac{x+iy}{1-z} \\ \text{Complex Plane to Sphere: } x &= \frac{2u}{u^2+v^2+1}, \quad y = \frac{2v}{u^2+v^2+1}, \quad z = \frac{u^2+v^2-1}{u^2+v^2+1} \end{aligned}$$

Proof. Left to Reader as an Exercise. □

Definition 7.2.2 (The Riemann Sphere and the Point at Infinity). Under stereographic projection, as a point $P(x, y, z)$ on the sphere moves closer and closer to the North Pole $N(0, 0, 1)$, its z -coordinate approaches 1. The denominator $(1 - z)$ in our mapping equation approaches 0, causing the projected complex coordinate w to expand infinitely outward ($|w| \rightarrow \infty$).

To complete this one-to-one mapping, we define a single ideal point called the **point at infinity**, denoted by ∞ . We map the North Pole N directly to this point. The set of all complex numbers combined with this point:

$$\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$$

is called the **extended complex plane** or the **Riemann Sphere**.

Example 7.2.1 (Numerical Test on Projective Mapping). Find the exact coordinates of the projected point $w = u + vi$ on the complex plane corresponding to the South Pole $S(0, 0, -1)$ of the unit sphere.

Solution. We are given the coordinates of the South Pole on the sphere:

$$x = 0, \quad y = 0, \quad z = -1.$$

Apply the Sphere-to-Complex projection equations:

$$\begin{aligned} u &= \frac{x}{1-z} = \frac{0}{1-(-1)} = \frac{0}{2} = 0. \\ v &= \frac{y}{1-z} = \frac{0}{1-(-1)} = \frac{0}{2} = 0. \end{aligned}$$

Thus, the projected complex number is:

$$w = u + vi = 0 + 0i.$$

This numerical test proves that the South Pole of the Riemann sphere maps directly to the origin $O(0, 0)$ of the complex plane. □

Example 7.2.2 (Algebraic Expression Derivation). Derive the projection equation $w = \frac{x+iy}{1-z}$ using the parametric equation of a straight line in 3D space passing through the North Pole $N(0, 0, 1)$ and a point $P(x, y, z)$ on the sphere.

Solution. The vector equation of the straight line passing through $N(0, 0, 1)$ and $P(x, y, z)$ can be written parametrically using a real parameter t :

$$L(t) = N + t(P - N) = (0, 0, 1) + t(x - 0, y - 0, z - 1) = (tx, ty, 1 + t(z - 1)).$$

We want to find the point where this line intersects the equatorial plane $z = 0$. Setting the vertical coordinate equal to zero:

$$1 + t(z - 1) = 0 \implies t(1 - z) = 1 \implies t = \frac{1}{1 - z}.$$

Substitute this value of t back into our parametric expressions for the horizontal coordinates u and v on the intersecting plane:

$$u = tx = \frac{x}{1 - z} \quad \text{and} \quad v = ty = \frac{y}{1 - z}.$$

Combining these horizontal coordinates into a single complex number $w = u + vi$ yields:

$$w = \frac{x}{1 - z} + i \frac{y}{1 - z} = \frac{x + iy}{1 - z}.$$

This completes the parametric derivation of the stereographic projection. □

Example 7.2.3 (Algebraic Expression Application). A point on the complex plane is given by the coordinate $w = 1 + i$. Find its corresponding 3-dimensional coordinate (x, y, z) on the surface of the unit sphere.

Solution. Identify the components from the complex coordinate:

$$u = \operatorname{Re}(w) = 1, \quad v = \operatorname{Im}(w) = 1.$$

First, compute the sum of the squares of these components:

$$u^2 + v^2 = 1^2 + 1^2 = 2.$$

Now, apply the Complex-to-Sphere conversion formulas:

$$\begin{aligned} x &= \frac{2u}{u^2 + v^2 + 1} = \frac{2(1)}{2 + 1} = \frac{2}{3}, \\ y &= \frac{2v}{u^2 + v^2 + 1} = \frac{2(1)}{2 + 1} = \frac{2}{3}, \\ z &= \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1} = \frac{2 - 1}{2 + 1} = \frac{1}{3}. \end{aligned}$$

To verify that this point lies on the unit sphere, check if $x^2 + y^2 + z^2 = 1$:

$$\left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = \frac{9}{9} = 1.$$

The calculation is correct. The coordinates on the sphere are $(\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$. □

Example 7.2.4 (Abstract Application: Mapping of the Equator). Show abstractly that the entire equator of the unit sphere maps to the unit circle on the complex plane under stereographic projection.

Solution. The equator of the unit sphere is the horizontal circle where the vertical coordinate $z = 0$.

Substituting $z = 0$ into our unit sphere equation yields:

$$x^2 + y^2 + 0^2 = 1 \implies x^2 + y^2 = 1.$$

Now, apply the Sphere-to-Complex mapping equation to find the modulus of the projected point $|w|$:

$$|w| = \left| \frac{x + iy}{1 - z} \right|.$$

Substitute $z = 0$:

$$|w| = \left| \frac{x + iy}{1 - 0} \right| = |x + iy| = \sqrt{x^2 + y^2}.$$

Since $x^2 + y^2 = 1$ along the equator, we substitute this value:

$$|w| = \sqrt{1} = 1.$$

This abstract proof shows that the modulus of the projected coordinate is always exactly 1, meaning the equator of the sphere maps perfectly to the unit circle $|w| = 1$ on the complex plane. $\square$

Example 7.2.5 (Failure of Flat Projections (Conformal Preservation)). Explain why standard flat map projections (like the Mercator projection) distort the relative sizes of landmasses near the poles, and discuss how stereographic projection handles angles to resolve this distortion.

Solution. It is mathematically impossible to flatten a 3-dimensional sphere onto a 2-dimensional plane without stretching, tearing, or compressing some parts of the shape. This is because a sphere has intrinsic positive curvature, whereas a flat sheet of paper has zero curvature.

Standard cylindrical projections (like the Mercator projection) stretch the regions near the poles horizontally and vertically to fit a rectangular grid. This causes extreme area distortion (for instance, making Greenland appear the same size as Africa, when Africa is physically 14 times larger).

Stereographic projection also distorts areas, expanding the region around the North Pole infinitely. However, it possesses a beautiful geometric property: it is **conformal**. This means that if two curves intersect at a certain angle on the surface of the sphere, their projected curves on the complex plane will intersect at the exact same angle.

While it fails to preserve sizes (areas), it perfectly preserves shapes (angles) locally. This makes stereographic projection highly valuable in conformal mapping, complex analysis, and navigation. $\square$

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Problems for Section 7.2

1. Find the projected complex coordinate w for the point $P(0, 1, 0)$ on the equator of the Riemann sphere.
2. Find the projected complex coordinate w for the point $P(1, 0, 0)$ on the equator of the Riemann sphere.

3. If a point on the unit sphere has coordinates $P\left(\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}\right)$, find its projected complex coordinate.
4. A complex coordinate is given by $w = 3$. Find its corresponding coordinates (x, y, z) on the unit sphere.
5. If $w = -i$, find the corresponding 3D coordinates on the unit sphere.
6. Show that any point in the Southern Hemisphere of the Riemann sphere ($z < 0$) maps to a point inside the unit circle ($|w| < 1$) on the complex plane.
7. Show that any point in the Northern Hemisphere of the Riemann sphere ($z > 0$) maps to a point outside the unit circle ($|w| > 1$) on the complex plane.
8. Find the projected complex coordinate for a point on the sphere that is extremely close to the North Pole, say $P(0.01, 0, 0.9999)$.
9. Prove that under stereographic projection, the reflection of a point through the equatorial plane ($z \rightarrow -z$) corresponds to taking the inversion reciprocal ($w \rightarrow \frac{1}{\bar{w}}$) on the complex plane.
10. Show that a straight line passing through the origin on the complex plane corresponds to a great circle (a circle of maximum diameter) passing through both the North and South Poles on the Riemann sphere.
11. Prove that any circle on the surface of the Riemann sphere that does not pass through the North Pole maps to a perfect circle on the flat complex plane.
12. Prove that any circle on the surface of the Riemann sphere that *does* pass through the North Pole maps to a straight line on the flat complex plane.
13. Find the 3D coordinates on the unit sphere corresponding to the complex number $w = \sqrt{3}i$.
14. If the complex coordinate w lies on the real axis ($\text{Im}(w) = 0$), show that its corresponding point on the Riemann sphere must satisfy $y = 0$.
15. If the complex coordinate w lies on the imaginary axis ($\text{Re}(w) = 0$), show that its corresponding point on the Riemann sphere must satisfy $x = 0$.
16. Let $z_1 = \cos \phi$ represent the vertical coordinate on the sphere. If we increase the polar angle ϕ starting from 0 (the North Pole) to π (the South Pole), describe the movement of the projected point w on the complex plane.
17. Explain the term "extended complex plane" and why it is topologically equivalent to a closed sphere.
18. If we perform a rotation of the Riemann sphere around its vertical z -axis by an angle θ , show that this corresponds to a simple rotation multiplier $w \rightarrow w \cdot e^{i\theta}$ on the complex plane.
19. Find the coordinates of the projected point w if the vertical coordinate on the sphere is $z = 0.5$ and the azimuthal angle is $\theta = 45^\circ$.
20. Discuss how stereographic projection bridges the continuous, infinite boundary of the complex plane with a compact, closed 3D object, providing an elegant way to visualize infinity.